



**PLANNING &
DEVELOPMENT**

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MEMO

To: Honorable Mayor Patrick Collins and Members of the City Council
From: Charles W. Bloom, AICP, Planning and Development Director
Subject: A resolution adopting a Comprehensive Climate Action Plan in accordance with the Environmental Protection Agency's Climate Pollution Reduction Planning Grant awarded to the City.
Date: September 14, 2026

Background:

This resolution formally adopts "Prospering Together: Laramie County's Path to a Sustainable Future" as the City of Cheyenne's Comprehensive Climate Action Plan (CCAP) pursuant to the \$1,000,000 EPA Climate Pollution Reduction Grant accepted under Resolution No. 6388 which was adopted on February 26, 2024.

This plan was developed in partnership with Lotus Engineering & Sustainability, LLC, and was previously managed by the City's prior Economic Resource Administrator. While the plan serves as a foundational policy guide, its adoption does not commit the City to any individual measure, leaving specific future implementation actions to be evaluated separately. Adoption of this plan gives the City a leg up on future funding opportunities for eligible projects related to reducing emissions and improving community resilience such as community solar, energy efficiency retrofits for municipal facilities, fleet electrification, active transportation infrastructure, and sustainable waste management initiatives.

The proposed Governing Body review schedule is as follows:

- | | |
|-----------------------------|--------------------|
| • Introduction and Referral | September 14, 2026 |
| • Public Services Committee | September 21, 2026 |
| • Final Consideration | September 28, 2026 |

Attachments:

1. Resolution and Plan (Exhibit A)

RESOLUTION NO. _____

ENTITLED: "A RESOLUTION ADOPTING A COMPREHENSIVE CLIMATE ACTION PLAN IN ACCORDANCE WITH THE ENVIRONMENTAL PROTECTION AGENCY'S CLIMATE POLLUTION REDUCTION PLANNING GRANT AWARDED TO THE CITY."

WHEREAS, on February 26, 2024, the Governing Body for the City of Cheyenne adopted Resolution No. 6388, accepting an Environmental Protection Agency Climate Pollution Reduction Grants (CPRG) Program planning grant in the amount of one million dollars (\$1,000,000.00) to reduce greenhouse gas emissions and other harmful air pollution;

WHEREAS, CPRG planning grant recipients can utilize funding to design climate action plans that incorporate a variety of measures to reduce greenhouse gas (GHG) emissions across six key sectors, including electricity generation, industry, transportation, buildings, agriculture/natural and working lands, and waste management;

WHEREAS, as a condition of the Phase 1 CPRG Program planning grant, the City of Cheyenne timely submitted a Priority Climate Action Plan (PCAP) to the EPA;

WHEREAS, the CPRG planning grant requires the City of Cheyenne to prepare and submit a Comprehensive Climate Action Plan (CCAP) and a subsequent Status Report evaluating the progress and implementation of the CCAP;

WHEREAS, on August 26, 2024, the City of Cheyenne entered into a Professional Services Agreement, Contract No. 8036, with Lotus Engineering & Sustainability, LLC (the "Consultant") for preparation of a CCAP; and

WHEREAS, the Consultant, with the assistance of the City and community partners, prepared "Prospering Together: Laramie County's Path to a Sustainable Future," a CCAP, which is attached hereto as Exhibit A and incorporated herein.

NOW, THEREFORE, BE IT RESOLVED BY THE GOVERNING BODY OF THE CITY OF CHEYENNE, WYOMING, that the Governing Body hereby adopts "Prospering Together: Laramie County's Path to a Sustainable Future," a Comprehensive Climate Action Plan, attached hereto as Exhibit A.

BE IT FURTHER RESOLVED, that the CCAP shall serve as a policy guide for the City's efforts to promote efficient use of resources, support sustainable practices, and enhance community resilience. Adoption of the CCAP does not commit the City to implement any specific measure or strategy identified within the CCAP, and any future actions will be considered separately as appropriate;

BE IT FINALLY RESOLVED, that the City shall submit a Status Report as required by the Environmental Protection Agency Climate Pollution Reduction Grants (CPRG) Program planning grant.

PRESENTED, READ AND ADOPTED THIS _____ DAY OF _____,
2026.

Patrick Collins, Mayor

(SEAL)

ATTEST:

Kylie Soden, City Clerk

PROSPERING * TOGETHER



Cheyenne MSA **Climate Action Plan**

Prospering Together: Laramie County's Path to a Sustainable Future

On behalf of all cities within Laramie County, Wyoming:

The Prospering Together Plan was written in 2026 to create a healthy and prosperous community that embraces our unique way of life through efficient utilization of resources, environmental stewardship, and collective action.



Prepared For:

The Climate Pollution Reduction Grant Program
State and Local Climate and Energy Program
U.S. Environmental Protection Agency

Prepared By:

City of Cheyenne
Lotus Engineering and Sustainability, LLC
With contribution and review by U.S. EPA



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1. Introduction/Executive Summary

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In This Chapter

1.1 Climate Pollution Reduction Grant Overview

1.2 Climate Change: What is it?

1.3 Community Values Connected to Climate Pollution Reduction

1.4 Comprehensive Climate Action Plan Purpose and Scope

1. Introduction/Executive Summary

The Cheyenne Metropolitan Statistical Area (MSA) received a **\$1 million planning grant** (assistance agreement #5D - 00110301-0) to develop a climate action plan to **reduce climate pollution emissions as part of the Climate Pollution Reduction Grants (CPRG) program**, an initiative of the U.S. Environmental Protection Agency (EPA). The City of Cheyenne spearheaded climate action planning efforts on behalf of all cities within Laramie County, Wyoming (the MSA).

This Comprehensive Climate Action Plan (CCAP) is the second deliverable for the Climate Pollution Reduction Grants program after the Priority Climate Action Plan (PCAP). This plan includes a **greenhouse gas (GHG) emissions inventory, quantified climate pollution reduction measures, and a workforce planning analysis.**

At its core, the Cheyenne MSA Comprehensive Climate Action Plan outlines investments in **policies, practices, and technologies that reduce greenhouse gas and other pollution emissions, create high-quality jobs, spur economic growth, and enhance the quality of life** for all communities and residents of Laramie County. The plan prioritizes equitable access to benefits and ensures that vulnerable communities receive a fair share of the opportunities arising from investment. The flexible strategies in the plan can be **customized by the local communities to meet their specific needs and priorities**, while encouraging local decision-makers to pursue initiatives that will benefit the entire planning area.

The contents of this document do not necessarily reflect the views and policies of the EPA, nor does the EPA or the City of Cheyenne endorse trade names or recommend the use of commercial products mentioned in this document.

This CCAP is organized into 6 sections:



Introduction/ Executive Summary



Climate Pollution Inventory



Quantified Climate Pollution Reduction Measures



Workforce Planning Analysis



Benefits Analysis



Stakeholder Engagement Activities

1.1. Climate Pollution Reduction Grant Overview

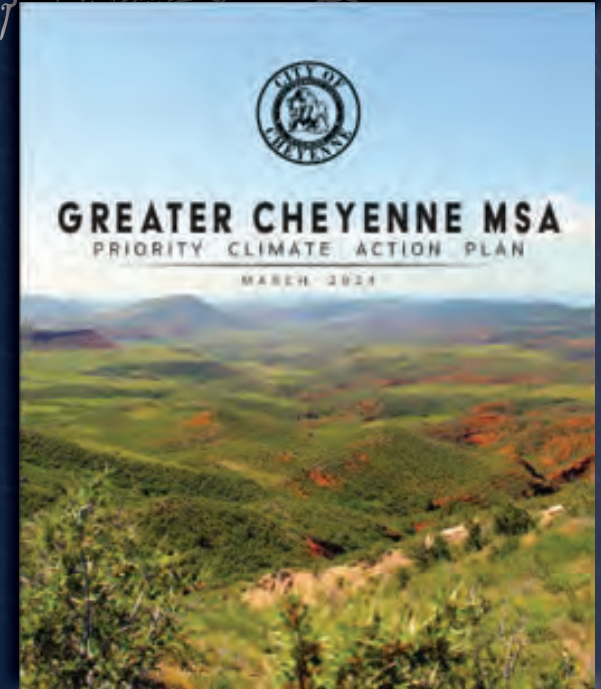
Authorized under Section 60114 of the Inflation Reduction Act and administered by the U.S. Environmental Protection Agency, the Climate Pollution Reduction Grants program provides \$5 billion in grants to states, local governments, tribes, and territories to develop and implement ambitious plans to reduce climate and other harmful air pollution. The first phase of the program provides \$250 million in noncompetitive planning grants to every state and the largest census-designated metropolitan statistical areas. The second phase designates approximately \$4.6 billion in funding for competitive implementation grants. Through the program, the U.S. Environmental Protection Agency seeks to **achieve three broad objectives**:

1. **Tackle climate pollution** while supporting the creation of good jobs and lowering energy costs for families.
2. **Accelerate work to address environmental injustices** and empower community-driven solutions in overburdened neighborhoods.
3. **Deliver cleaner air by reducing harmful air pollution** in places where people live, work, play, and go to school.

Through the Climate Pollution Reduction Grants program, the Cheyenne MSA was allocated a \$1 million planning grant to develop a climate action plan in coordination with local stakeholders throughout the region. U.S. Environmental Protection Agency requirements dictate that the planning grant funds be designated for the completion of the following products:

- Priority Climate Action Plan, due March 1, 2024.
- Comprehensive Climate Action Plan, due December 1, 2025.
- Status Report, due at the close of the 4-year grant period, due Aug. 1, 2027.

Participation in the CPRG program will allow the Cheyenne Metropolitan Service Area to **assess, develop, and implement plans for reducing greenhouse gas emissions and other harmful air pollution** through policy, collaboration, and innovation in an **equitable, just, and sustainable way**.



Cover of the Greater Cheyenne MSA Priority Climate Action Plan (PCAP) from 2024.

1.2. Climate Change: What is it?

Climate change refers to the **long-term shifts in the Earth's temperature, precipitation patterns, and other atmospheric conditions**. These processes are mainly driven by human activities, including the burning of fossil fuels, deforestation, and industrial processes.¹

Human activities leading to detrimental climate change impacts produce greenhouse gas emissions. Greenhouse gases (GHGs) trap heat in the atmosphere, leading to a rise in Earth's temperature and are significant contributors to climate pollution. Greenhouse gases, such as **carbon dioxide (CO₂)**, **methane (CH₄)**, and **nitrous oxide (N₂O)**, have different effects on the planet's warming.² The higher the GHG's ability to absorb energy is, the more it's able to warm the atmosphere over time. Conversely, the longer a GHG remains in the atmosphere, the longer it can exert warming effects.

Increasing amounts of GHG emissions creates an imbalance in the atmosphere, intensifying its harmful effects to the Earth's climate. Climate change is a recurring aspect of our lives, disproportionately impacting vulnerable communities. Simultaneously, the benefits of climate action are increasingly clear in **expanding green jobs, cleaner air, safer homes, and better mobility**. Preparing, and responding, to the risks of climate change signifies improvements to the daily lives of residents across the Cheyenne MSA region and a future that is resilient, equitable, and healthy.

¹ NASA. (2024, October 21). What Is Climate Change? Science.nasa.gov; NASA. <https://science.nasa.gov/climate-change/what-is-climate-change/>

² United States Environmental Protection Agency. (2025, January 16). Overview of Greenhouse Gases. US EPA. <https://www.epa.gov/ghgemissions/overview-greenhouse-gases>



1.3. Community Values Connected to Climate Pollution Reduction

The City of Cheyenne staff identified the following community values/priorities:

- Transportation and connectivity.
- Economic opportunity and development.
- Wildlife and water conservation.
- Air quality.
- Access to nature.
- Protecting rural landscapes.

Across the state, residents ranked the values of **recreation, aesthetics, community, and biodiversity** most highly.³

1.4. Comprehensive Climate Action Plan Purpose and Scope

The Comprehensive Climate Action Plan is a list of **near and long-term measures to reduce climate pollution**, as well as an analysis of emissions reductions that would be achieved through implementation.

³ *Public Values and Preferences Relating to Wyoming's Energy Future*

Over 20 stakeholders
from community groups,
and local or state
agencies attended the
March 2025 Information
Session and shared these
core values for Cheyenne
and Laramie County:

“FREEDOM”

“TOUGH”

“INDEPENDENCE”

“WILDLIFE”

“FAMILY FRIENDLY”

The plan includes:

- **A greenhouse gas inventory** to quantify and track the amount of greenhouse gases (carbon dioxide, methane, nitrous oxide) released into the atmosphere from human activities in the planning area.
- **A list of quantified climate pollution reduction measures** that achieve significant greenhouse gas reductions while considering other relevant planning goals.
- **A benefits analysis** that assesses how the greenhouse gas reduction measures positively impact the region's communities, including, but not limited to, co-pollutant emissions reductions, increased climate resilience, improved access to services and amenities, jobs created, workforce development and reduced costs from energy efficiency improvements
- **A workforce planning analysis** that assesses anticipated workforce shortages, identifying possible solutions and identifying funding and programs that can support current and future workforce needs.

The "Howdy from Cheyenne" rural by Jordan Dean, located at Lincoln and Pioneer.



2. Greenhouse Gas Inventory

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In This Chapter

2.1 Inventory Trends and Analysis

2.2 Near-Term and Long-Term GHG
Reduction Targets

2. Greenhouse Gas Inventory

A greenhouse gas (GHG) emissions inventory is a historical accounting method that **measures the amount of human-caused greenhouse gases emitted or removed** from the atmosphere. GHG inventories are within a predefined time frame, like a certain fiscal or calendar year, and across specific sectors. The GHG inventory covered in this section complies with the Global Protocol for Community-scale Greenhouse Gas Emission Inventories. The inventory includes a blend of measured and estimated activity data for the calendar year 2023 for the MSA. Staff calculated emissions for the emissions sectors highlighted in the table to the right. The GHGs included in the inventory cover carbon dioxide (CO₂), methane (CH₄), nitrous oxide (N₂O) and hydrofluorocarbons (HFC-134a). This section of the plan includes an overview of sources, methodology, and results from the inventory.

Table 1. *Greenhouse Gas Inventory sectors and descriptions.*

Inventory Sector	Description
Building Energy	Electricity generation and consumption: Black Hills Energy and Tri-State Generation and Transmission Association. Staff also collected and estimated usage data for natural gas, stationary diesel, propane, and wood.
Transportation	Gas and diesel powered on-road vehicles, electric vehicles, gas-powered public transit systems, jet fuel and aviation gas for airplanes, and fuel usage for off-road vehicles and equipment.
Waste	Landfilled waste emissions.
Wastewater Treatment	Fugitive and process emissions from wastewater treatment and emissions from septic tanks.
Industrial Processes & Product Use	Emissions from ammonia manufacturing and nitric acid production from Cheyenne-based operations.
Agriculture & Forestry	Livestock, urea fertilization and manure, soil fertilizer and chemicals. It also includes emissions and removals from forests and urban forests.

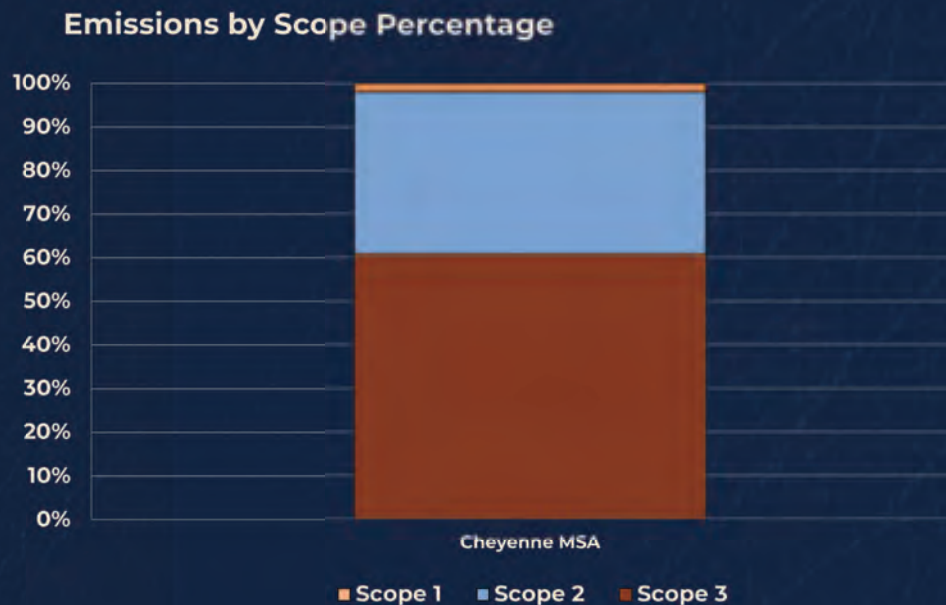
2.1. Inventory Trends and Analysis

The Cheyenne MSA region's greenhouse gas emissions totaled **2,879,148 metric tons of carbon dioxide equivalent (mt CO2e)** in 2023. This value includes the total net carbon removed from the atmosphere by trees and forests. To put the total emissions perspective, the amount of emissions is comparable to the electricity used by over 600,000 homes in a year or the CO2 released from driving a gasoline powered passenger vehicle for more than 7 billion miles.¹ Building energy use makes up over half of the MSA's emissions, followed by transportation, agriculture & forestry, industrial processes & product use (IPPU), waste, and wastewater treatment. **With a population of approximately 100,984 in 2023, the emissions per capita of Cheyenne MSA is 28.5 mt CO2e.**

Greenhouse gas emissions are further categorized into **three scopes (scope 1, 2, and 3)**. Scopes organize emissions sources by a community's boundary and areas of control. This allows for a deeper understanding of specific emissions and pinpoints the greatest emission reduction opportunities.² Scope 1 emissions come from sources occurring within a community's boundary, such as space and water heating fuel use in homes and businesses. Scope 2 emissions are indirect and result from the electricity that is consumed within the community boundary. Scope 3 emissions are all

¹ <https://www.epa.gov/energy/greenhouse-gas-equivalencies-calculator#results>

² <https://www.nationalgrid.com/stories/energy-explained/what-are-scope-1-2-3-carbon-emissions>



Cheyenne's MSA scope breakdown (left) reveals that Scope 1 constitutes the largest portion at 1,749,905 (61%). Scope 2 accounts for 1,073,423 (37%), while Scope 3 represents the smallest share of emissions at 55,820 (2%).

Figure 1. GHG Emissions in the Cheyenne MSA broken out by Scope.

other emissions that occur outside a community's boundary as a result of activities occurring within the community's boundary (i.e., waste generated in homes and businesses disposed of at a landfill in a neighboring community).³

2.1.1. Building Energy

Building energy use contributed 63% of the total emissions, or 1,803,426 mt CO₂e. The largest portion of this energy consumption comes from building electricity use at 37%. Natural gas usage in buildings follows next at 19%. Fugitive emissions from the local natural gas distribution system make up 5%, electricity transmission and distribution losses (power source to consumer) makes up 2%, while stationary diesel, propane, and wood make up less than 1%.

2.1.2. Transportation

In the transportation sector, on-road vehicles with internal combustion engines have the highest share of emissions at **12% of total MSA emissions.** The next largest transportation related emissions source are railways, which generated 3% of emissions, and off-road vehicle fuel use (ATVs, UTVs, 4x4, snowmobiles, etc.) which accounts for 2%. Other sources include: electric vehicles, electric vehicle transmission and distribution losses, public transit, public transit electricity use, and local and itinerant aviation. These sources accounted for less than 1% of total emissions.

³ <https://www.nationalgrid.com/stories/energy-explained/what-are-scope-1-2-3-carbon-emissions>

Emissions by Community and Sector (mt CO₂e)



Figure 2. GHG Emissions (mt-CO₂e) in the Cheyenne MSA broken out by Sector.

2.1.3. Remaining Sectors

In total, **agriculture, forestry, and other land uses contributed 8% of the total emissions, or 240,861 mt CO₂e** with livestock providing 5% and fertilized farm soils 3%. Within the analysis, removals from forestry and urban trees were included. The net GHG flux was positive, due to tree canopy loss.

Industrial processes and product use (IPPU) were responsible for 8% of the total MSA emissions, with industrial processes within the boundary accounting for 99% and refrigerant leakage responsible for 1% of the IPPU sector emissions.

Emissions from waste and wastewater treatment plants accounted for 3% of the total emissions, with **landfilled waste contributing to 96% of the total waste sector emissions due to the presence of the Cheyenne Landfill in the MSA boundary.**

2.1.4. Statewide Resource Conservation Efforts

Wyoming is a major producer of coal, natural gas, and crude oil. **Wyoming has been the top coal-producing state since the mid 1980s, responsible for approximately 40% of all coal mined in the United States in 2024.**⁴

⁴ <https://www.eia.gov/state/?sid=WY>



A herd of steer, which expel methane gas, in a ready-to-be-grazed pasture.

Based out of Casper, the Natural Resource Conservation Service (NRCS), provides programs and services that help landowners and ranchers plan and implement voluntary conservation activities throughout Wyoming.⁵ The NRCS, along with the Wyoming Natural Resource Trust, Conservation Fund, and the United States Department of Agriculture (USDA), have developed different programs with conservation in mind. These programs invest into local communities to ensure working lands continue their operations, **boost rural economies, increase agricultural competitiveness, and improve soil health and quality of life.**^{6, 7}

Utility companies, such as Rocky Mountain Power and Black Hills Energy, are also making sustainable strides to the environment and customers. These companies offer **consumer rebates and incentives for batteries, heating and cooling, smart accessories, heat pumps, thermostats, switches, plugs, etc.**^{8, 9}

2.1.5. State of Wyoming GHG Inventory

Wyoming demonstrates a promising direction in managing their GHG emissions. While a formal statewide GHG inventory has not yet been conducted, the Wyoming Department of Environmental Quality (DEQ) actively monitors industrial emissions from major and minor industries.¹⁰ Additionally, data from the US Environmental Protection Agency showed an **emissions reduction of over 5 million metric tons in Wyoming's total consumption-based (goods and services) GHG emissions between 2012 and 2019.**¹¹ This momentum provides a foundation for future sustainability and conservation efforts and continued and new emission reduction actions.

5 <https://www.nrcs.usda.gov/contact/state-office-contacts/wyoming-state-office>

6 <https://www.nrt.wyo.gov/projects/project-types>

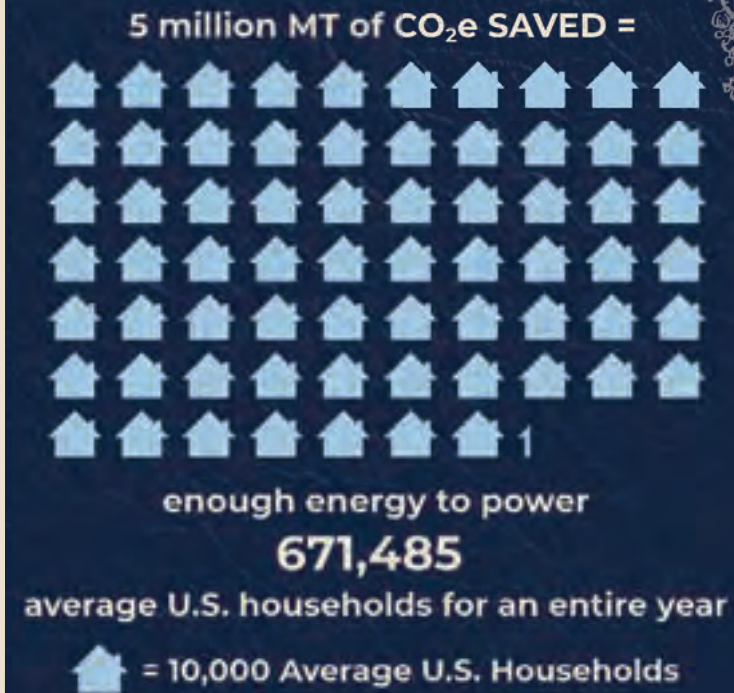
7 <https://www.nrcs.usda.gov/state-offices/wyoming>

8 <https://www.blackhillsenergy.com/efficiency-and-savings/residential-rebates/cheyenne-electric-laramie-county-residential-rebates>

9 <https://www.rockymountainpower.net/savings-energy-choices/business/wattsmart-efficiency-incentives-wyoming/wy-incentive-lists.html>

10 <https://deq.wyoming.gov/aqd/emissions-inventories/>

11 <https://pasteur.epa.gov/uploads/10.23719/1531923/WY.html>



Equivalency data calculated using <https://www.epa.gov/energy/greenhouse-gas-equivalencies-calculator>

2.2. Near-Term and Long-Term GHG Reduction Targets

During the 2024 Wyoming legislative session, several conservation-focused house bills were passed. Key outcomes include the sale of the Kelly Parcel, containing crucial big game habitat and migration routes, to Grand Teton National Park. Approximately **\$9 million was allocated for the removal of non-native, invasive grasses**. Legislation also authorized the Wyoming Wildlife Trust to allocate matching funds for conservation projects including habitat enhancement for various big game species and a project on fish passages.¹²

In addition to Wyoming's state level actions, in 2020, the Laramie City Council adopted **a goal to achieve net-zero carbon emissions by 2050**.¹³ Although Cheyenne currently lacks overarching sustainability goals, the city and its surrounding region is advancing initiatives outlined in the previous and current Priority Climate Action Plan (PCAP)/ Comprehensive Climate Action Plan (CCAP). These initiatives include **residential and commercial building electrification, vehicles miles traveled (VMT) reduction, electric vehicle (EV) adoption, etc.** Furthermore, within the City of Cheyenne, the Cheyenne Botanic Gardens has been implementing sustainable measures, including on-site solar and wind turbines, and water conservation, since 1977.¹⁴ These local efforts and initiatives demonstrate Cheyenne MSA's commitment to environmental stewardship and climate action.

¹² <https://www.trcp.org/2024/03/25/wyoming-conservation-wins-from-the-2024-legislative-session/>

¹³ <https://www.cityoflaramie.org/1385/Emissions-Goals>

¹⁴ <https://www.botanic.org/about/sustainability/>

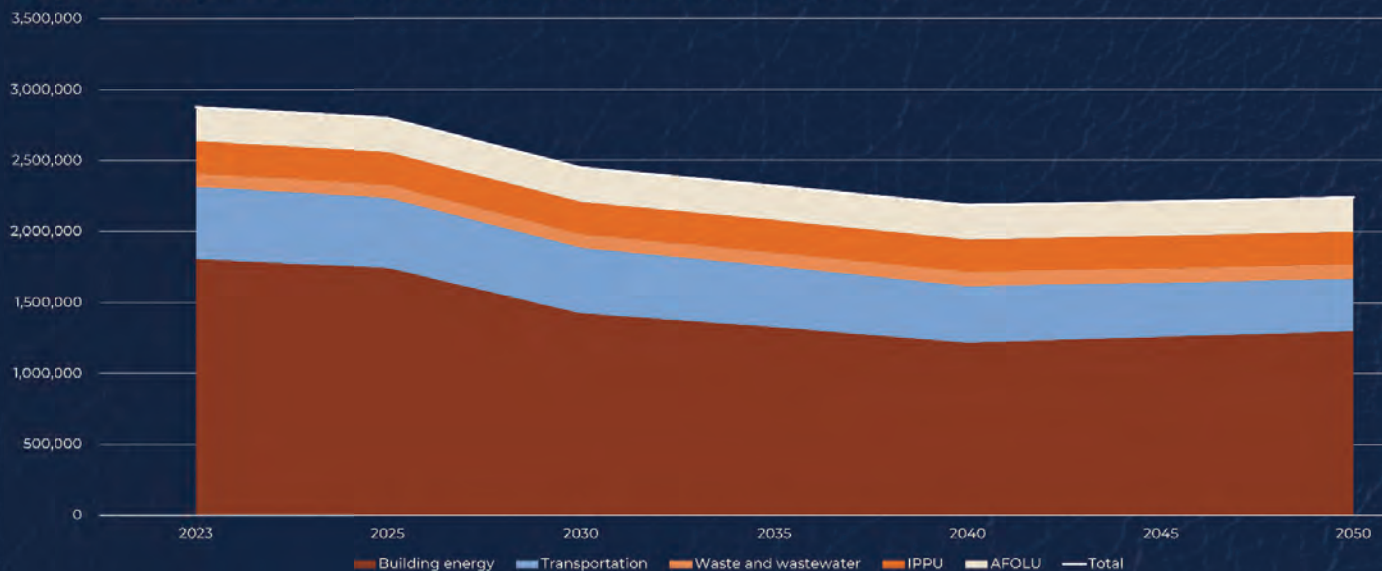


Cheyenne residents enjoying a restaurant patio after participating in Bike to Work Day.

2.2.1. Business-as-usual Projections

The business-as-usual scenario (BAU) illustrates what would happen **if Cheyenne MSA did not make any changes in policies and/or investments to reduce emissions. This would be considered as a “worse-case scenario”** related to the amount of intentional intervention and illustrates the region's energy and buildings, on-road transportation, workforce, waste management and water, and natural and working lands emissions in 2050 if there is no additional regional climate pollution reduction.

Cheyenne Metropolitan Statistical Area Business-as-Usual Emissions Forecast



Important note: At the time of the Greenhouse Gas Inventory and BAU projections, the actual and projected energy use from Data Centers in the region were not calculated due to a lack of data.

Figure 3. GHG Emissions (mt-CO2e) projections for the Cheyenne MSA through 2050.

3. Quantified Climate Pollution Reduction Measures

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In This Chapter

- 3.1 Energy and Buildings
- 3.2 Transportation
- 3.3 Waste Management and Water
- 3.4 Natural and Working Lands
- 3.5 Workforce Planning



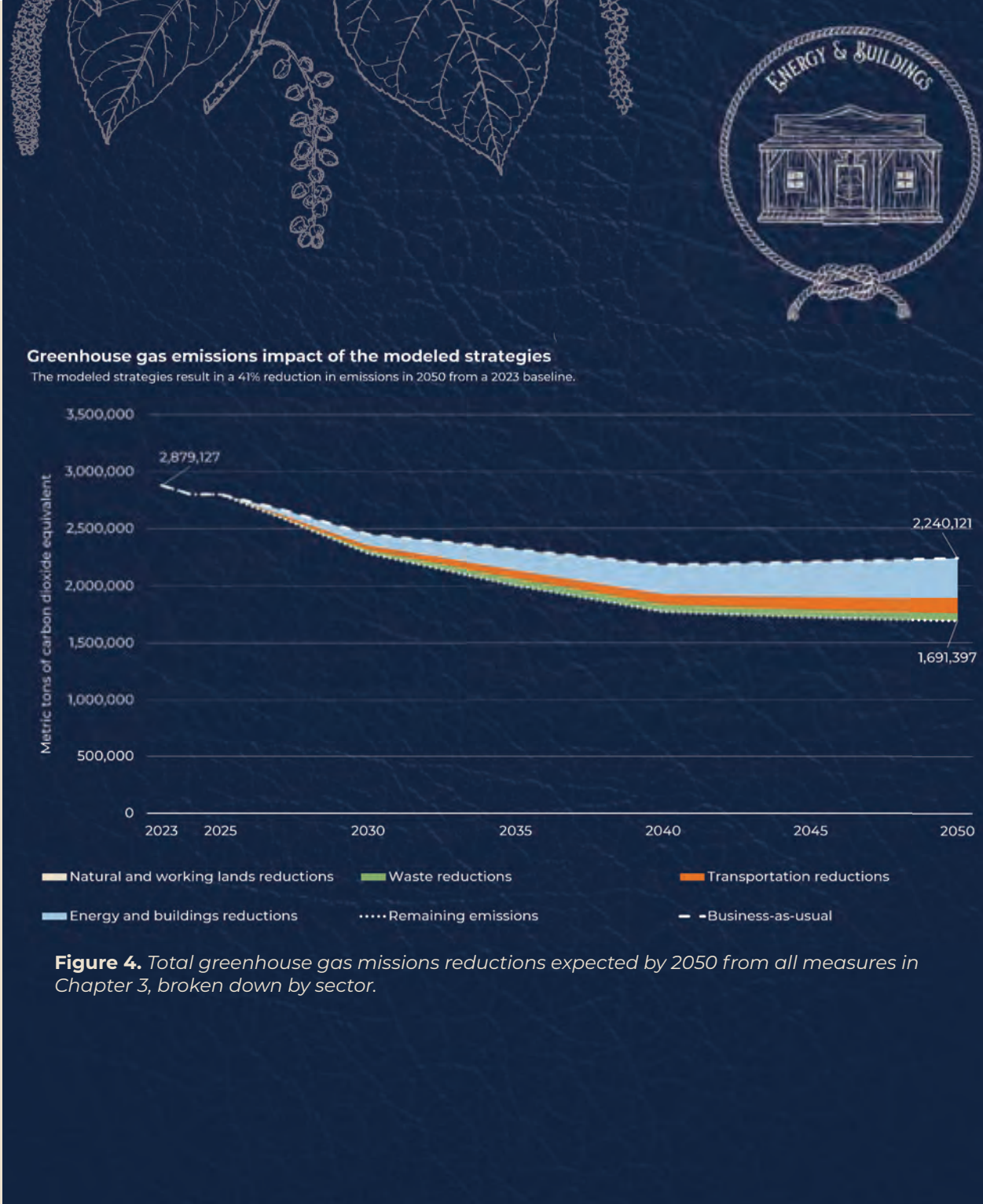
3. Climate Pollution Reduction Measures

3.1. Energy and Buildings

3.1.1. Introduction

The energy and buildings sector covers **emissions from electricity, natural gas, stationary diesel, and wood use**. This sector also includes natural gas fugitive emissions, which are natural gas leaks that occur during production, storage, and transportation, and losses from energy transmission and distribution, which happens when electricity is lost as it is transported from power plants to homes and businesses.

Buildings energy use is the largest source of emissions in the MSA region, accounting for roughly 62% of total emissions. Tackling emissions in this area is important, not only for the environment and wildlife, but **helps residents and businesses save money on energy bills, while also building a more resilient energy grid.** The goals and measures for this sector focus on cutting emissions while supporting the region's economic growth and stability.



Goals:

- Selectively expand clean energy development (i.e., solar, geothermal, wind, nuclear) to diversify the region's economy and reduce emissions.
- Increase education, incentives, access to, and adoption of energy efficiency measures to reduce ongoing energy bills.
- Mitigate emissions of businesses.

3.1.2. Energy and Buildings Measures

Measure 1: Increase energy efficiency and electrification in residential buildings.

An easy way to **save money on energy and reduce emissions** from the residential energy use sector is by using **less energy overall**.

This measure aims to reduce energy use in the residential buildings sector. One way to do that is by weatherizing residential homes and apartments; this can be done through **additional insulation, thicker windows, sealing cracks, and more. Also, electric appliances like heat pumps and heat pump water heaters, are proven to be more efficient** at using energy than natural gas powered appliances. Heat pumps also come with the added benefit of providing air conditioning in addition to heating. Strategies under this measure include taking advantage of existing **rebate and incentive programs, community education, and continued updates to building energy and construction codes.**

Energy and Buildings Measure 1 Costs and Benefits:

GHG Impact Potential	Government Cost	Resident/Business Cost	Government Benefit	Resident/Business Benefit
	\$\$	\$\$-\$\$\$	None	\$



The Wyoming State Capitol Building, decked out in red, white, and blue ahead of America's 250th Independence Day.



Strategies:

- Adopt the latest building energy codes.
 - Convene a code panel to review the latest version of the International Energy Conservation Code.
 - Provide education to the public, builders, contractors, engineers, and architects to help them understand code changes.
- Incentivize energy efficient upgrades and all-electric equipment in residential buildings.
 - Promote existing rebate programs through utilities and the state and provide additional incentives for energy efficiency and electrification to the extent allowed under applicable laws to help fill gaps.
 - Lower the cost of building permits for buildings that implement energy efficiency measures above and beyond what the code requires.
 - Provide grant funding to encourage landlords to implement energy efficiency measures.
 - Establish the administrative framework and partnerships necessary to launch the PACE program that was authorized through resolution 6135 on April 2, 2021.
 - Explore implementing an energy efficiency pilot program in South Cheyenne.
- Provide education on energy-efficient appliances and how they can lower home energy bills.
 - Encourage contractors
 - and builders to educate themselves on residential energy code compliance.
 - Provide educational materials for residents on energy efficiency measures and their costs and benefits, using effective messaging that calls on people's motivation around environmental stewardship.



Strategies Continued:

- Support energy efficiency projects in low-income and affordable housing.
- Leverage the Wyoming Community Development Authority (WCDA) Spruce Up Program.
- Expand weatherization retrofits across the region.

Implementation Timelines and Milestones:

- Code panel review is completed.
- Latest building energy codes are adopted.

Metrics for Tracking Progress:

- Change in power bills or electricity rates.
- Energy usage before and after upgrades.
- Participation in utility energy efficiency rebate programs.
- Number of weatherization retrofits in the region.
- Usage rate of incentives.
- Quantified GHG Emission Reduction

Energy and buildings measure 1: Cumulative GHG emission reductions

This strategy results in a 9% reduction in building emissions in 2050 from a 2023 baseline.

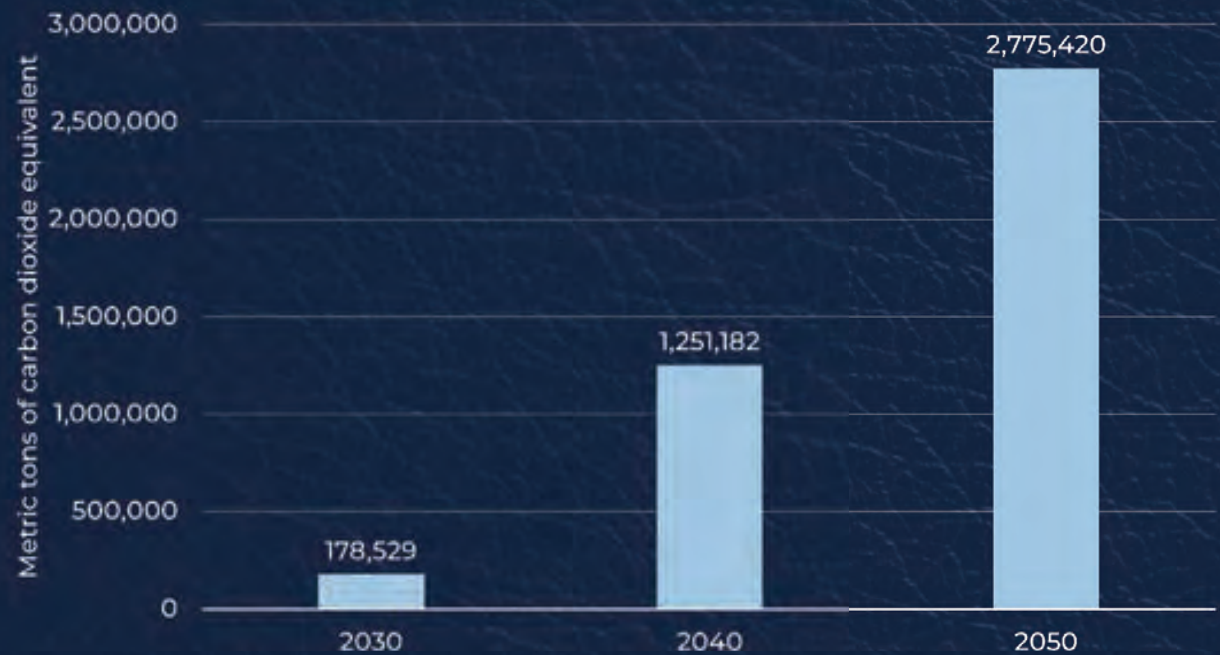


Figure 5. Cumulative greenhouse gas emissions reduction from Energy and Buildings Measure 1 through 2050. Scenario: 75% improvement in new home efficiency, 50% improvement in existing home efficiency, 50% heating equipment replaced with electric by 2040.

Energy and buildings measure 1: Cumulative reduction in co-pollutants in 2050

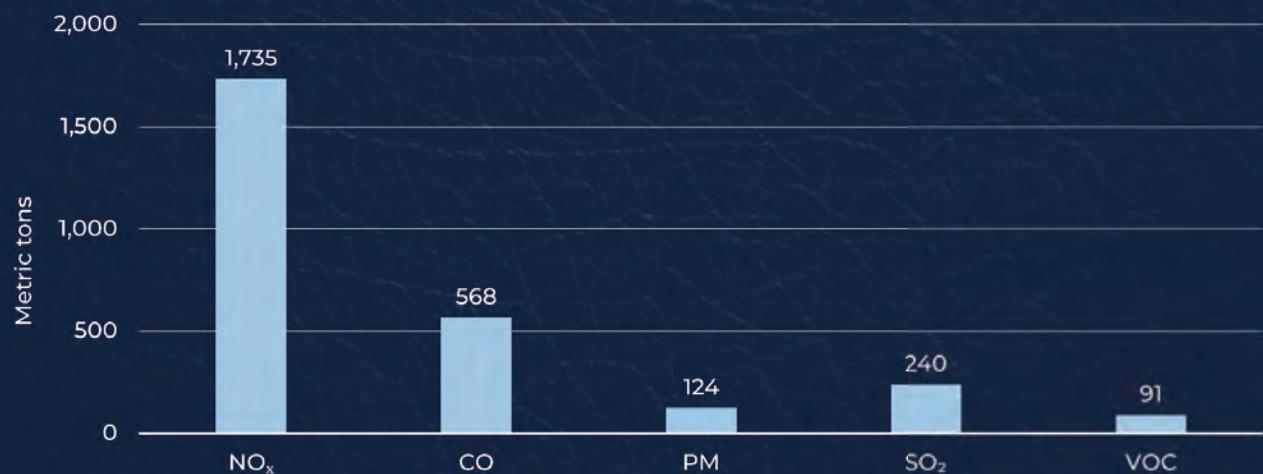


Figure 6. Co-pollutant emissions reductions from Energy and Buildings Measure 1 through 2050.

Measure 2: Increase energy efficiency and electrification in commercial and industrial buildings.

Similar to the residential space, **a simple way for industries and companies to save money on energy is to use less.** This measure aims to increase energy efficiency and electrification in commercial and industrial buildings. Energy efficiency efforts can be supported through upgrading to **LED lighting, occupancy lighting sensors, more insulation in walls and roofs, and energy efficient windows.** Strategies under this measure include incentivizing and educating energy efficient appliances and equipment and adopting the latest energy codes.

Strategies:

- Adopt the latest building energy codes pursuant to Wyoming Statute § 35-9-121(a)(ii) .
 - Pursue grants to achieve LEED certification in new public buildings.
- Incentivize energy efficient upgrades and all-electric equipment to the extent allowed under applicable laws.
 - Provide tax credits for businesses to implement energy efficiency and electrification.
- Provide education on energy-efficient measures to help businesses learn about and navigate opportunities.
 - Showcase projects that have been completed, such as energy efficiency and renewable energy projects at agricultural producers and rural small businesses funded through the Rural Energy for America Program.
 - Encourage contractors and builders to educate themselves on commercial energy code compliance.

Energy and Buildings Measure 3 Costs and Benefits:

GHG Impact Potential	Government Cost	Resident/Business Cost	Government Benefit	Resident/Business Benefit
	\$\$	\$\$-\$\$\$	None	\$



The Wyoming Capitol building in Cheyenne, framed by snowy branches.



Implementation Timelines and Milestones:

- Tax credits are created.
- Project showcases are hosted.

Metrics for Tracking Progress:

- Number of tax credits provided.
- Number of project showcases hosted.
- Engagement methods, like number of builders educated, number of residents reached via educational campaigns, etc.

Energy and buildings measure 2: Cumulative GHG emission reductions

This strategy results in a 10% reduction in building emissions in 2050 from a 2023 baseline.

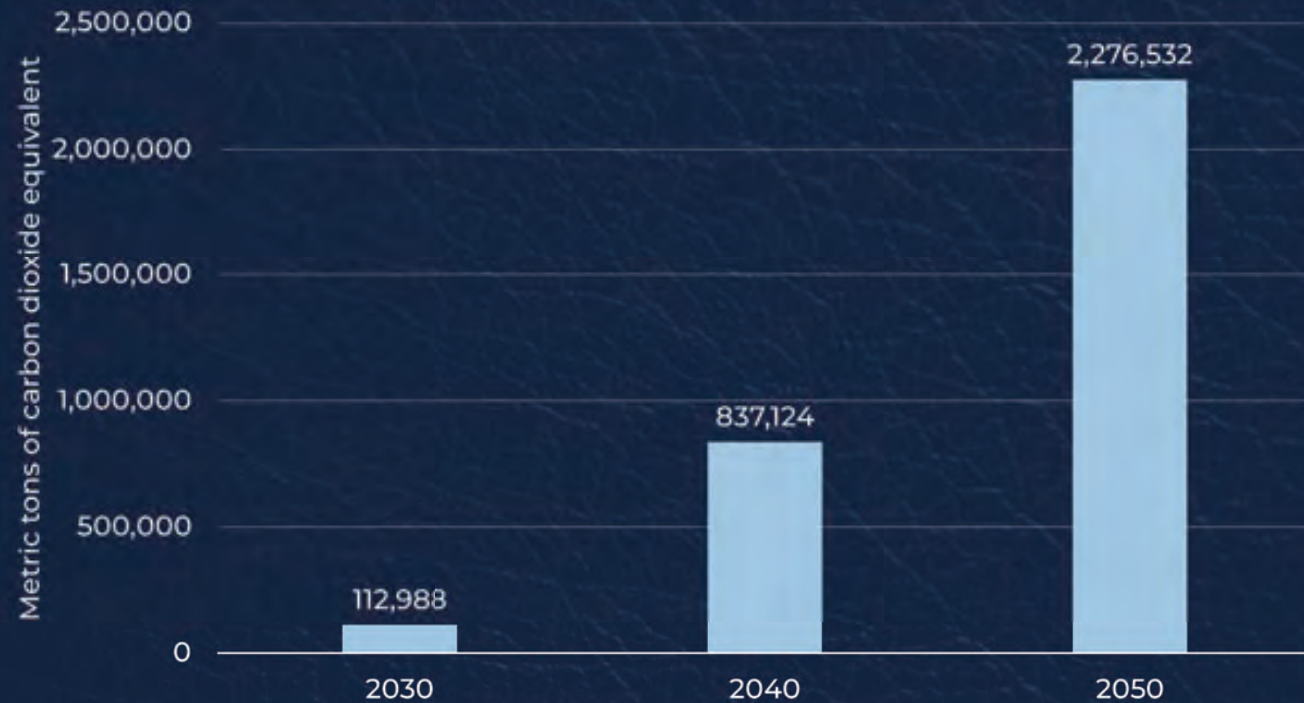
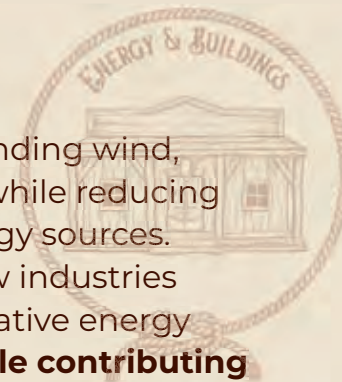


Figure 7. Cumulative greenhouse gas emissions reduction from Energy and Buildings Measure 2 through 2050. Scenario: 75% improvement in new commercial efficiency by 2050, 25% improvement in existing commercial efficiency by 2040, 100% heating equipment replaced with electric by 2050



Measure 3: Encourage adoption of alternative energy.

Wyoming is the nation's leading energy producer, yet the state heavily relies on fossil fuels. Expanding wind, solar, and other alternative energy technologies will diversify the Cheyenne region's energy portfolio, while reducing GHG emissions. This measure aims to increase adoption, awareness, and education of alternative energy sources. By integrating alternative energy, industries can support grid resilience and the region can attract new industries and businesses. Strategies under this measure include education and outreach for residents on alternative energy resources. This energy transition **positions Wyoming to continue leading in energy production, while contributing to the state's sustainability goals.**

Energy and Buildings Measure 3 Costs and Benefits:

GHG Impact Potential	Government Cost	Resident/Business Cost	Government Benefit	Resident/Business Benefit
	\$\$\$\$	\$-\$\$\$\$	\$	\$-\$\$

See more details in
Appendix C: Workforce
Analysis.



Strategies:

- Encourage the adoption of clean energy through leading by example at municipal facilities.
- Continue piloting municipal clean energy projects to reduce operating costs, such as solar parking lots, wind turbines, or a project at the landfill.
- Showcase completed projects such as solar, geothermal and hydroelectric projects at public facilities.
- Expand biogas capture at the wastewater treatment facility so it can be used for fuel or electricity production.
- Collaborate with Laramie County and the State to showcase wins from energy efficiency programs.

Energy and buildings measure 3: Cumulative GHG emission reductions

This strategy results in a 1% reduction in building emissions in 2050 from a 2023 baseline.

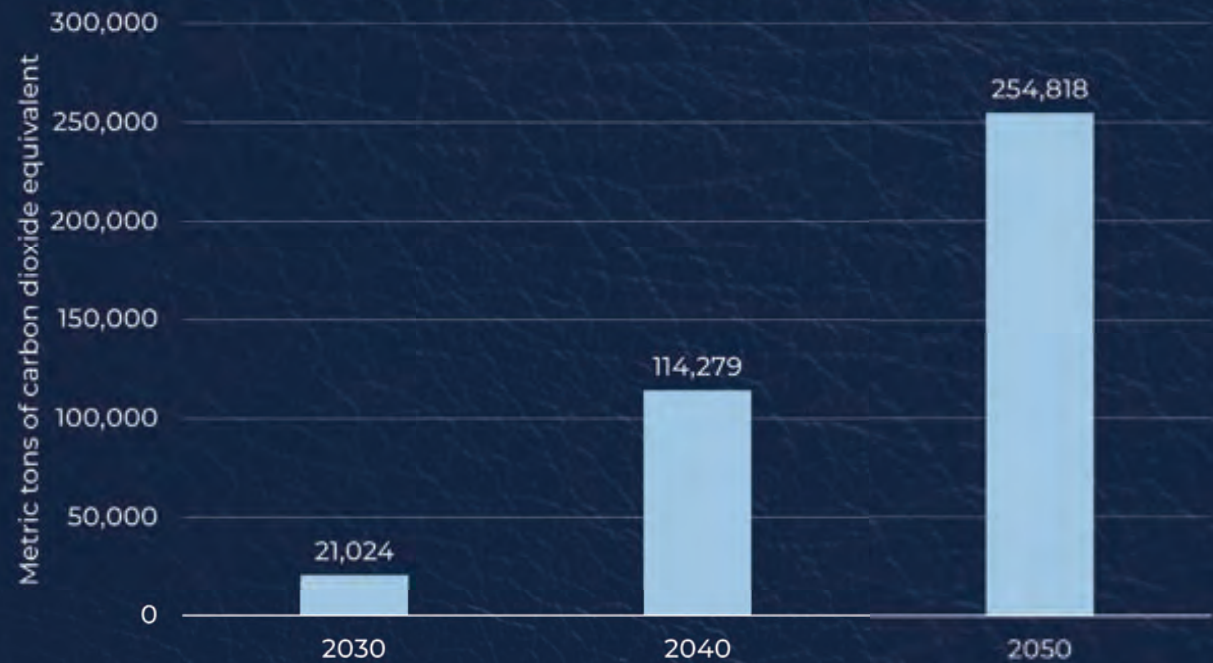


Figure 8. Cumulative greenhouse gas emissions reduction from Energy and Buildings Measure 3 through 2050. Scenario: Target of 50% installed solar capacity by 2050.

Energy and buildings measure 3: Cumulative reduction in co-pollutants in 2050

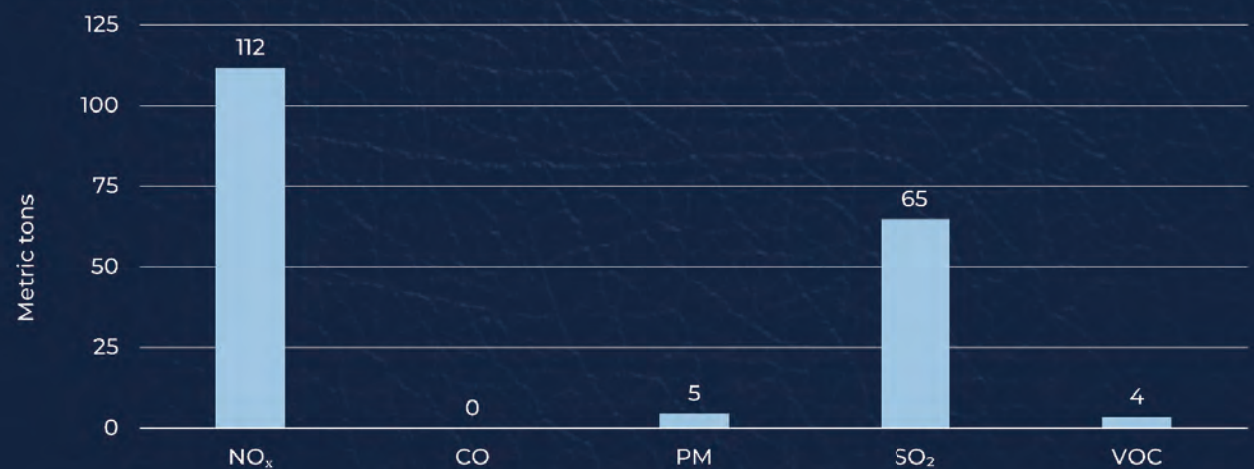


Figure 9. Co-pollutant emissions reductions from Energy and Buildings Measure 3 through 2050.

Strategies Continued:

- Educate and empower residents and businesses to implement clean energy projects.
 - Provide educational programs demonstrating new technologies like solar and battery storage at the Children's Museum or a similar business to encourage solar on homes.
 - Provide information on net metering to remove concerns and uncertainty relating to solar.
 - Provide education on renewable energy systems, especially ground-mounted home solar, and its lifecycle costs and benefits.
- Explore innovative projects to maintain grid reliability.
 - Encourage data centers to create on-site renewable energy.
 - Implement land use policies to regulate data centers, such as co-locating data centers near more nimble energy users that can quickly scale their energy use up or down.
 - Explore the feasibility of microgrids for rural areas.

Implementation Timelines and Milestones:

- Municipal pilot projects are completed.
- Educational programs and materials are created.
- Microgrid research is completed.
- Land use policies have been implemented

Metrics for Tracking Progress:

- Amount of renewable energy generation on public facilities.
- Operating costs reduced from alternative energy projects.
- Energy bill savings.
- Amount of on-site renewable energy produced at data centers.
- Number of data centers using on-site renewables.



Historic buildings on W Lincoln Way in downtown Cheyenne. This street hosts many successful local businesses.

Measure 4: Implement policies and programs that reduce the overall impact of buildings.

Generally, the built environment is one of the largest contributors of energy use, but has the easiest efficiency opportunities. By promoting high performance building standards, expanding incentives for retrofits, and improving energy management practices, the Cheyenne MSA can significantly reduce electricity and heating demands. This measure not only reduces long-term utility costs for residents and businesses, but also supports cleaner building operations. Strategies that support this measure include managing the impacts of new and existing buildings and the environmental building benefits.

Energy and Buildings Measure 4 Costs and Benefits:

GHG Impact Potential	Government Cost	Resident/Business Cost	Government Benefit	Resident/Business Benefit
	\$\$	\$\$-\$\$\$\$	\$	\$

Strategies:

- Address the impacts from retrofits, new construction, and demolition of buildings.
 - Right-size the use of space when retrofitting or repurposing existing buildings.
 - Explore different funding opportunities or apply building permit fee reduction incentives, or similar, to preserve or redevelop historical buildings in a sustainable way.
 - Explore ways to reduce the costs of adaptive reuse and demolition requirements.
 - Implement a construction & demolition program that requires the reuse and recycling of building materials.
- Encourage building features that provide environmental benefits.
 - Require new developments to reflect the natural land that exists in the region.
 - Promote green roofs on large new developments, especially downtown.
 - Adopt dark sky building requirements to limit light pollution.
 - Continue to limit light pollution in city parks and facilities through strategic lighting design.
 - Map migratory bird routes to increase bird window safety for public buildings and data centers.



Implementation Timelines and Milestones:

- Construction and demolition (C&D) diversion and recycling program is implemented.
- Dark sky friendly requirements are adopted.
- Incentivization programs are continued, created, or expanded to promote rehabilitation of historic properties or rehabilitation of existing structures.

Metrics for Tracking Progress:

- Permits for re-use site plans (applied and completed).
- SHPO/National Trust for Historic Preservation and National Main Street Center have metrics and research on preservation costs and energy savings.
- Cost reduction of adaptive reuse and demolition requirements.
- Amount of C&D materials recycled.
- Number of historical buildings preserved or redeveloped sustainably.

Emissions reductions and Co-pollutant impacts were not modeled for this set of strategies.

Measure 5: Implement policies and programs that increase the availability of affordable housing and market rate housing.

Affordable or low-income housing supports community resilience and the **long-term economic health of residents**. Expanding **incentives for mixed-income developments, limiting urban sprawl, and supporting renovations of existing units** extends housing options and reduces mobility and/or opportunity burdens. This measure aims to increase **affordable and accessible housing options** for residents and visitors throughout the Cheyenne MSA region. Expanding affordable housing **supports a diverse workforce, strengthens the MSA's economy, and ensures growth continues to be sustainable and equitable**. Key strategies for this measure include working with local organizations to ensure and **encourage redevelopment programs and improve and/or renovate aging or existing buildings**.



The Atlas Theatre, built in 1887, is one of Cheyenne's many historic buildings and is on the National Registry of Historic Places.



Strategies:

- Improve the aging infrastructure in downtown Cheyenne while encouraging housing redevelopment projects and incentivizing infill growth.
- Promote the development of high-density housing through density bonuses or other mechanisms.
- Work with the Downtown Development Authority on grant programs for downtown redevelopment projects.
- Work with the Urban Renewal Authority for downtown redevelopment projects.
- Partner with Habitat for Humanity, Wyoming Housing Network, and other housing developers.

Implementation Timelines and Milestones:

- Grant programs for redevelopment projects are created.
- Metrics for Tracking Progress
- Number and location of redevelopment projects implemented.
- Number of market and affordable units added to the housing inventory.

Emissions reductions and Co-pollutant impacts were not modeled for this set of strategies.

Energy and Buildings Measure 5 Costs and Benefits:

GHG Impact Potential	Government Cost	Resident/Business Cost	Government Benefit	Resident/Business Benefit
	\$\$	None	\$\$	\$\$

Energy and buildings: Cumulative GHG emission reductions

Modeled measures result in a 19% reduction in building emissions in 2050 from a 2023 baseline.

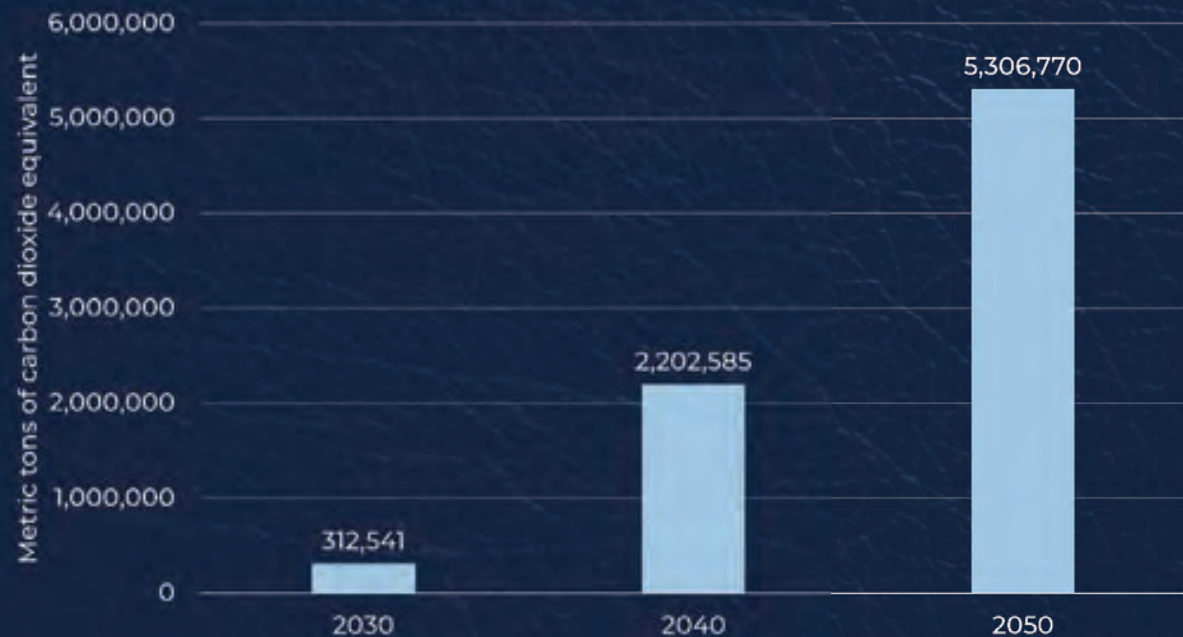


Figure 10. Cumulative greenhouse gas emissions reduction from all Energy and Buildings Measures.

Letter on Data Centers

Dear Community,

This Comprehensive Climate Action Planning process began back in late 2023, made possible by a grant from the US Environmental Protection Agency's (EPA) Climate Pollution Reduction Grant program. **Using best available data from local, state, and federal resources, greenhouse gas emissions were calculated, following the Global Protocol for Community-scale Greenhouse Gas Emissions Inventories.** From those results, strategies were created and prioritized to help the Cheyenne and Laramie County region achieve sustainability goals, as vetted by City staff and the public. The EPA required the plan to have strategies that addressed emissions produced in the following sectors: commercial and residential buildings, transportation, waste and wastewater treatment, industry, electricity generation, agriculture, forestry, and other land uses, and natural and working lands.

What was not included in that original community-wide greenhouse gas emissions inventory was emissions from planned or under-construction data centers in the region. As these facilities were not completed, actively consuming energy, and creating emissions, their **data was not available** for inclusion into the region's baseline emissions inventory.

The construction of data centers and permit requests to build these facilities have increased quickly across the United States and locally here in the Laramie County region. In fact, in 2026 alone the US is anticipated to nearly double the quantity of data centers, the majority of which are being proposed and built in rural areas ([Pew Research](#)). We recognize that **these facilities can be physically large and may consume large amounts of electricity, water, and refrigerants** to keep the servers up and running. This can lead to increased greenhouse gas emissions and other environmental impacts locally.

Going forward, **so long as data is made available, the region will include energy use and greenhouse gas emissions from data centers in the regional greenhouse gas emissions inventory.** It is important to continue to track and understand these facilities' impacts locally, both in creating jobs as well as their use of electricity, water, and other resources. This plan is meant to be a living document that can be revisited at any time to address new sources of emissions or environmental challenges that occur over time and to achieve **the plan's ultimate goal to create a healthy and prosperous community that embraces our unique way of life** through efficient utilization of resources, environmental stewardship, and collective action.

Sincerely,

The City of Cheyenne

3.2. Transportation

3.2.1. Introduction

Transportation is a big part of everyday life in the Cheyenne MSA region. People rely on transportation to go to work, school, shops, and ultimately shapes how the community functions and grows. **The transportation sector is the second largest source of emissions in the MSA region, accounting for approximately 18% of total emissions.** This sector includes emissions from on-road fossil fuel use, electric vehicles, electric vehicles transmission and distribution losses, public transit electricity use, aviation, off-road vehicles (i.e., UTVs/ATVs), and railways.

Addressing the emissions not only **helps the region reach their sustainable goals, but improves community health, eases traffic congestion, and supports a connected and efficient community.** The goals for this sector focus on having fewer vehicles on the road, creating safer places for residents to walk and bike, and providing reliable transportation options that connect people to the places they need or want to go.

Goals:

- Reduce cars on the road while ensuring it is safe to bike and walk.
- Remove barriers to active transportation.
- Provide easy access to transportation options that connect people from their homes to where they work, shop, etc.

3.2.2. Transportation Measures

Measure 1: Improve the active transportation network in Cheyenne to reduce traffic-related incidents and encourage walking and biking.

When communities invest in protected bike lanes and well-designed sidewalks and street crossings, they **expand transportation opportunities and reduce the reliance on personal vehicles.** This measure aims to improve the transportation system to increase safety. These improvements not only **reduce traffic-related accidents but also make travel more enjoyable for community members** who depend on alternatives to vehicles for transport. By having a well-known active transportation system, the Cheyenne MSA connects and engages residents and visitors to **promote healthier lifestyles and builds a community that values safety.** This measure includes strategies such as expanding and promoting the active transportation network throughout the MSA's region.

Transportation Measure 1 Costs and Benefits:

GHC Impact Potential	Government Cost	Resident/Business Cost	Government Benefit	Resident/Business Benefit
	\$\$\$	\$	\$	\$\$



Strategies:

- Expand the active transportation network.
- Prioritize the active transportation network.
- Include bicycle facilities in road construction and restriping projects.
- Promote the active transportation network so it's more widely used.
- Increase the safety of active transportation citywide

Implementation Timelines and Milestones:

- All sidewalks, bike lanes, and Greenway segments in Cheyenne are connected.
- Bike signage is implemented.
- All bus stops have paths and sidewalks leading to it.

Metrics for Tracking Progress:

- Traffic incidents.
- Miles of bike routes, designated bike lanes, and protected and separated bike lanes added.
- Miles of multi-use pathways and Greenway segments added.

Transportation measure 1: Cumulative GHG emission reductions

This strategy results in a 0.04% reduction in transportation emissions in 2050 from a 2023 baseline.

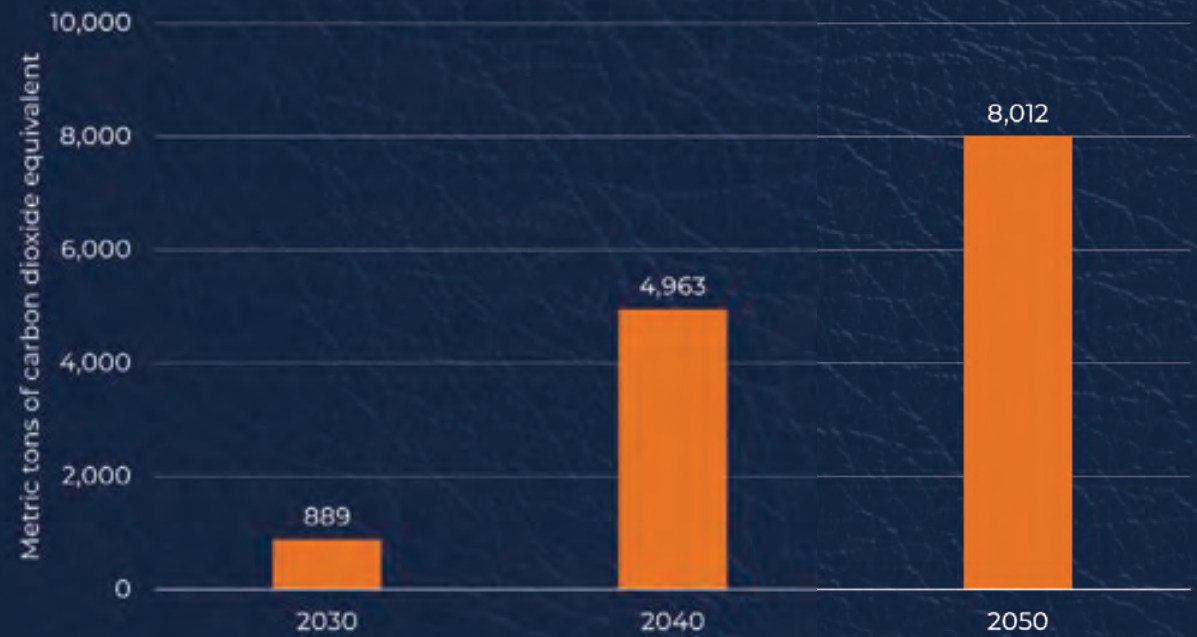


Figure 11. Cumulative greenhouse gas emissions reduction through 2050 from Transportation Measure 1. Scenario: 10% of vehicle trips of 2 miles or less converted to walking/biking between by 2040

Transportation measure 1: Cumulative reduction in co-pollutants in 2050

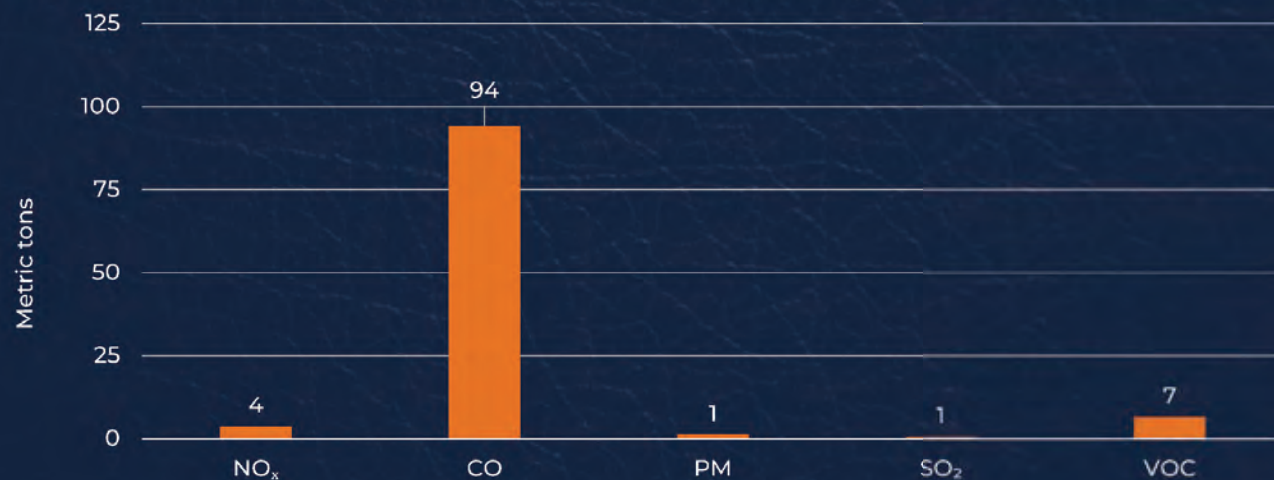


Figure 12. Cumulative co-pollutant emissions reduction through 2050 from Transportation Measure 1. Scenario: 10% of vehicle trips of 2 miles or less converted to walking/biking between by 2040.

Measure 2: Promote the adoption of alternative fuel vehicles.

Fossil fuel powered vehicles create not only GHG emissions, but also pollute the local air quality via tailpipe emissions. There are increasingly more alternative fuels to power vehicles, which also help to **improve local air quality** through lower tailpipe emissions. This strategy aims to increase awareness in the Cheyenne MSA of adoption, and education around non-fossil fuel-powered vehicles. Strategies aimed at achieving the measure include **expanded electric vehicle charging infrastructure in the MSA as well as having the City lead by example.**

Over the years, **electric vehicles continue to grow in popularity and affordability**, as newer models enter the market with higher driving ranges and lower prices. This measure aims to increase adoption, awareness, and education on non-fossil fuel-powered vehicles. When communities offer free/low-cost charging stations and incentives for new and used electric vehicles, they give the residents real alternatives to gas and diesel-powered vehicles. These steps **reduce tailpipe emissions, while improving local air quality.** Strategies that support this measure include community and government efforts.

Transportation Measure 2 Costs and Benefits:

GHG Impact Potential	Government Cost	Resident/Business Cost	Government Benefit	Resident/Business Benefit
	\$\$	\$\$\$	\$	\$

Strategies:

- Increase the proportion of alternative fuel vehicles in Cheyenne's fleet.
- Encourage the use of hybrids, compressed natural gas, or hydrogen vehicles for the City fleet.
- Identify viable funding sources to purchase hybrids, compressed natural gas, or hydrogen vehicles for public fleets.
- Promote the adoption of alternative fuel vehicles within the community.



An electric vehicle fueling up at one of the many charging stations available in and around Cheyenne.



Strategies Continued:

- Install EV chargers at locations desired by residents and visitors in partnership with the utility to ensure sufficient power supply.
- Demystify EVs through public education events such as EV demo days.
- Install more EV chargers outside of the City to provide more options and reduce range anxiety.
- Develop workforce training programs for EVs.

Implementation Timelines and Milestones:

- City secures funding for and purchases hybrids, compressed natural gas, or hydrogen vehicles for the City fleet.
- EV chargers at locations desired by residents are installed.

Metrics for Tracking Progress:

- Usage rate of EV chargers.
- EV adoption rate.
- Number of alternative fuel vehicles in the City fleet.

Transportation measure 2: Cumulative GHG emission reductions

This strategy results in a 25% reduction in transportation emissions in 2050 from a 2023 baseline.

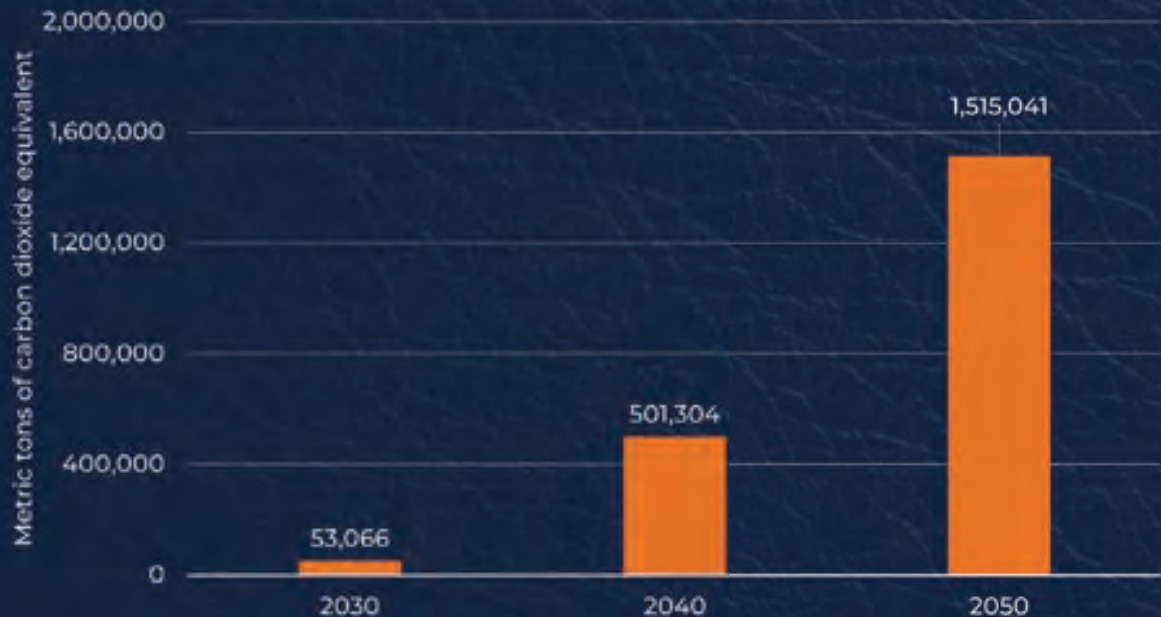


Figure 12. Cumulative greenhouse gas emissions reduction through 2050 from Transportation Measure 2. Scenario: 100% of light-duty vehicles converted to alternative fuel vehicles by 2050.

Transportation measure 2: Cumulative reduction in co-pollutants in 2050

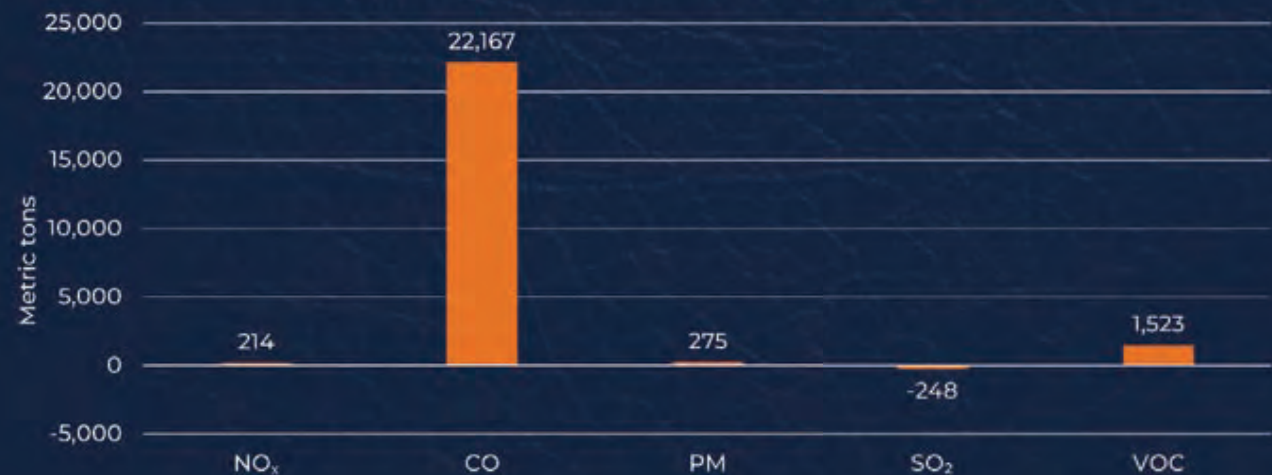


Figure 13. Cumulative co-pollutant emissions reduction through 2050 from Transportation Measure 2. Scenario: 100% of light-duty vehicles converted to alternative fuel vehicles by 2050.

Measure 3: Increase connectivity across the county and across the region, and into Colorado and Nebraska.

The Cheyenne MSA is located in Southeast Wyoming, with close access to cities in Colorado, Nebraska and Wyoming. **Improving access to I-25 and I-80** strengthens connections to the two neighboring states while coordinating transit and transportation planning within its nearby communities. Increasing connectivity across the county and into neighboring states expands travel efficiency throughout the region. When communities increase connectivity, it **reduces travel barriers, eases congestion, and improves transportation related safety**. Strategies under this measure include rural access to transit and enhanced community transportation connections.

Transportation Measure 3 Costs and Benefits:

GHG Impact Potential	Government Cost	Resident/Business Cost	Government Benefit	Resident/Business Benefit
				

Strategies:

- Increase rural transit across and throughout the county.
 - Offer on-demand transportation outside of city limits through turn-key services.
- Enhance connections between Laramie (Albany County), Laramie County, Colorado, and Nebraska.
 - Advocate for a commuter train from Wyoming to Colorado.
 - Advocate for regional bus service to Fort Collins or Loveland, Colorado.
 - Implement a multi-jurisdictional commuter program with the North Front Range MPO's VanGo vanpooling service to reduce single-occupancy vehicle travel along Interstate 25 between Cheyenne and northern Colorado communities like Fort Collins and Loveland. Develop an MOU with Kimball County to improve connectivity through shared bus stops.



Digital mockup of "CoCo," the Colorado Connector passenger rail, set to start service in 2029. There are plans to eventually expand service into Wyoming and New Mexico.



Strategies Continued:

- Develop an MOU with the Albany County Transportation Authority (ACTA) to create regional connectivity between Laramie and Cheyenne.

Implementation Timelines and Milestones:

- Establishment of on-demand transportation services.
- Regional bus service is established between Cheyenne and Front Range Colorado communities.
- A partnership with the North Front Range MPO is created and VanGo service is implemented between Cheyenne and northern Front Range communities.
- MOU with Kimball County is developed.
- MOU with ACTA is developed.
- Commuter train from Wyoming to Colorado is created.

Metrics for Tracking Progress:

- Radius of on demand transportation outside of city limits.
- MOU completed.
- Quantified GHG Emission Reduction

Transportation measure 3: Cumulative GHG emission reductions

This strategy results in a 0.1% reduction in transportation emissions in 2050 from a 2023 baseline.

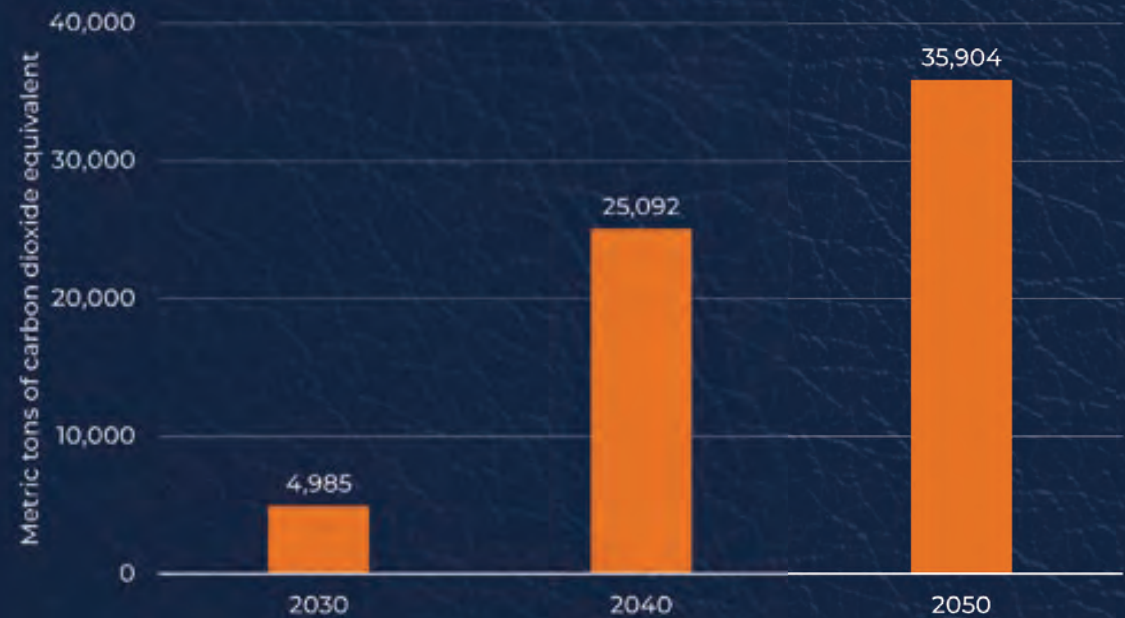


Figure 14. Cumulative greenhouse gas emissions reduction through 2050 from Transportation Measure 3. Scenario: 23% of cross-state commuter trips shifted from light-duty vehicles to transit by 2035.

Transportation measure 3: Cumulative reduction in co-pollutants in 2050

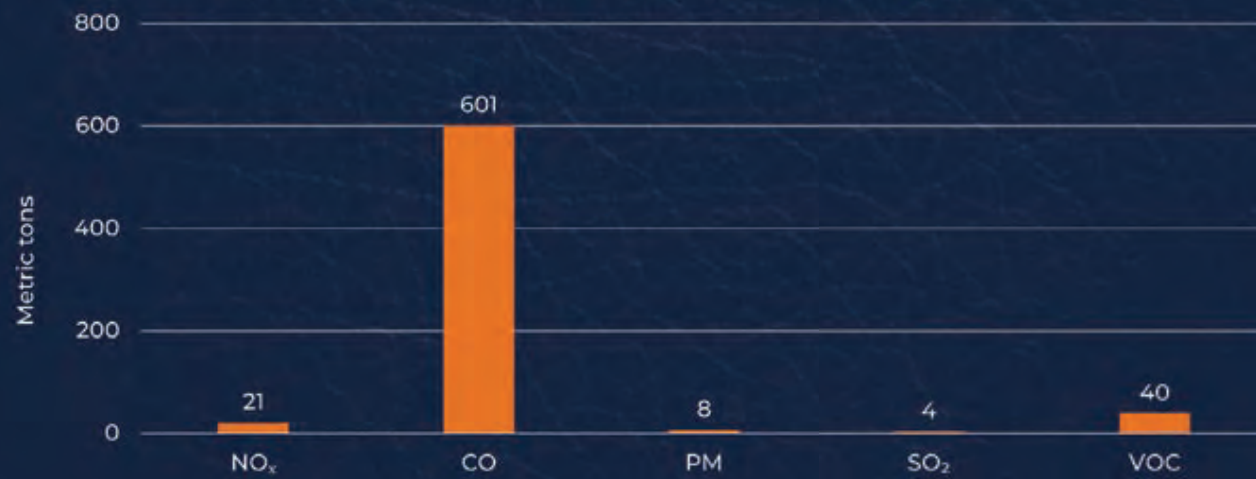


Figure 15. Cumulative co-pollutant emissions reduction through 2050 from Transportation Measure 3. Scenario: 23% of cross-state commuter trips shifted from light-duty vehicles to transit by 2035.

Measure 4: Shift commuter patterns from single occupancy vehicles to transit.

A way to **cut personal transportation costs and reduce emissions is by relying on less single-occupancy vehicles** for everyday travel. This measure focuses on helping commuters choose options other than driving alone. **Strengthening and promoting the Cheyenne MSA's transit system** plays a huge role in this shift. **More frequent and expanded service, easy to navigate and accessible routes and stops** makes transit a reliable and practical transportation choice. Emphasizing local **transit eases congested roads and helps residents save** on fuel and vehicle wear and tear. Strategies under this measure include increasing public transportation participation.

Transportation Measure 4 Costs and Benefits:

GHG Impact Potential	Government Cost	Resident/Business Cost	Government Benefit	Resident/Business Benefit
	\$\$\$	\$	\$	\$\$

Strategies:

- Increase public transportation ridership.
 - Add more stops and increase bus frequency to improve accessibility to stores and businesses.
 - Build awareness of transit options and services with public education.
 - Understand commuting patterns and how they relate to the Cheyenne workforce.
 - Establish a park and ride facility downtown as part of the transit loop.
 - Improve transit shelters to provide shade.
 - Continue identifying challenges transit system users face, especially the downtown workforce.



One of Cheyenne Transit Program's many purple buses.



Strategies Continued:

- Work with the Cheyenne Transit Program to analyze current transit ridership and assess unmet needs for bus stops and fixed routes.
- Increase visibility of transit stops through signage.
- Enhance the passenger experience by deploying real-time bus tracking and interactive route mapping across mobile and web platforms, providing riders with instant access to arrival times and service alerts.

Implementation Timelines and Milestones:

- Downtown park and ride facility is created.
- All transit shelters offer shade.

Transportation measure 4: Cumulative GHG emission reductions

This strategy results in a 0.8% reduction in transportation emissions in 2050 from a 2023 baseline.

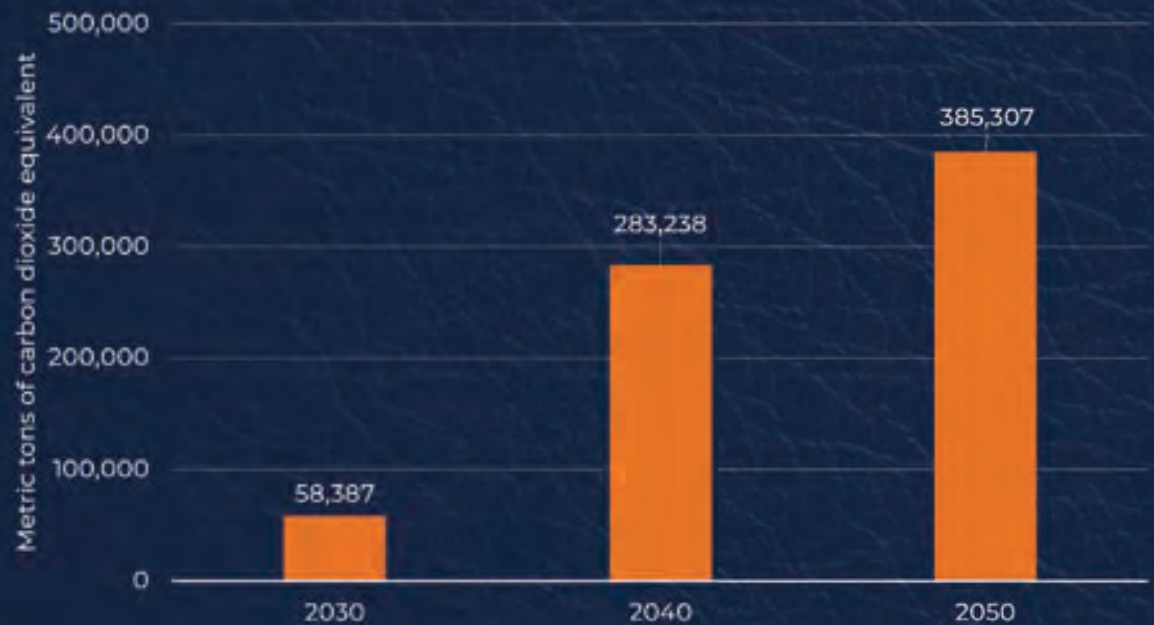


Figure 16. Cumulative greenhouse gas emissions reduction through 2050 from Transportation Measure 4. Scenario: 20% of trips shifted from light-duty vehicles to transit by 2035.

Transportation measure 4: Cumulative reduction in co-pollutants in 2050

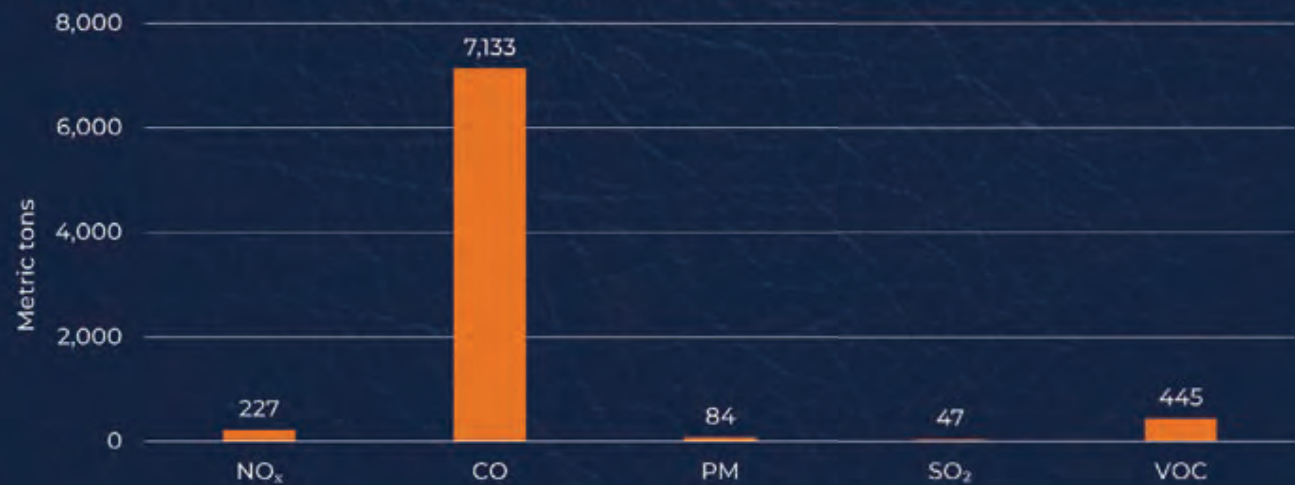


Figure 17. Cumulative co-pollutant emissions reduction through 2050 from Transportation Measure 4. Scenario: 20% of trips shifted from light-duty vehicles to transit by 2035.

Metrics for Tracking Progress:

- Increase in “Ridership” of public transit.
- Number of bus stops added.
- Number of shade implements/structures added at bus stops.
- Bus stop distance from main shopping hubs.
- Launch a bus-tracking app and website with live maps and schedules, aiming for at least 10% of active users in the first six months.

Transportation: Cumulative reduction in co-pollutants in 2050

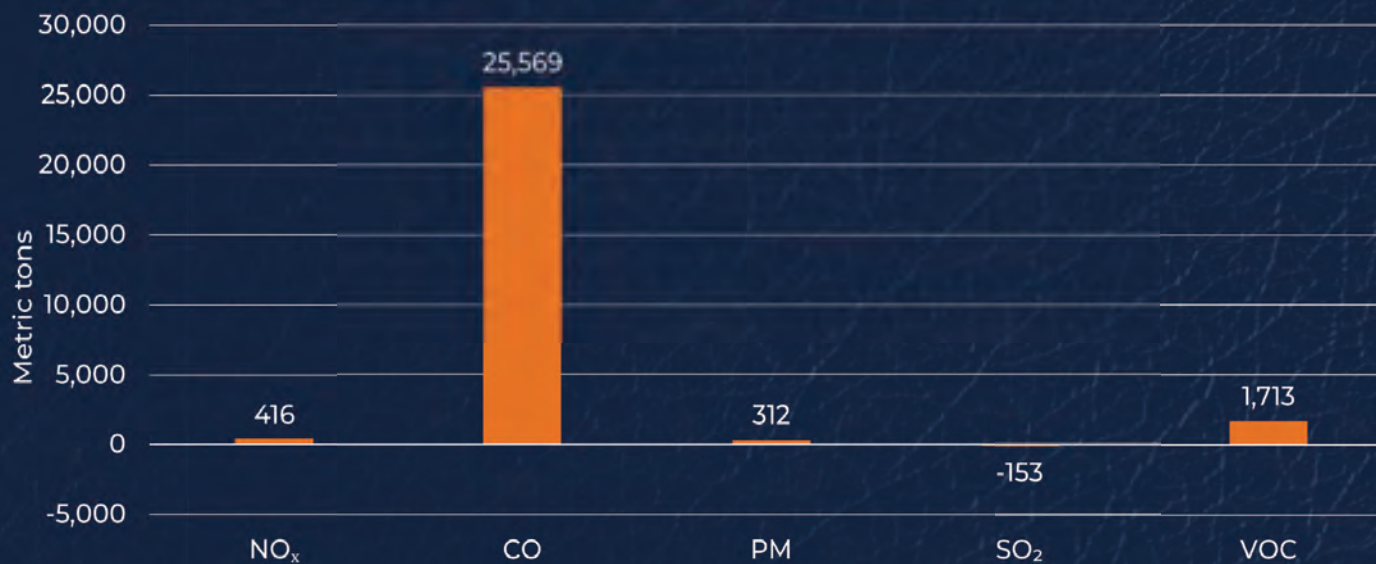


Figure 18. Cumulative co-pollutant emissions reduction through 2050 from all Transportation Measures.

3.3. Waste Management and Water

3.3.1. Introduction

As the community grows, so does the amount of waste produced and demand on local water treatment and distribution systems. These sectors **directly affect environmental health and public services**. The waste management sector includes **emissions from landfilled waste and compost**, while the water resource sector includes **emissions from wastewater treatment processes**.

Landfills generate the greenhouse gas, methane, and filling landfills up too quickly shortens the lifespan of the landfill and increases long-term costs. At the same time, for the water resources sector, **climate change impacts in Wyoming are likely to cause increases in the likelihood of droughts, water shortages**, and changing weather patterns. Ensuring that water is used sustainably and waste is managed to limit contamination and extend infrastructure, **helps protect community health and keeps costs low**. The goal for this sector focuses on **increasing the longevity of the landfill, preparing for water shortages and droughts, and water public outreach**.

Goals:

- Increase the waste diversion rate to ensure the longevity of the landfill.
- Provide a resilient and reliable water supply.
- Promote water conservation and responsible water use to prepare for potential water supply shortages and droughts.



3.3.2 Waste Measures

Measure 1: Increase the proportion of waste that is recycled and composted while reducing contamination levels.

One way to **cut down on waste and lower emissions** is by correctly **recycling and composting** instead of throwing everything away in a trash can. This not only ensures that trash is properly sorted and taken care of, but **reduces the amount headed to the landfill to create emissions**.

This measure's goal is to increase the amount of reused and/or composted materials. Trash is something that everyone can see, whether it's through overflowing trash cans or items on the side of the road. Strategies in this measure include **running research, including more waste collection options, and increasing public education and awareness on waste**.

Waste Management and Water Measure 1 Costs and Benefits:

GHG Impact Potential	Government Cost	Resident/Business Cost	Government Benefit	Resident/Business Benefit
	\$\$\$	\$	\$\$\$	\$

Strategies:

- Increase public engagement and awareness.
 - Develop informational campaigns to educate the community on proper waste disposal practices.
 - Promote buy local campaigns.
 - Provide clear signage or fact sheets with information about what can and can't be recycled.
- Add more options for waste collection across Cheyenne.
 - Increase composting capacity.
 - Implement recycling in parks, events, and municipal operations.
- Conduct research to identify new solutions, policies, and practices for waste management.
 - Research single-use plastics policies, including opt-in services for single-use plastics at restaurants and delivery services.



A City of Cheyenne waste bin with Cheyenne Depot standing tall in the background.



Strategies Continued:

- Research new technologies for waste disposal, including incinerators or more landfill gas capture.

Implementation Timelines and Milestones:

- Recycling signage or fact sheets are created and distributed.
- Composting capacity/participation is increased.
- Recycling in parks, events, and municipal operations is implemented.

Metrics for Tracking Progress:

- Increase in composting capacity/participation.
- Increase in waste diverted from the landfill.

Co-pollutant emissions were not modeled for this measure due to lack of data.

Waste measure 1: Cumulative GHG emission reductions

This strategy results in a 73% reduction in waste emissions in 2050 from a 2023 baseline.

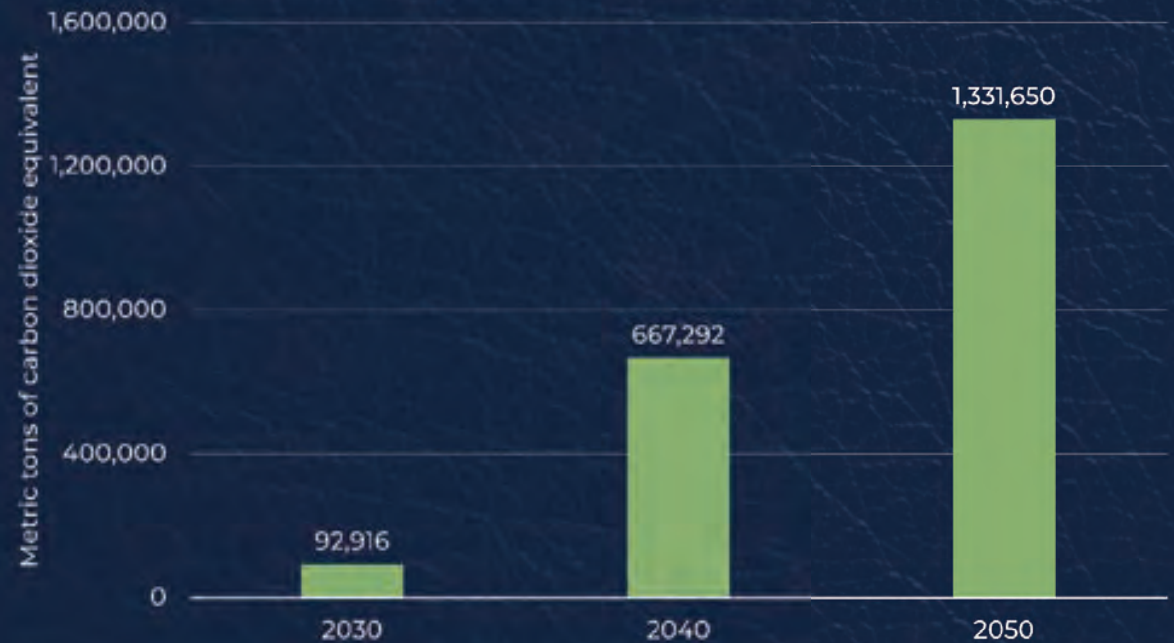
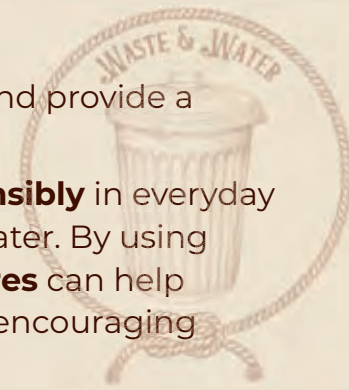


Figure 19. Cumulative greenhouse gas emissions reduction through 2050 from Waste Management and Water Measure 1. Scenario: 100% of recyclable waste and 100% of compostable waste diverted from landfill by 2035.



Measure 2: Promote water conservation and responsible water use to protect existing water sources and provide a resilient and reliable water supply.

Protecting the MSA’s water resources and supply starts with **using water more efficiently and responsibly** in everyday efforts. This measure focuses on not only helping the region, but residents and businesses conserve water. By using **smart-irrigation systems, planting native or drought-tolerant plants, and installing low-flow fixtures** can help not only conserve water but protect the quality and quantity of local water sources. Key strategies are encouraging residents and businesses to conserve water.

Waste and Water Management Measure 2 Costs and Benefits:

GHG Impact Potential	Government Cost	Resident/Business Cost	Government Benefit	Resident/Business Benefit
None	\$\$\$	\$-\$\$	\$\$	\$



Kids and families enjoying the splash pad in Downtown Cheyenne on a warm day.

Strategies:

- Protect existing water sources.
- Promote water conservation at the municipal level.
- Promote residential water conservation.
- Encourage the commercial and industrial sectors to adopt water conservation practices.

Implementation Timelines and Milestones:

- Groundwater levels are increased.
- Single-family residents use less water.
- Requirements for irrigators are implemented.
- Data centers reduce water consumption.

Metrics for Tracking Progress:

- Increase in sustainable water supply.
- Decrease in water use after conservation measures taken.
- Increase in number of household greywater systems installed.
- Monitoring and stabilization of groundwater levels.
- Reduction in single family water usage by 5% within five years.

Emissions reductions and Co-pollutant impacts were not modeled for this set of strategies due to lack of data.



The Paul Smith Children's Village at the Cheyenne Botanical Gardens.



Measure 3: Improve water quality and enhance stormwater management.

Protecting waterways and keeping stormwater infrastructure clean starts with **improving runoff management and treating wastewater** before it reaches rivers and streams. This measure focuses on helping residents, businesses, and the city to adopt actions like installing **rain gardens, using permeable pavement, and maintaining green infrastructure**. Cleaner stormwater **protects local water systems, supports wildlife, and reduces the cost of water treatment**. Strategies in this measure include improving stormwater quality near developments and in community parks and open spaces.

Waste Management and Water Measure 3 Costs and Benefits:

GHG Impact Potential	Government Cost	Resident/Business Cost	Government Benefit	Resident/Business Benefit
None	\$\$\$	\$\$-\$\$\$	\$\$\$	\$

Strategies:

- Enhance water management of community parks and open spaces to improve water quality.
 - Continue to use Integrated Pest Management practices in public parks instead of using pesticides.
 - Implement projects identified in the City’s Drainage Plans, prioritizing projects that improve water quality and enhance public open spaces.
 - Include green infrastructure measures in complete streets projects, such as bioswales and permeable pavement.
- Improve stormwater quality in commercial and residential development.
 - Reduce sediment pollution from new developments through implementation of best management practices.



A portion of Cheyenne’s Greenway, which offers more than 21 miles of path for walking, biking, skating, and running

Strategies Continued:

- Improve stormwater pollution prevention at mobile businesses such as food trucks.
- Ensure infill developments comply with water quality requirements.
- Establish an ongoing water quality monitoring and testing program.

Implementation Timelines and Milestones:

- Drainage Plan projects are implemented.
- All public works and roadway projects include green infrastructure.
- Water quality monitoring and testing program is created.

Metrics for Tracking Progress:

- Percent reduction of sediment pollution.
- Number of water quality projects from the Drainage Plans implemented.
- Establishment of an ongoing water quality monitoring program.
- Number of green infrastructure measures implemented.
- City time to dredge drainage ways and detention ponds is reduced by 10% in the next five years.

Emissions reductions and Co-pollutant impacts were not modeled for this set of strategies.



A walking and biking path circling Sloan's Lake in Lions Park.

3.4. Natural and Working Lands

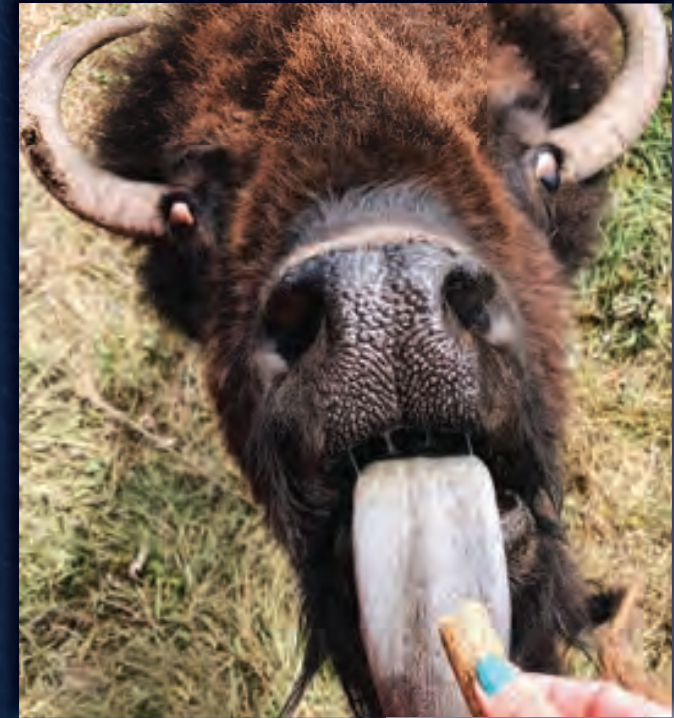
3.4.1. Introduction

Natural and working lands are a defining part of the Cheyenne Region, **shaping its character, economy, and sense of place.** These lands include rangelands, agricultural lands, open spaces, parks, and the ecosystems that support local wildlife. They provide opportunities for **cleaner air and water, expanding wildlife habitat, recreation opportunities, and spaces for agriculture and ranching.**

Protecting these landscapes is important not only for ecological reasons but also for community well-being. Sensitive and rare habitats can easily be disrupted. When these areas are lost, the impacts lead to reduced biodiversity, increased erosion, decreased water quality, and fewer opportunities for residents and visitors to enjoy the outdoors. The goal for this sector focuses on **protecting and preserving natural areas by supporting landowners and managers** and ensuring that important habitats remain healthy and connected.

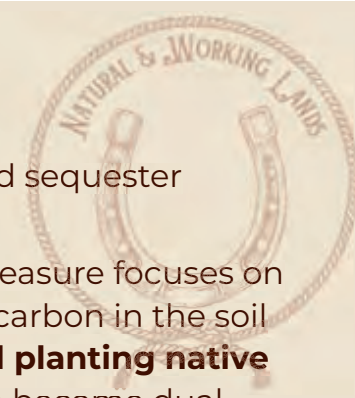
Goal:

- Preserve and protect sensitive, rare, and unique natural areas.



A Bison at Terry Bison Ranch extends its tongue to eat a piece of feed.





3.4.2. Natural and Working Lands Measures

Measure 1: Encourage agriculture lands and range lands to protect and conserve natural resources and sequester carbon.

Working lands and natural areas play an important role in protecting Cheyenne's environment. This measure focuses on **helping agricultural lands, rangelands, and open spaces conserve natural resources** while storing carbon in the soil and vegetation. Practices such as **rotational grazing, improved soil management, and restoring and planting native vegetation** encourages carbon sequestration compared to traditional practices. Natural areas can also become dual-purpose, by offering places for people to recreate while understanding the importance of preserving Cheyenne's unique landscapes. Strategies under this measure include **encouraging and intentional partnerships for water conservation and water quality protection**.

Natural and Working Lands Measure 1 Costs and Benefits:

GHG Impact Potential	Government Cost	Resident/Business Cost	Government Benefit	Resident/Business Benefit
	\$	\$-\$\$	None	\$

Strategies:

- Encouraging water conservation and water quality protection of agricultural lands and rangelands, especially in riparian, wetland, and overgrazed areas.
 - Educate landowners and managers on irrigation best practices.
 - Encourage Regenerative Agriculture practices.
 - Encourage farmers to utilize manure and compost over fertilizers.
- Leverage regional collaborations to promote resources and best practices.
 - Continue to collaborate with Laramie County Conservation District, the Natural Resources Conservation Service (NRCS) and the Farm Service Agency (FSA) to promote best practices.
 - Continue the partnership to apply biosolids from wastewater treatment at local ranches.
 - Collaborate with the University of Wyoming Extension on agriculture and integrated pest management programs.
 - Collaborate with Wyoming State Parks and Cultural Resources to implement educational campaigns.

Implementation Timelines and Milestones:

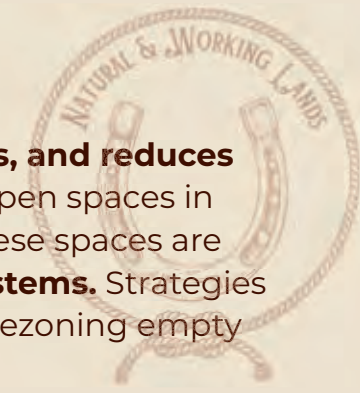
- Water friendly crop manual/educational materials are developed and distributed.
- Additional ranchers identified and receive wastewater treatment plant (WWTP) biosolids.
- Creation of and opening enrollment for agriculture and integrated pest management programs.

Metrics for Tracking Progress:

- Number of farmers utilizing manure and compost over fertilizers.
- Bird and pollinator surveys on improved lands.

Emissions reductions and Co-pollutant impacts were not modeled for this set of strategies.





Measure 2: Manage development to protect natural lands and wildlife.

Effective land management helps **safeguard wildlife corridors, preserves native plant communities, and reduces the spread of invasive species.** This measure focuses on strengthening natural areas, habitats, and open spaces in order for these areas to remain resilient to changes and accessible for everyone. By improving how these spaces are maintained and monitored, **increased biodiversity helps protect Cheyenne’s most sensitive ecosystems.** Strategies under this measure consist of focusing development in existing urban areas and repurposing and/or rezoning empty buildings.

Natural and Working Lands Measure 2 Costs and Benefits:

GHG Impact Potential	Government Cost	Resident/Business Cost	Government Benefit	Resident/Business Benefit
	\$	None	\$\$	\$

Strategies:

- Promote urban infill redevelopment to reduce urban sprawl.
- Encourage new development to reuse already disturbed sites (brownfields) before developing open undeveloped land.
- Support Urban Renewal Plan incentives.
- Review and revise the urban growth boundary.
- Conduct a comprehensive review of zoning and land use policies to ensure they conform with sustainability goals.
- Identify unique, sensitive, and rare natural areas worthy of protection and protect as open space.
- Create community garden regulations and use certain lots to create community gardens.
- Work with Wyoming State Parks and Cultural Resources, the Historic Preservation Board, and the Downtown Development Authority to align historic preservation goals.

Implementation Timelines and Milestones:

- Brownfields are the main land area being used for development instead of open undeveloped land.
- Land use policies are updated.
- Revise urban growth boundary.
- Review of zoning and land use policies is completed.

Metrics for Tracking Progress:

- Number of vacant properties transformed.
- Population density increase.
- Track changes/needs for change to the land use policies.
- Percentage of brownfields used for developments.

Emissions reductions and Co-pollutant impacts were not modeled for this set of strategies.



Measure 3: Improve the management of natural spaces.

A simple way to protect and improve natural areas is through more intentional management. This measure focuses on **balancing conservation and recreation, to better benefit both people and wildlife.**

Practices like restoring and planting native vegetation, maintaining trails, and eliminating invasive vegetation. Well-managed natural areas support wildlife habitat, like pronghorns, deer and various species of birds, **improves air and water quality, and reduces erosion and flooding.**

Strategies under this measure include maintaining and monitoring tree canopies around the Cheyenne MSA and empowering the community to become active environmental stewards.

Natural and Working Lands Measure 3 Costs and Benefits:

GHG Impact Potential	Government Cost	Resident/Business Cost	Government Benefit	Resident/Business Benefit
CO ₂	\$	None	None	\$

Strategies:

- Maintain a healthy tree canopy in the region.
 - Continue to replace dying trees and expand the tree canopy with diverse and climate-specific species.
 - Maintain brush and wooded spaces in targeted areas to reduce wildfire risk.
 - Conserve the tree canopy and native vegetation in riparian areas during development.
 - Educate builders, developers, and landscapers on appropriate planting and irrigation when new developments are being built.
- Educate and empower residents and policy makers to become environmental stewards.
 - Update zoning policies to allow and encourage native plants and grasses in residential development.
 - Improve natural resources education of decision and policy makers.



A mountain biker enjoying Cheyenne's beautiful natural spaces.



Strategies Continued:

- Involve youth in community garden programming.
- Develop educational programs about native gardens and plants, native plant maintenance, and tree health.

Implementation Timelines and Milestones:

- Incorporation of tree plotter software.
- Zoning policies have been updated.
- Educational programs are developed and open for enrollment.

Metrics for Tracking Progress:

- Maintain a tree plotter software inventory for residents to expand the city's tree inventory.
- Number of dying trees replaced each year.
- Number of planting and irrigation trainings offered.

Natural and working lands measure 3: Cumulative GHG emission sequestration

This strategy increases carbon sequestration and does not reduce emissions in a particular sector.

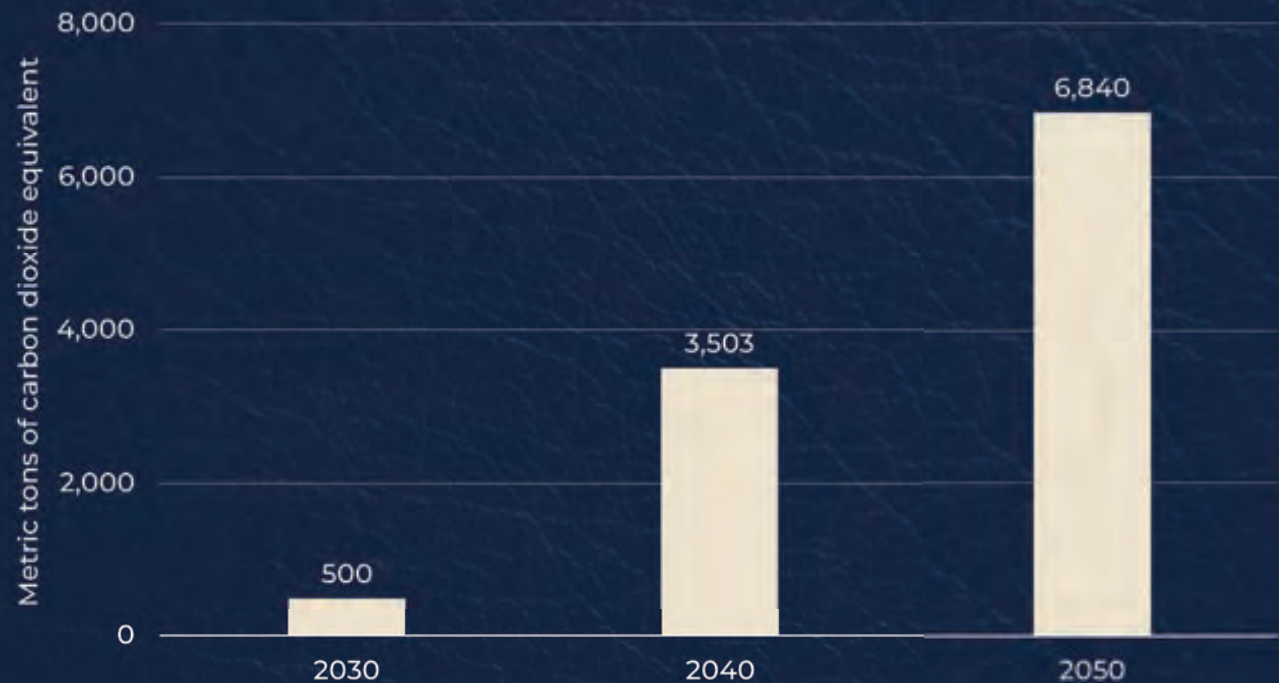


Figure 20. Cumulative greenhouse gas emissions reduction through 2050 from Natural and Working Lands Measure 3. Scenario: 20% increase in land area preserved by 2035.

3.5 Workforce Planning

3.5.1 Introduction

Cheyenne's ability to grow and prosper depends on how well the region develops and supports its current and future workforce. This sector focuses on **how well the community prepares people for jobs, supports local employers, and creates pathways that help residents build strong and lasting careers.**

Addressing workforce needs helps the region stay and become competitive and resilient. **High quality jobs that sustain families** not only strengthens the local economy, but also improves the overall quality of life. At the same time, **aligning worker skills with emerging trends** across key industries helps the Cheyenne MSA region stay adaptable in an ever changing job market. The goals for this sector focus on making the region an appealing place to live and work, **helping attract new talent, while giving young residents a reason to stay.**

Goals:

- Promote the creation and accessibility of high-quality jobs that will help sustain families.
- Build a pipeline of skilled workers that aligns with workforce trends across key industries to promote economic vitality.
- Ensure that as the Cheyenne MSA grows into a workforce hub, environmental impacts are taken into account.
- Improve the quality of life to attract new workers and retain the young population.



The outdoor coffee bar at Paramount Cafe, a locally owned and operated coffee shop.

3.5.2 Workforce Measures

Measure 1: Provide training and workforce development programs that promote the creation and accessibility of high-quality jobs.

Investing in training and workforce development supports a strong local economy and creates more opportunities. This measure focuses on building skills that align with current and future job needs and retaining the workforce, making high-quality jobs more accessible throughout the community. Offering training, apprenticeships, and career pathways helps residents gain hands-on experience and find stable and well-paying jobs. Strategies in this measure include training workers in emerging technologies and strengthening a robust workforce.

Workforce Measure 1 Costs and Benefits:

GHG Impact Potential	Government Cost	Resident/Business Cost	Government Benefit	Resident/Business Benefit
None	\$	\$	\$	\$-\$-\$-\$

Strategies:

- Ensure workers are trained in emerging industries and technologies by working with the Laramie County Community College and other agencies.
 - Provide non-degree programs that focus on sustainability and environmentally conscious practices.
 - Create training programs on EV repair and maintenance.
 - Provide contractor training for clean energy technologies like heat pumps to increase familiarity.
 - Promote training and classes using messaging that will resonate with various industries.
 - Provide training for preservation and adaptive reuse.



A firefighter performs the traditional spray-down of the new engine during the department's wet-down ceremony.

Strategies Continued:

- Ensure a robust workforce exists to support growth in the region.
 - Partner with Cheyenne LEADS, the University of Wyoming, Laramie County Community College and the Chamber of Commerce to attract clean energy businesses.
 - Explore offering incentives to recruit businesses and workers.
 - Leverage demand for more public transportation and a light rail to bring in new workers.
 - Recruit more long-term drivers for public transportation by offering good pay and benefits.

Implementation Timelines and Milestones:

- Training programs are created.
- Contractor training is provided.
- Partnership between the City, Cheyenne LEADS, the University of Wyoming, Laramie County Community College and the Chamber of Commerce is established.
- Incentives are created to the extent allowed under applicable laws.

Metrics for Tracking Progress:

- New clean energy businesses added each year.
- Amount of incentives offered.
- Addition of community college programs to support local employers.

Emissions reductions and Co-pollutant impacts were not modeled for this set of strategies.



*Dedicated workers serving meals
at the Hell on Wheels Rodeo.*

Measure 2: Support businesses in adopting resource conservation practices.

An easy way to reduce resource use, save money, and support long-term sustainability is by helping businesses adopt conservation practices. This measure focuses on providing businesses with the tools and support to use energy and water more efficiently throughout their operations. **Changes like upgrading equipment, reducing consumption, and improving water efficiency reduces environmental effects, while also lowering operation costs.** Strategies for this measure include educating business owners on resource efficient resources and technical assistance.

Natural and Working Lands Measure 2 Costs and Benefits:

GHG Impact Potential	Government Cost	Resident/Business Cost	Government Benefit	Resident/Business Benefit
None	\$-\$\$	None	None	\$

Strategies:

- Promote resource-efficient business operations through education and technical assistance.
 - Increase awareness of best practices to conserve resources and how businesses can benefit from it.
 - Continue to work with data centers to reduce their energy use, water use, and generate solar and wind power.
 - Provide 1:1 support to small businesses to help them navigate energy efficiency opportunities.
 - Explore whether it is more efficient to transport cargo by rail instead of trucks.
 - Work with the Wyoming Department of Workforce Services, the Wyoming Small Business Development Center and Laramie County Community College to provide workforce training.

Implementation Timelines and Milestones:

- Creation of 1:1 technical assistance program.
- Completion of transport cargo research.
- Workforce training is provided.

Metrics for Tracking Progress:

- Number of businesses participating in 1:1 navigation services.
- Reduction in water use.
- Number of workforce training enrollment.

Emissions reductions and Co-pollutant impacts were not modeled for this set of strategies.



Measure 3: Ensure that families moving into Cheyenne have access to essential resources nearby.

A simple way to support families moving into Cheyenne is by making sure there are **essential and recreational resources available** across neighborhoods. This measure focuses on improving access to every need such as grocery stores, schools, healthcare, parks, etc. When families have access to these resources, it provides a better quality-of-life for them and the community as a whole. Better access also **strengthens neighborhoods and supports local businesses**. Strategies under this measure include improving local infrastructure and services to attract families to the Cheyenne area.

Natural and Working Lands Measure 3 Costs and Benefits:

GHG Impact Potential	Government Cost	Resident/Business Cost	Government Benefit	Resident/Business Benefit
None	\$\$\$	\$	\$	\$\$

Strategies:

- Improve local infrastructure and services.
 - Create a “third space” in South Cheyenne, such as a recreation/community center.
 - Provide access to childcare, such as at a community center or library.
 - Develop a recreational center that the Cheyenne community can use year round.
 - Support more grocery store options for the community.
 - Provide indoor recreational opportunities like mini golf and indoor parks.

Implementation Timelines and Milestones:

- Third space is created.
- Recreation centers developed.
- More grocery stores are developed.
- More indoor recreational opportunities like mini golf and indoor parks are created.

Metrics for Tracking Progress:

- Number of programs offered at the recreational center.
- Average distance to the nearest grocery store per household decreased
- Number of indoor amenities offered.
- Usage rates of indoor recreational opportunities.

Emissions reductions and Co-pollutant impacts were not modeled for this set of strategies.



4. Benefits Analysis

WY



In This Chapter

4.1 Household Efficiency, Electrification, and Rooftop Solar

4.2 Commercial Efficiency, Electrification, and Rooftop Solar

4.3 Transportation

4.4 Municipal Support



4. Benefits Analysis

4.1 Household Efficiency, Electrification, and Rooftop Solar

- Residential Efficiency
- Residential Building Electrification
- Residential Rooftop Solar

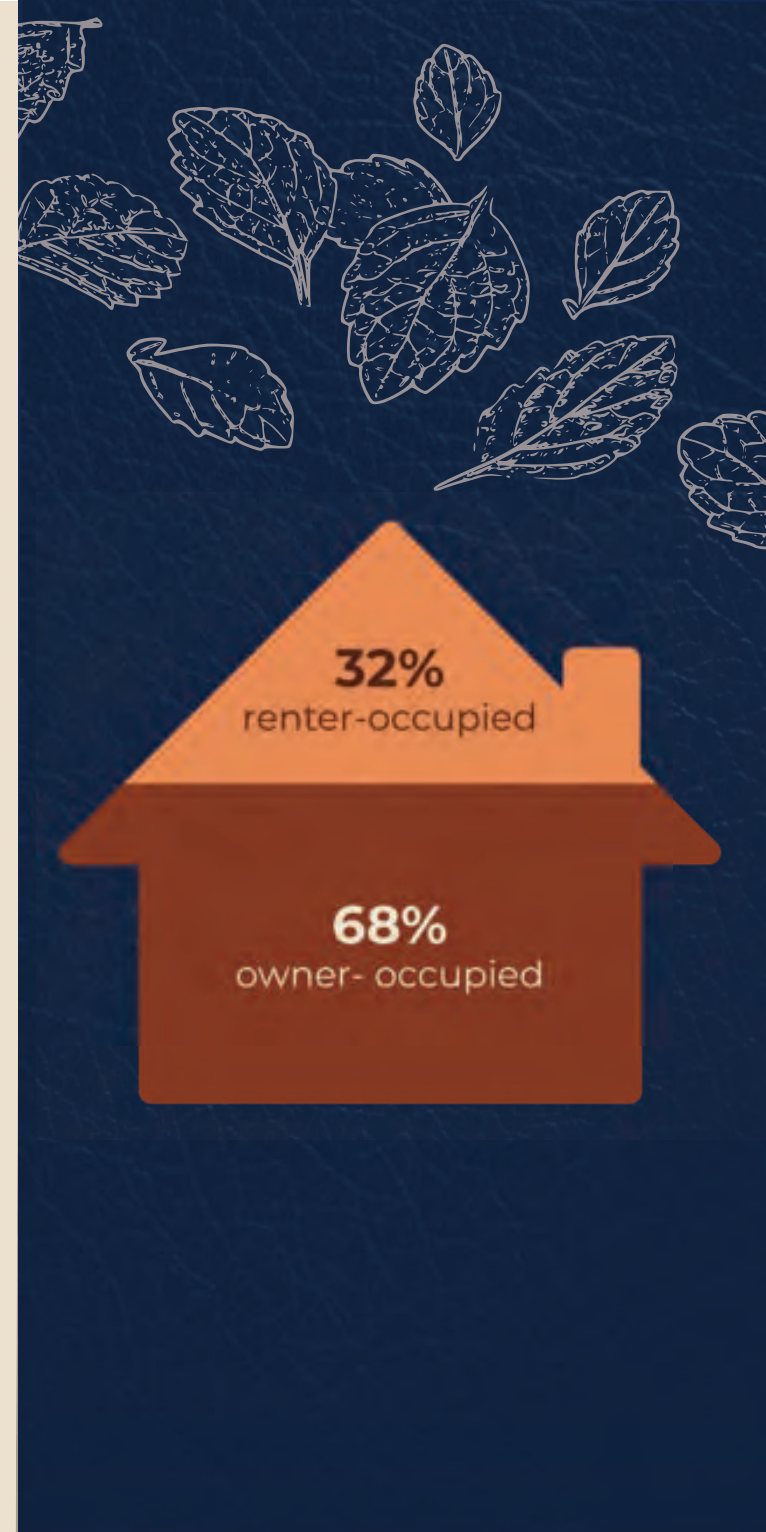
More households in Laramie County and Cheyenne are likely to be built prior to 2010. This demonstrates the opportunity for energy efficiency upgrades and electrification programs. In the Cheyenne metropolitan statistical area (MSA). This impacts the types of resources and programs that community members will need to reduce their energy usage. Black Hills Energy, one of the energy providers in the county, currently has the capacity to provide 29% of its energy from wind, solar, and storage resources and is projected to have capacity to provide 49% of its energy from wind, solar, and storage by 2030. This follows a national trend of sourcing cleaner energy sources.

4.2 Commercial Efficiency, Electrification, and Rooftop Solar

- Commercial Efficiency
- Commercial Building Electrification
- Commercial Rooftop Solar

4.3 Transportation

- 10% Gas VMT Reduction
- EV Adoption - Gas & Diesel



4.4 Municipal Support

- Solar on Municipally-Owned Land (examples include Cheyenne Landfill and Belvoir Ranch) and Energy-Intensive Municipal Facilities (examples include Cheyenne's Wastewater Treatment Plants)



Belvoir Ranch, one potential location for municipally-owned solar arrays.

Benefits		Energy and Buildings					Transportation			
		Measure 1	Measure 2	Measure 3	Measure 4	Measure 5	Measure 1	Measure 2	Measure 3	Measure 4
Economic	Promote green economy	★	★	★	★		★	★	★	★
	Job growth/creation	★	★	★	★			★	★	★
	Cost savings	★	★	★		★			★	★
Social	Enhance quality of place	★	★	★	★	★	★	★	★	★
	Improve transportation access						★	★	★	★
	Enhance regional collaboration			★		★	★	★	★	★
Environmental	Improve air and public health outcomes	★	★	★	★		★	★	★	★
	Shift to more sustainable behavior	★	★	★	★	★	★	★	★	★
	Improve climate resilience	★	★	★	★		★	★	★	★

Figure 21. Table of benefits from each measure proposed in Chapter 3. Note: half stars indicate indirect benefits while full stars indicate direct benefits.

Benefits		Waste Management & Water			Natural & Working Lands			Workforce Planning		
		Measure 1	Measure 2	Measure 3	Measure 1	Measure 2	Measure 3	Measure 1	Measure 2	Measure 3
Economic	Promote green economy									
	Job growth/creation									
	Cost savings									
Social	Enhance quality of place									
	Improve transportation access									
	Enhance regional collaboration									
Environmental	Improve air and public health outcomes									
	Shift to more sustainable behavior									
	Improve climate resilience									

Figure 22. Table of benefits from each measure proposed in Chapter 3. Note: half stars indicate indirect benefits while full stars indicate direct benefits.

5. Appendices

WY



In This Chapter

Appendix A: Stakeholder Engagement Activities

Appendix B: Glossary

Appendix C: Workforce Analysis

Appendix D: Inventory and Inventory Methodology

Appendix E: BAU Modeling Methodologies



5. Key Takeaways

Appendix A: Stakeholder Engagement Activities

Overview

The engagement processes from the PCAP led to the following reflections to improve our approach for the CCAP:

- **More education and awareness are needed for community members** to make informed decisions and suggestions regarding climate change efforts in the Cheyenne MSA.
- **More discussion and alignment opportunities for stakeholders from diverse sectors** and backgrounds to ensure equitable and comprehensive GHG pollution reduction initiatives now and in the future.
- **Expand our ability to collect and analyze community level data** regarding GHG emissions and environmental changes so we can gain a clearer picture of our role in reducing GHG emissions in the world.

During the development of the CCAP, we implemented the following tactics to gain further insights for this plan:

Create an Environmental Equity Steering Committee (EESC) for this grant and long-term environmental equity

work: This committee will be composed of stakeholders from diverse sectors and backgrounds, ensuring strong advocacy from LIDACs and underrepresented areas in the MSA, leaders from the private sector, and bipartisan representation. This will establish support systems for steering committee members to promote participation. We will include representation from higher education institutions, Laramie County Community College and University of Wyoming, particularly for workforce development insights. Although the City of Cheyenne reached out to these institutions and they expressed support, their involvement will be more effective during the CCAP development.

Create a Comprehensive Community Outreach Plan with the EESC: Built in partnership with stakeholders and the EPA's Planning for Equitable Development tool, this Plan will assign tasks to ensure outreach work is completed effectively and timely, and will focus on outreach materials that inform and educate stakeholders and the general public, attendance of community events, organization of town halls, and of course, continued gathering of key measures and data regarding direct and indirect emissions reductions.



Appendix B: Glossary

Greenhouse gas (GHG) inventory: The information used was based on version 1.2 of the U.S. Community Protocol for Accounting and Reporting GHG Emissions and additional activities/sources are considered in accordance with the Global Protocol for Community-Scale GHG Emissions Inventories. The GHG inventory includes a list of emission sources and sinks, and the associated emissions quantified using standard methods. The PCAP includes a “simplified” inventory. The CCAP will include a comprehensive inventory of emissions and sinks for the following sectors: industry, electricity generation/use, transportation, commercial and residential buildings, agriculture, natural and working lands, and waste and materials management.

GHG emission projections: This PCAP uses a Business-as-Usual projection from 2021 (baseline year) to 2030 and 2050. This projection was carried out for every activity and source based on various activity growth rates and carbon intensity growth rates and is based on national data due to minimal current data for the Cheyenne MSA.

GHG reduction measures: There are eleven GHG Reduction Measures in this PCAP. They are Residential Efficiency (Electricity/Natural Gas), Residential Building Electrification, Commercial Efficiency (Electricity/Natural Gas), Commercial Building Electrification, Residential Rooftop Solar, Commercial Rooftop Solar, a 10% Gasoline VMT Reduction, EV Adoption for Gasoline-powered vehicles, EV Adoption for Diesel-powered vehicles, a Landfill Solar project, and a Ranch Solar project.

Benefits analysis: The LIDAC benefits analysis focuses primarily on how energy efficiency and renewable energy sources will provide cost savings for residential energy bills. Another benefit will be improved air quality in LIDAC areas from reduced car exhaust, as well as improved multi-modal transportation opportunities, increasing walking, biking, and other healthy activities. LIDACs are communities with residents that have low incomes, limited access to resources, and disproportionate exposure to environmental or climate burdens. Although the Inflation Reduction Act does not formally define LIDACs, EPA strongly recommends grantees use the Climate and Economic Justice Screening Tool and the Environmental Justice Screening and Mapping Tool to identify LIDACs in their communities. These tools identify LIDACs by assessing indicators for categories of burden: air quality, climate change, energy, environmental hazards, health, housing, legacy pollution, transportation, water and wastewater, and workforce development.

Review of authority to implement for each measure: As stated in Section 3.4.2., the City of Cheyenne has the authority to implement these reduction measures in partnership with executing organizations. These executing organizations will be further identified in the development of the CCAP.

Intersection with Other Funding Availability: This PCAP anticipates leveraging other sources of funding from federal, state, and private sectors. See funding measures listed in the GHG Reduction Strategies table in Section 3.4.

Next steps: This PCAP provides a foundation for more detailed analysis in the CCAP, including projected energy and emissions across the region and within specific geographies. It highlights how specific actions can benefit LIDAC neighborhoods both directly and indirectly. The CCAP will construct a model of the Cheyenne MSA and systematically evaluate actions and policies and their GHG impacts, financial impacts, and implementation mechanisms, supported by an extensive engagement process.

Priority Climate Action Plan (PCAP): a narrative report that includes a focused list of near-term, high-priority, and implementation-ready measures to reduce GHG pollution and an analysis of GHG emissions reductions.

Comprehensive Climate Action Plan (CCAP): a narrative report that provides an overview of the grantees' significant GHG sources/sinks and sectors, establishes near-term and long-term GHG emission reduction goals, and provides strategies and identifies measures that address the highest priority sectors to help the grantees meet those goals.

MSA: metropolitan statistical areas as defined by the U.S. Census 2020 MSA population. A list of eligible MSAs can be found in Appendix 15.2 in EPA's CPRG: Formula Grants for Planning, Program Guidance for States, Municipalities, and Air Control Agencies.

State: all 50 U.S. states and the District of Columbia and Puerto Rico. All other Tribes or U.S. territories (the Virgin Islands, Guam, American Samoa, and the Commonwealth of the Northern Mariana Islands) should follow CRPG guidance for Tribes and Territories.



Appendix C: Workforce Analysis

Introduction

In 2023, the City of Cheyenne (Cheyenne, the City) became the lead agency for the Cheyenne Metropolitan Statistical Area's (MSA) Climate Pollution Reduction Grant (CPRG) work. Cheyenne contracted Lotus Engineering and Sustainability, LLC (Lotus) to assist the development of a Comprehensive Climate Action Plan (CCAP) detailing pollution reduction measures. Through those measures, the CCAP aims to improve local infrastructure, protect rural landscapes and the environment, increase resilience, and spur economic growth.

As part of the CCAP, the project team (Cheyenne and Lotus) conducted a workforce development analysis to:

- Delineate current clean energy workforce trends, nationally and locally.
- Identify current barriers and challenges in Cheyenne and Wyoming at large that impede clean energy workforce expansion.
- Identify opportunities to expand the clean energy workforce in Cheyenne and Wyoming at large.

Measure	Job (Job Code)
Increase energy efficiency and electrification in residential buildings.	Construction Laborers (47-2061) , Electricians (47-2111), Heating, Air Conditioning, and Refrigeration Mechanics and Installers (49-9021), Insulation Workers: Floor, Ceiling, and Wall (47-2131), Mechanical (47-2132).
Increase energy efficiency and electrification in commercial and industrial buildings.	Electricians (47-2111), Construction Laborers (47-2061), Heating, Air Conditioning, and Refrigeration Mechanics and Installers (49-9021), Insulation Workers: Floor, Ceiling, and Wall (47-2131), Mechanical (47-2132).
Encourage adoption of alternative energy.	Civil Engineers (17-2050), Construction Equipment Operators (47-2070), Construction Laborers (47-2060), Electrical Engineers (17-2071), Electricians (47-2111), Electrical and Electronics Repairers, Powerhouse, Substation, and Relay (49-2095), Electrical Power-Line Installers and Repairers (49-9051), Power Plant Operators, Distributors, and Dispatchers (51-8010), Pipelayers, Plumbers, Pipefitters, and Steamfitters (47-2150), Structural Iron and Steel Workers (47-2220), Welding, Soldering, and Brazing Workers (51-4120), Wind Turbine Service Technicians (49-9080).
Promote the adoption of alternative fuel vehicles.	Electrical Power-Line Installers and Repairers (49-9051), Electricians (47-2111), Automotive Service Technicians and Mechanics (49-3023).

Table 2. Measures in the CCAP, jobs required to implement, and job codes.

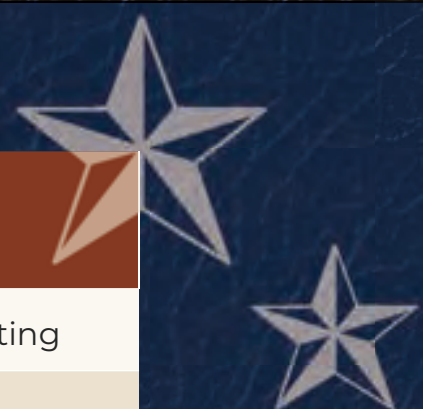
Climate Pollution Strategies in the Comprehensive Climate Action Plan

Pollution reduction measures developed for the CCAP will require additional workforce to ensure successful implementation. Consistent with U.S. Environmental Protection Agency guidance and CPRG grant requirements, this workforce analysis evaluates the job types, codes, and sectors expected to emerge from climate action investments. Table 1 lists the clean energy workforce related measures developed, the jobs, and Bureau of Labor Statistics (BLS) job codes critical to implementation.

The findings from this workforce analysis are derived from informational interviews with workforce and energy providers and a literature review of existing national, regional, and local studies and reports about energy and transportation workforce development.

Stakeholder Engagement

Cheyenne worked with Lotus to identify stakeholders to engage with in conversations about workforce development. The project team was able to host informational interviews with key leaders in the space. Conversations centered around current challenges, initiatives, and opportunities for stakeholders, and identifying opportunities for collaboration.



Workforce and Energy Providers	Stakeholders
High West Energy	Brooke Darden, Communications and Marketing
Laramie County Community College	Tonya Hacker, Outreach and Workforce Development Dr. Joseph Schaffer, Laramie County Community College President

Table 3. Stakeholders interviewed for the workforce analysis.

Literature Review

In addition to stakeholder engagement, the project team reviewed existing national, regional, and local studies and reports about workforce development. The research served as another method to determine the state of workforce development in both Cheyenne and Wyoming.

Cheyenne Metropolitan Statistical Area Workforce

According to the U.S. Census Bureau, Cheyenne has had an annual growth rate of 0.6% since the 1970s. Laramie County, on the other hand, has grown quicker than Cheyenne with an annual growth rate of 1% since the 1970s. However, Wyoming has been experiencing an exodus of younger populations, specifically from those in their 20s and 30s, resulting in an aging population.¹⁸ The state is already creating an educated, talented pool of soon to be professionals, but struggles to retain those students to expand and diversify the Wyoming workforce.

¹⁸ https://wyomingbusiness.org/wp-content/uploads/2024/07/WY_Workforce_Note.pdf

Report	Publishing Organization	Year
Clean Jobs America 2025	E2	2025
Clean Jobs America 2024	E2	2024
US Energy and Employment Jobs Report (USEER)	US Department of Energy	2024
US Energy and Employment Jobs Report (USEER)	US Department of Energy	2025
Wyoming Workforce Annual Report	Wyoming Workforce Development Council	2024
Wyoming Energy Authority Annual Report	Wyoming Energy Authority	2024
Wyoming’s Energy Transformation: Insights from Federal Engagement with Coal Communities	Resources for the Future	2024
Diagnosing Wyoming’s Workforce Challenges	Growth Lab	2024
How Wyoming’s Exodus of Young Adults Holds Back Economic Diversification	Growth Lab	2024
Economic Indicators for Greater Cheyenne	Center for Business and Economic Analysis at Laramie County Community College	2024
Housing in Wyoming: Constraints and Solutions	Growth Lab: Pathways to Prosperity Project	2023

Table 4. Studies and reports included in the literature review.

Wyoming Clean Energy Jobs and Trends

In the past few years, alternative energy job numbers and trends show that there has been gradual growth in this sector in Wyoming. The data reviewed in this section is not exhaustive. Some reports were excluded from the trends and projections presented below due to their age or limited relevance to the MSA's goals and planning efforts.

Report	Total Clean Energy Jobs	Sectoral Jobs	Trends in Job #s	Other trends
<u>Clean Jobs America (2024)</u>	8,497 jobs in Wyoming	Energy efficiency: 6,939 jobs. Renewables: 518 jobs. Other (storage and grid, clean vehicles, and biofuels): 1,041.	3.2% total job growth in clean energy employment, ranked 42nd nationally. Energy efficiency: 1.4% job growth. Renewables: 21.9% job growth. Storage and grid: 5.0% job growth. Clean vehicles: 11.5% job growth. Biofuels: 5.0% job growth.	58% of all new U.S. energy sector jobs in 2023 were in clean energy.
<u>Clean Jobs America (2025)</u>	8,735 jobs in Wyoming	Energy efficiency: 7,054 jobs. Renewables: 595 jobs. Other (storage and grid, clean vehicles, and biofuels): 1,085.	3.2% total job growth in clean energy employment, ranked 42nd nationally. Energy efficiency: 1.7% job growth. Renewables: 15% job growth. Storage and grid: 11.8% job growth (top 10 in the nation). Clean vehicles: -5.4% job growth. Biofuels: 2.6% job growth.	82% of all new U.S. energy sector jobs in 2024 were in clean energy, up from the previous year. Women and people of color are overall underrepresented in the clean energy workforce. 27% of workers are people of color, 28% of workers are women.
<u>USEER (2024)</u>	8,559 jobs in Wyoming.	None reported.	2% growth in clean energy jobs in Wyoming, up from the previous year. Fastest rate of growth nationally in the renewable energy sector (23.3%). 3rd highest state percentage of total energy efficiency jobs (2.4%).	Employers reported 37.6% overall hiring difficulty, down from the previous year. Wyoming is the top state in energy jobs as a percentage of total jobs (16.6%). This is largely due to the heavy presence of fuels.

Table 5. Clean energy jobs and trends (continued on the next page).

Report	Total Clean Energy Jobs	Sectoral Jobs	Trends in Job #s	Other trends
<u>USEER (2025)</u>	No clean energy jobs reported.	None reported.	Wyoming was one of the top states that had the most energy employment per 100,000 workers.	Average wages: Electric power generation: \$65,430, Fuels: \$62,780, Transmission, distribution, and storage: \$59,840, Energy efficiency: \$59,390, Motor vehicles and component parts: \$53,620.
Wyoming Workforce Annual Report (2025)	No clean energy jobs reported.	5.7% of state employment is in construction, which includes the construction of wind farms.	1.1% employment increase from 2023 to 2024. Over the next 10 years (2022-2032), Wyoming is projected to: Add more than 33,000 jobs overall. Add 3,968 (18.7%) construction jobs and 2,668 (21.0%) transportation and warehousing jobs. Add more than 2,700 new jobs in science, technology, engineering, and mathematics (STEM) occupations.	Wyoming's job growth was driven by construction(5.7%), health care and social assistance (1.6%), and professional & business services (1.7%). Mining, including oil and gas, lost jobs (-6.1%). Wyoming's unemployment rate was 3.2% in 2024, up from 2.9% in 2023. Electricity generated through renewable resources: 18.2%.

Report	Projected Growth	Subsector Growth	Projection Year	Percent Growth Over Baseline
<u>Wyoming's Clean Energy Jobs Potential Through 2030 (2022)</u>	2,483 clean energy jobs.	Solar: 279 jobs. Wind: 143 jobs. Battery storage: 641 jobs. Energy efficiency: 1,420.	2030 (low end projection)	200% over 2020

Table 6. National Laboratory of the Rockies's (NLR) projected clean energy job growth in Wyoming.

National Landscape: Clean Energy Workforce

To provide context for Wyoming's current and projected clean energy workforce, this section summarizes national employment and industry trends in the energy and transportation sectors.

National Energy Sector Trends

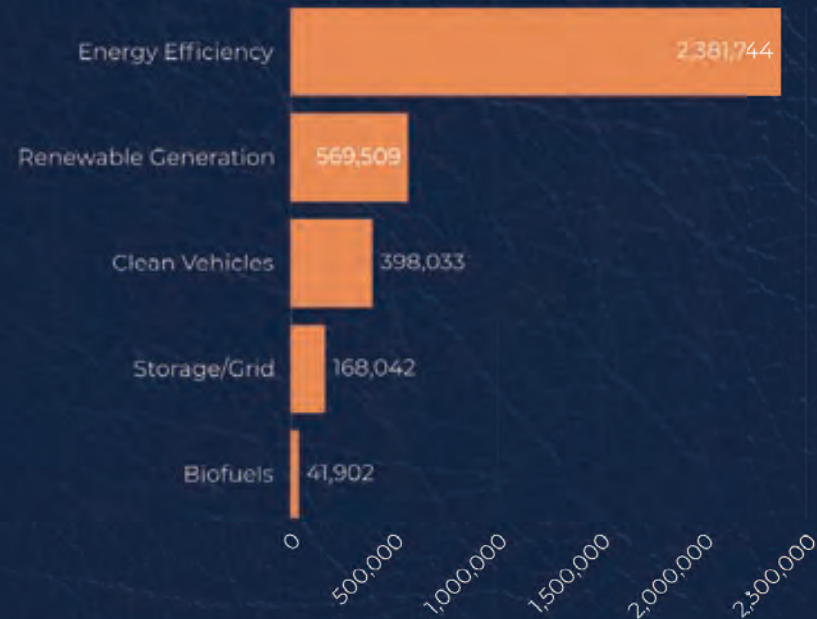
USEER and Clean Jobs America, reported clean energy jobs increased in 2023 and 2024 and continue to outpace U.S. employment. Solar and wind keep expanding, increasing job availability in those electricity generating sub-sectors. There is notable growth and opportunity in the construction and manufacturing industries, as these industries employ about half of the energy sector's workforce. Despite its dynamism and continued growth, clean energy saw a slower rate of growth in 2024 and beyond due to federal policy decisions. According to E2's report,¹⁹ in the first half of 2025, companies canceled more than \$22 billion in clean energy factories and projects, eliminating 16,500 announced jobs; 830,000 jobs are expected to be lost by 2030. Because of this, the clean energy workforce is at a challenging juncture in time. Job growth is expected to continue, but at a slower rate than in past years.

For a more detailed look into key findings from the USEER and Clean Jobs America reports, view Table 6 on the following page.

¹⁹ https://cleanjobsamerica.e2.org/wp-content/uploads/2025/09/E2-2025-Clean-Jobs-America-2025_final.pdf



U.S. Clean Energy Jobs By Sector
Total Clean Energy Jobs: 3,559,050



Source: https://cleanjobsamerica.e2.org/wp-content/uploads/2025/09/E2-2025-Clean-Jobs-America-2025_final.pdf



Report	Overall Growth	Key Findings
<u>USEER 2024</u>	3% growth in 2023 = 149,000 new clean energy jobs. Outpaced overall national growth by 50%.	Clean energy jobs accounted for 59% of new energy sector jobs (4.9% growth rate). Since 2020, jobs in clean energy have grown by 400,000 (12.8% growth rate). Clean vehicle jobs increased by 11.4% (24,826 jobs); jobs in gas and diesel vehicles, by 2% (39,305 jobs). Wind and solar jobs increased the number and share of Hispanic/Latino workers by 9,400. 32% of total new energy jobs, and 33% of new clean energy jobs in 2023 were filled by Hispanic or Latino workers, representing 18% of total energy jobs. Union employers were 50% more likely to offer or require diversity and/or inclusion training programs than non-union employers (36% and 24%, respectively). Women held 26% of energy jobs, but represent 47% of the U.S. workforce. Black or African American help 9% of energy jobs but they represent 13% of the U.S. workforce.
<u>USEER 2025</u>	Energy use jobs, specifically for energy efficiency, were 2,381,700 in 2024. This accounts for 47% of the energy use employment landscape. Detailed numbers for clean energy jobs are: Solar: 370,600 jobs, Wind: 133,000 jobs, Battery storage: 78,800, EV charging: 2,900.	Traditional fuel production offered the highest wages at \$65,400. The construction and manufacturing industries employ almost 50% of the energy sector workforce. 39.7% of jobs in the electric power generation (EPG) sector are for solar and 14.2% for wind, making them the two largest sub-sectors in EPG.
<u>Clean Jobs America 2024</u>	6.4% in 2023 = 149,170 new clean energy jobs.	Clean energy jobs accounted for nearly 60% of all jobs in the entire energy sector. 3.5 million Americans working in clean energy at the start of 2024. 1 in 16 new jobs nationwide were in clean energy. 42,000 clean energy manufacturing jobs added in 2023. Clean energy jobs grew 200% faster than job growth economywide. Manufacturing and other services accounted for almost 60% of the jobs added.
<u>Clean Jobs America 2025</u>	2.8% in 2024 = 100,000 new clean energy jobs. Outpaced U.S. employment more than three times over.	Clean energy jobs exceed the number of oil, gas, and coal by more than 3:1. Over 3.5 million Americans worked in clean energy at the start of 2025. 82% of all net new energy sector jobs in 2024 were clean energy. Nearly 60% of all clean energy jobs are in construction and manufacturing today.

Table 7. National energy sector trends.

Barriers to Clean Energy Workforce Expansion in Cheyenne

Barriers for clean energy jobs were identified from the literature review and engagement process. Many of these challenges, specifically the emigration of young adults and lack of workforce availability, are greatly interconnected. These challenges are explored in greater detail below.

Emigration from Young Adults

Wyoming has been experiencing an exodus of its young adult population for years. Nearly two-thirds of the population leaves by their thirties, impacting the state's efforts to diversify its economy and expand its workforce.²⁰ Historically, in Laramie County, more people moved in than out of the county. 2020 and 2021 were the two most balanced years in the last 20 years.²¹ In addition to this, during the engagement process, stakeholders expressed that employee retention is increasingly becoming more challenging for direct and indirect reasons. The following sub-sections discuss these factors.

Lack of Affordable Housing

Growth Lab, a research organization housed at Harvard Kennedy School, has been conducting a study to identify growth diagnostics and statewide challenges.²² Part of that study included identifying constraints and solutions to housing in Wyoming.²³ Wyoming's housing market is more constrained than much of the United States. Home prices consistently exceed what economic fundamentals predict and housing supply has not been responsive to demand. These constraints have limited population growth, reduced density near city centers, and reduced workforce availability. The primary causes of these constraints are excessive regulations on housing density and types and insufficient investment in critical arterial infrastructure, particularly water systems.

Wyoming has strict regulations regarding residential zoning rules including minimum lot size, height regulations, parking regulations, etc. These restrict the development of denser and smaller housing that would help meet housing needs and demands. Studies conducted by the Wyoming Planning Association (WYOPASS) and the CATO Institute²⁴ revealed that Wyoming ranks highly in zoning and land use constraints, which impact the availability of housing supply. Building height and parking regulations further limit density and multi-family development due to how the requirements impact developer profitability.

Another contributing factor is limited land supply and availability. The federal government owns and manages a large amount of land in Wyoming, and in Cheyenne specifically, the presence of F.E. Warren Air Force Base further limits developable land.

²⁰ <https://growthlab.hks.harvard.edu/sites/projects.iq.harvard.edu/files/2024-10-glwp-233-wyoming-growth-diagnostic.pdf>

²¹ <https://www.wyomingeconomicdata.com/wp-content/uploads/Annual-Trends-Report-2024-1.pdf>

²² <https://growthlab.hks.harvard.edu/project/pathways-prosperity-wyoming/>

²³ <https://growthlab.hks.harvard.edu/wp-content/uploads/2023/09/2023-04-cid-wp-435-wyoming-housing-note.pdf>

²⁴ <https://www.cato.org/sites/cato.org/files/pubs/pdf/pa-823.pdf>

Lack of Amenities

Recreation opportunities and amenities are methods to attract new residents and retain existing ones. Housing, building, and parking regulations discussed above reduce foot traffic in downtown core areas, which impact the amount of urban amenities. As many young adults prefer living in walkable and densely populated areas, many might choose to leave Wyoming in search of places that have more amenities to offer.

Lack of Occupations Requiring a Tertiary Degree

In 2022, Wyoming had the sixth highest score in the nation and highest in the West in the National Assessment of Education Progress.²⁵ However, Wyoming's economy and workforce are comprised of the mining, accommodation, and food services industries, which often don't require a tertiary degree. In turn, many Wyomingites who have a tertiary degree end up moving out of state in search of employment opportunities and higher paying jobs.²⁶

Lack of Workforce Availability

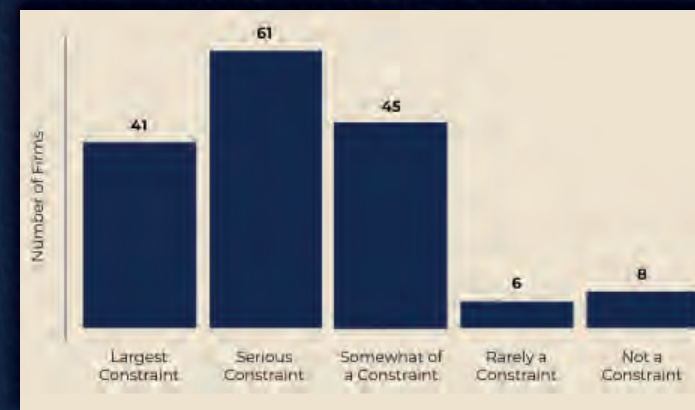
In recent years, Wyoming has been experiencing a decrease in workforce availability. This is supported by the fact that, in 2023, there were more than two job openings for every unemployed person in Wyoming.²⁷ Many businesses have expressed that the lack of workforce availability is hindering company growth. According to a survey conducted by the Wyoming Department of Workforce Services, more than 50% of companies identified workforce availability as a constraint, as shown in Figure 1 below. In addition to this, from 2021-2022, Wyoming had an employment growth rate of 1%, the second worst in the nation; the national rate was 13.8%.²⁸

Business leaders and public officials have also identified the lack of workforce availability as the main barrier to expansion. This has caused businesses to abandon plans to expand to the state over the fear of not being able to hire workers.²⁹

²⁵ <https://ignitewy.org/wp-content/uploads/2023/01/Wyomings-National-Rank-Quality-Counts-and-NAEP.pdf>

^{26, 27, 28} https://wyomingbusiness.org/wp-content/uploads/2024/07/WY_Workforce_Note.pdf

²⁹ <https://growthlab.hks.harvard.edu/wp-content/uploads/2024/11/2024-10-glwp-233-wyoming-growth-diagnostic.pdf>



Workforce availability as a constraint identified through a survey for businesses.

Economic Reliance on Resource Extraction

It is important to note that Wyoming has a unique economic landscape. The state has had a long-standing presence and economic dependence on resource extraction, specifically coal, oil, and natural gas.

- As of 2024, Wyoming is the largest producer of coal, 7th largest producer of oil, and 10th largest producer of natural gas in the United States.³⁰
- In 2022, 21 of Wyoming's 23 counties produced oil and/or natural gas.³¹
- Wyoming is the second largest producer of oil and natural gas on federal lands.³²
- In 2022, the state levied approximately 80% of taxes, equaling \$1,092,942,314 for oil, gas, coal, and other minerals.³³
- Wyoming has a state severance tax for the extraction of oil and gas. Much of the tax revenue is stored in a trust fund, the Permanent Wyoming Mineral Trust Fund, that goes directly to the state's General Fund to help pay for state operations. In 2022, oil and gas paid over \$835 million in severance taxes, equating to 80% of severance taxes collected.³⁴
- As of August 2024, coal mining employs around 4,400 people and pays more than \$100,000 annually, equating to 1.5% of state nonfarm employment and approximately .07% of the state population.³⁵

The economic importance of extractive practices in Wyoming can therefore not be overlooked or dismissed. Historically, extractive practices have provided significant employment and funding to the state and its communities. It is imperative to understand this as it may impact the rate and support of clean energy adoption.

³⁰ https://wyoenergy.org/wp-content/uploads/2025/01/24_WyomingEnergy_Mag.pdf

³¹ <https://pawyo.org/facts-figures/>

³² <https://pawyo.org/facts-figures/>

³³ <https://pawyo.org/facts-figures/>

³⁴ <https://pawyo.org/facts-figures/>

³⁵ <https://www.rff.org/publications/reports/wyomings-energy-transformation/>



Low Wage Growth

Between 2021-2022, Wyoming saw the fourth lowest wage growth nationally.³⁶ One of the contributing factors is slow wage increases within occupations. From 2012-2022, median wages for management, business and financial operations, computer and mathematics, and legal fields were 8-25% lower than previous national levels and have continued to decline since. The only industry that has seen a continuous increase is the healthcare industry. Another contributing factor is the changes in the occupational structure of the economy. The state's occupational structure is mainly comprised of construction and extraction; life, physical, and social sciences; and installation and maintenance. This compounds with low industry productivity. Workers in Wyoming, on average, have a lower output per person than workers in the same industries across the U.S.³⁷ Almost 83% of businesses fall in this category. Because each worker produces less, businesses also tend to pay lower wages. The only outliers for productivity are jobs related to resource extraction, where Wyoming workers are more productive than the national average.³⁸

Opportunities to Expand the Clean Energy Workforce

Economic Diversification

One of Wyoming's most significant challenges is the lack of economic diversity; the five industry sectors with the most employment are government, leisure and hospitality, retail trade, educational and health services, and construction.³⁹ However, efforts are underway across the state to capitalize on emerging trends and industries and diversify and strengthen the state's economy. Within clean energy in particular, carbon capture, utilization, and storage (CCUS) and wind energy present major opportunities for Wyoming to expand its workforce and increase job availability.

Carbon Capture, Utilization, and Storage (CCUS) in Wyoming

Wyoming is one of the first states with Class VI Well Primacy. This means the state is certified by the Environmental Protection Agency's (EPA) Underground Injection Control (UIC) program to issue and enforce CO2 injection well permits for underground storage. Thus, Wyoming, with private sector partners, worked to create the Integrated Technology Center (ITC), a research and innovation center for carbon capture and conversion technologies. The ITC receives \$15 million of investment from the state of Wyoming, \$5 million from private industry, and an additional \$1 million from the National Rural Electric Cooperative Association.⁴⁰

The ITC is not the only state-led effort. The University of Wyoming hosts the Center for Carbon Capture and Conversion (CCCC) and operates the Carbon Storage Assurance Facility Enterprise (CarbonSAFE). At CCCC, research scientists work

³⁶ https://wyomingbusiness.org/wp-content/uploads/2024/07/WY_Workforce_Note.pdf

³⁷ https://wyomingbusiness.org/wp-content/uploads/2024/07/WY_Workforce_Note.pdf

³⁸ https://wyomingbusiness.org/wp-content/uploads/2024/07/WY_Workforce_Note.pdf

³⁹ https://doe.state.wy.us/LMI/trends/0126/ss_proj.htm

⁴⁰ <https://wyoenergy.org/portfolio/carbon-capture-utilization-storage-ccus/>

to develop uses for coal, including new uses, converting coal into products, and carbon capture. CarbonSAFE, funded through the Department of Energy (DOE), is one of 13 original CCUS project sites ensuring safe deployment for CCUS systems.

Wyoming's exploration of CCUS isn't stopping any time soon. Through CCUS work, Wyoming is expected to add an annual average of up to 3,340 project jobs over 15 years and 1,964 operations jobs.⁴¹ Recent efforts showcase the state and private entities' commitment to expanding efforts and becoming a national leader in carbon capture. Moreover, the state's Class VI primacy gives Wyoming an advantageous position to continue developing CCUS projects.

Wind Energy

Wyoming is the eighth state in the country with the most wind energy generation potential.⁴² Geographically, Wyoming's mountain passes move steady winds across its prairie lands, giving the state strong wind energy generation potential. Because of this, wind energy is an abundant resource in Wyoming, currently making it the state's second-largest source of electricity generation.

Wind energy projects are already underway to harness this potential and expand renewable electricity generation across the state and create new jobs within Wyoming's clean energy workforce. One of these projects, the Chokecherry and Sierra Madre Wind Energy Project (CCSM Project), is a utility-scale windfarm that is set to be completed in 2029. The CCSM Project is expected to be the largest wind power project in North America, create thousands of construction jobs and around 114 permanent operations and maintenance jobs, and contribute millions in property taxes and sales and use taxes.⁴³

In addition, Southeast Wyoming has an average wind speed greater than 23 MPH, making it an ideal place for wind energy generation. Given its geographic location, Cheyenne and Laramie County at large should prioritize supporting and expanding wind energy development and attracting investment that will lead to economic growth and workforce expansion.

Conclusion

Based on this landscape review of Wyoming's workforce, the Cheyenne Metro Statistical Area (MSA) is at a critical juncture for its workforce development ecosystem. The MSA, and Wyoming as a whole, are facing the challenge of an aging population and low workforce availability. This can prevent businesses from wanting to move or open in the area, and hinders the state's economic output. The clean energy economy has major potential in Wyoming, specifically for wind energy development. The Cheyenne MSA has an opportunity to lead the charge in the clean energy workforce expansion by leveraging existing resources.

⁴¹ <https://wyoenergy.org/portfolio/carbon-capture-utilization-storage-ccus/>

⁴² <https://wyomingrenewables.org/renewable-technologies/wind/>

⁴³ <https://www.powercompanyofwyoming.com/about/index.shtml>

Appendix C1: Existing Programs and Partnerships

Laramie County Community College (LCCC) has been partnering with the school district to work together on developing a talent pipeline and helping kids earn a significant number of credits before going to college.⁴⁴

Partners: LCCC, Laramie County School District

Laramie County Community College offers students the opportunity to enroll in the LCCC Accelerate Program. The program allows high school students to earn up to 24 college credits, making it free to earn an associates degree.⁴⁵

Partners: LCCC

Laramie County Community College offers a one semester wind energy program where students learn the ins and outs of operating and repairing wind turbines.⁴⁶

Partners: LCCC

The Department of Workforce Services (DWS) provides training services for those seeking employment. They have a location in Cheyenne.

Partners: DWS

The Wyoming Innovation Partnership (WIP) brings together state agencies and education partners to strengthen Wyoming's workforce and economy through better coordination and collaboration. WIP has an energy program and a workforce development program that trains and assists students in recruitment and job placement.⁴⁷

Partners: WIP, Wyoming Business Alliance (WBA), Wyoming Heritage Foundation (WHF)

Wyoming's Workforce Development Council leads the state's NextGen Sector Partnerships program, which aims to identify and address pressing needs in key industries.⁴⁸

Partners: DWS

The Wyoming State Energy Office facilitates the State Energy Program (SEP), a federally funded program. The program helps communities save money on electricity bills and provides technical assistance.⁴⁹

Partners: Wyoming State Energy Office

⁴⁴ Interview with Laramie County Community College representative.

⁴⁵ <https://www.lccc.wy.edu/accelerate/>

⁴⁶ <https://www.lccc.wy.edu/pathways/tradesTechnical/windEnergyProgram/index.aspx>

⁴⁷ <https://wip.wyo.gov/home>

⁴⁸ <https://dws.wyo.gov/dws-division/councils-and-commissions/wwdc/next-generation-sector-partnerships/>

⁴⁹ <https://wyoenergy.org/state-energy-office/>

The Wyoming Integrated Test Center (ITC) provides a space for carbon capture, utilization and sequestration (CCUS) testing and research.⁵⁰

Partners: ITC

The University of Wyoming hosts the Center for Carbon Capture and Conversion (CCCC) and operates the Carbon Storage Assurance Facility Enterprise (CarbonSAFE).

Partners: University of Wyoming

⁵⁰ <https://wyomingitc.org/>



Laramie County Community College students perform maintenance on a wind turbine on F.E. Warren Air Force Base near Cheyenne in December, 2019. Photo Credit: Laramie County Community College.

Appendix D: Authority to Implement

Energy and Buildings

Measure 1: The City of Cheyenne and Laramie County have authority to adopt and enforce local building codes pursuant to WY Stat § 35-9-121. The City can also implement certain residential energy efficiency and electrification incentive methods through WY Stat § 15-1-103 (a)(xiii), which allows governing bodies to license and regulate certain business activities, building contractors and professional trades. Under Resolution 6135, the City of Cheyenne has the authority to initiate, develop, and execute a Property Assessed Clean Energy (PACE) program.

Implementation of incentive-based programs may require consideration of Article 16, Section 6 of the Wyoming Constitution.

Measure 2: The City of Cheyenne and Laramie County have authority to adopt and enforce local building codes pursuant to WY Stat § 35-9-121. The City can also implement certain commercial and industrial energy efficiency and electrification incentive methods through WY Stat § 15-1-103 (a)(xiii), which allows governing bodies to license and regulate certain business activities, building contractors and professional trades. Under Resolution 6135, the City of Cheyenne has the authority to initiate, develop, and execute a Property Assessed Clean Energy (PACE) program.

Implementation of incentive-based programs may require consideration of Article 16, Section 6 of the Wyoming Constitution.

Measure 3: The City of Cheyenne maintains authority over municipal facilities, procurement, and operations pursuant to WY Stat § 15-1-103. The City has existing authority to implement land use, zoning, and development policies pursuant to WY Stat § 16-1-601. The City's Unified Development Code further establishes regulatory authority to support implementation of land use policies related to Measure 3.

Measure 4: The City of Cheyenne and Laramie County have authority to adopt and enforce local building codes pursuant to WY Stat § 35-9-121. The City can also implement certain commercial and industrial energy efficiency and electrification incentive methods through WY Stat § 15-1-103 (a)(xiii), which allows governing bodies to license and regulate certain business activities, building contractors and professional trades. The City has existing authority to implement land use, zoning, and development policies pursuant to WY Stat § 16-1-601. The City's Unified Development Code further establishes regulatory authority to support implementation of policies that impact subdivisions, zoning, and design elements of development.

Implementation of incentive-based programs may require consideration of Article 16, Section 6 of the Wyoming Constitution.

Measure 5: The City of Cheyenne and Laramie County have authority to adopt and enforce local building codes pursuant to WY Stat § 35-9-121. The City can also implement certain commercial and industrial energy efficiency and electrification incentive methods through WY Stat § 15-1-103 (a)(xiii), which allows governing bodies to license and regulate certain business activities, building contractors and professional trades. The City has existing authority to implement land use, zoning, and development policies pursuant to WY Stat § 16-1-601. The City's Unified Development Code further establishes regulatory authority to support implementation of policies that impact subdivisions, zoning, and design elements of development.

Implementation of incentive-based programs may require consideration of Article 16, Section 6 of the Wyoming Constitution.

Transportation

Measure 1: The City of Cheyenne maintains authority over transportation infrastructure, public streets, sidewalks, public safety, and related public works activities pursuant to WY Stat § 15-1-103. The City also maintains authority to regulate land use, subdivision standards, transportation systems, connectivity, and development patterns pursuant to WY Stat § 15-1-601 et seq. and WY Stat § 9-8-301 et seq. The City's Unified Development Code further establishes regulatory authority to support implementation of policies that impact subdivisions, zoning, and design elements of development. This encapsulates infrastructure improvements related to active transportation, pedestrian safety, bicycle safety, etc.

Measure 2: The City of Cheyenne maintains authority over municipal facilities, procurement, and operations pursuant to WY Stat § 15-1-103, as well as authority over transportation infrastructure that allows for fleet vehicle purchases and public EV chargers.

Measure 3: The City of Cheyenne maintains authority over transportation infrastructure, public streets, public safety, and public transit coordination, and intergovernmental cooperation pursuant to WY Stat § 15-1-103, WY Stat § 15-1-601 et seq., and WY Stat § 9-8-301 et seq. The City's existing land use, transportation, and infrastructure planning authority may also support transit-oriented development, multimodal infrastructure improvements, and coordination of regional mobility initiatives intended to improve transportation access and reduce transportation-related congestion and emissions.

Measure 4: The City of Cheyenne maintains authority to support redevelopment and downtown revitalization activities pursuant to the Wyoming Urban Renewal Code, WY Stat § 15-9-101 et seq., and the Downtown Development Authority, WY Stat § 15-9-201 et seq. The City of Cheyenne also maintains authority over transportation infrastructure, public streets, sidewalks, public safety, and related public works activities pursuant to WY Stat § 15-1-103. The Cheyenne Transit Program (CTP) is a municipal entity that maintains authority to provide transit services. Regional coordination efforts may be supported by the CTP.

Waste

Measure 1: The City of Cheyenne maintains authority over solid waste collection and public sanitation, via its Public Works department, pursuant to WY Stat § 15-1-103 and related municipal powers authorities. The City may also regulate and contract for municipal services associated with waste collection and waste diversion activities. WY Stat § 15-7-101 et seq gives Cheyenne the authority to plan, create, construct, and equip liquid and solid waste facilities.

Where applicable, implementation of waste management measures may also involve coordination with the Wyoming Department of Environmental Quality (WDEQ) pursuant to the Wyoming Environmental Quality Act, WY Stat § 35-11-101 et seq., including applicable solid waste management and disposal regulations.

Measure 2: The City of Cheyenne and the Cheyenne Board of Public Utilities (BOPU) maintain authority over municipal water supply, water utility operations, water infrastructure, rate-setting, and water conservation activities pursuant to WY Stat § 15-7-101 et seq., WY Stat § 15-1-103, and related municipal utility authorities.

The City also maintains authority to regulate land use, subdivision standards, transportation systems, connectivity, and development patterns pursuant to WY Stat § 15-1-601 et seq. and WY Stat § 9-8-301 et seq., which may support water conservation efforts included but not limited to: to low-impact development and landscaping and irrigation requirements.

Measure 3: The City of Cheyenne and the Cheyenne Board of Public Utilities (BOPU) maintain authority over municipal water supply, water utility operations, water infrastructure, rate-setting, and water conservation activities pursuant to WY Stat § 15-7-101 et seq., WY Stat § 15-1-103, and related municipal utility authorities. WY Stat § 15-7-101 et seq also allows the City to enact rules and regulations for the protection of public parks.

The City also maintains authority to regulate land use, subdivision standards, transportation systems, connectivity, and development patterns pursuant to WY Stat § 15-1-601 et seq. and WY Stat § 9-8-301 et seq., which may support water conservation efforts included but not limited to: to low-impact development, landscaping and irrigation requirements, stormwater management, etc.

In 2023, the City adopted the 2023 Dry Creek Drainage Master Plan, which outlines projects to mitigate flooding.

Natural and Working Lands

Measure 1: The City of Cheyenne maintains authority to participate in land use planning, watershed protection, public education, intergovernmental coordination, and natural resource planning activities pursuant to WY Stat § 15-1-103, WY Stat § 9-8-301 et seq., and related municipal powers authorities. This authority may support implementation of policies and programs related to water conservation, watershed protection, public outreach, and promotion of

best management practices associated with agricultural lands. Implementation of agricultural and water resource protection measures may also involve coordination with the Wyoming Department of Environmental Quality (WDEQ), the Wyoming State Engineer's Office, local conservation districts, and other state and federal agencies.

Measure 2: The City of Cheyenne maintains authority to regulate land use, subdivision standards, development patterns, and zoning pursuant to WY Stat § 15-1-601 et seq. and WY Stat § 9-8-301 et seq., which may support water conservation efforts included but not limited to: to low-impact development, landscaping and irrigation requirements, stormwater management, etc. The City also maintains authority to support redevelopment and downtown revitalization activities pursuant to the Wyoming Urban Renewal Code, WY Stat § 15-9-101 et seq., and the Downtown Development Authority, WY Stat § 15-9-201 et seq. The City's Unified Development Code further establishes regulatory authority to support implementation of policies that impact subdivisions, zoning, and design elements of development.

Measure 3: WY Stat § 15-7-101 et seq. gives the City of Cheyenne authority to enact rules and regulations that support public improvement projects.

The City also maintains authority to regulate land use, subdivision standards, zoning, and development patterns pursuant to WY Stat § 15-1-601 et seq. and WY Stat § 9-8-301 et seq.

Workforce

Measure 1: The City of Cheyenne maintains authority to support economic development, public welfare, workforce initiatives, community programs, and intergovernmental partnerships pursuant to WY Stat § 15-1-103 and related municipal powers authorities. This authority may support implementation of workforce development programs, job training initiatives, and public education efforts.

Implementation of incentive-based programs may require consideration of Article 16, Section 6 of the Wyoming Constitution.

Measure 2: The City of Cheyenne maintains authority to regulate and support local business activities, economic development initiatives, public education programs, and municipal services pursuant to WY Stat § 15-1-103, including WY Stat § 15-1-103(a)(xiii).

Measure 3: The City of Cheyenne maintains authority to regulate and support local business activities, economic development initiatives, public education programs, and municipal services pursuant to WY Stat § 15-1-103, including WY Stat § 15-1-103(a)(xiii).

Appendix D: Inventory and Inventory Methodology

WY



In This Appendix

Introduction

Inventory Details

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Greenhouse Gas Emissions Inventory Management Plan for Community-Wide Emissions

Cheyenne, Wyoming

Calendar Year 2023

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Introduction

Lotus Engineering and Sustainability, LLC (Lotus) was contracted by the City of Cheyenne, Wyoming to create a greenhouse gas (GHG) emissions inventory for the City of Cheyenne and Cheyenne's Metropolitan Statistical Area (MSA) for calendar year 2023. The purpose of the GHG emissions inventory is to create a clear picture of current GHG emission sources in Cheyenne and the MSA and provide an updated inventory that will guide climate action planning.

To support this work, Lotus completed the following tasks:

- Developed a 2023 GHG inventory and incorporated GHG accounting best practices, including data collection and methodology that are consistent with accepted national and international standards. Lotus created a customized inventory management workbook to keep a record of data, emission factors, contacts, and assumptions.
- Created this accompanying Inventory Management Plan (IMP) for the inventory tool providing guidance so that staff can continue to track emissions. The IMP is contained herein and provides community information, inventory boundary and scope, GHG emission factors, emission quantification methods, data management methods, and an overview of the inventory tool.

Inventory Details

Protocol

The GHG inventory uses the Global Protocol for Community-Scale Greenhouse Gas Emission Inventories (GPC protocol) methodology.¹ The GPC provides a transparent and consistent GHG accounting methodology for reporting community GHG emissions. There are two reporting levels for the community framework:

- **BASIC:** The BASIC level includes stationary energy, in-boundary transportation, and community-generated waste and wastewater.
- **BASIC+:** The BASIC+ level includes BASIC emission sources, as well as a more comprehensive coverage of emissions sources such as transboundary transportation; electricity transmission and distribution

¹ For more information regarding GPC see: <https://ghgprotocol.org/ghg-protocol-cities>.

losses; industrial processes and product use; and agriculture, forestry, and other land uses.

The Cheyenne 2023 MSA GHG inventory was developed in compliance with the GPC partial BASIC+ reporting level. Relevant emissions sources included in the BASIC+ level were reported in addition to BASIC emissions as supplement.

Boundary

Per the GPC, communities shall establish a geographic boundary that identifies the spatial dimensions or physical perimeter of the inventory's boundary. The boundary for the Cheyenne MSA inventory reflects the County boundary (Figure 1).



Figure 1. Cheyenne MSA (Laramie County) Limits

Time Period

The 2023 inventory is based on the calendar year of 2023 (January 1, 2023 through December 31, 2023). If data specific to 2023 was not available, the most recent year was included and noted in the workbook and this IMP.

Emission Scopes

Per the GPC, BASIC emission scopes are defined as follows. The items in **bold** are included in the inventory as part of a BASIC inventory. The items listed in ***italics and bold*** are applicable to jurisdictions as part of a BASIC+ inventory. The other items are not applicable to the jurisdictions because they were confidential (C), not occurring (NO) or the emissions were included elsewhere

(IE).² Emission activities that are occurring but produce negligible amounts of emissions and were consequently excluded from the inventory are marked as *de minimis*.

- Scope 1: GHG emissions from sources located within the County boundary, including:
 - o Stationary Energy
 - **Building energy fuel combustion.**
 - **Fugitive emissions (i.e., leakage from natural gas consumption).**
 - **Energy industries energy consumption.**
 - Non-specified sources energy consumption. (NO)
 - Energy consumption from agriculture, forestry, and other land use (AFOLU). (IE)
 - o Transportation
 - **On-road transportation (including transit) fuel combustion.**
 - **Railways fuel combustion.**
 - Waterborne navigation. (NO)
 - **In-boundary aviation fuel combustion.**
 - **Off-road vehicles fuel combustion.**
 - o Waste
 - **Municipal solid waste generated in the boundary and disposed of within the boundary.**
 - **Composted waste generated in the boundary and disposed of within the boundary.**
 - **Municipal wastewater generated inside the boundary and disposed of within the boundary.**
 - Municipal wastewater generated outside the boundary and disposed of within the boundary. (NO)
 - Incinerated waste. (NO)
 - Emissions from closed landfills within the boundary. (NO)
 - o IPPU
 - **Emissions from industrial processes occurring in the boundary.**

² The GPC uses notation keys to note the reason why an emission source is not applicable to a community. The notation keys include: Not Occurring (NO), Included Elsewhere (IE), Confidential (C) and Not Estimated (NE).

- **Emissions from product use occurring within the boundary.**
- o AFOLU
 - **Net flux of carbon emissions and removals.**
 - **Fertilizer use and livestock emissions.**
- Scope 2: GHG emissions occurring as a result of the use of grid-supplied electricity, heat, steam and/or cooling within the city boundary.
 - o **Building electricity consumption.**
 - o **On-road transportation (including transit) electricity combustion.**
 - o Other industry and transportation activities consuming electricity. (NO)
- Scope 3: GHG emissions that occur outside the city boundary as a result of activities taking place within the county boundary, including:
 - o **Municipal solid waste generated in the boundary and disposed of outside the boundary.**
 - o Composted waste generated in the County and disposed of outside the County. (NO)
 - o **Trans-boundary aviation fuel consumption.**
 - o **Transmission and Distribution Losses.**
 - o Consumption-based: Food Emissions. (NO)
 - o Incinerated waste. (NO)

Note on Information-Only Items

The GPC provides guidance on emissions avoided through the purchase of renewable energy credits (RECs), local installation of renewable energy systems (including solar and wind), or recycling.³ In most cases, the Renewable Energy Credit would need to be “bundled” with the electricity purchase and not part of the utility’s energy portfolio to be accounted for within a community’s inventory. However, communities frequently want to understand the full potential impact of these activities. For communities in Cheyenne’s MSA, these items are calculated as “information-only” and include avoided emissions from:

- Recycling.
- Renewable Energy not bundled with the REC.

³ See GPC section 4.3 (page 44).

Reported Greenhouse Gases

The following GHGs are included in the inventory:

- Carbon dioxide (CO₂).
- Methane (CH₄).
- Nitrous oxide (N₂O).

The following fluorinated GHGs are typically used in industrial processes.

- Hydrofluorocarbons (HFCs)
- Perfluorocarbons (PFCs).
- Sulfur hexafluoride (SF₆).
- Nitrogen trifluoride (NF₃).

Only refrigerant leakage was reported for the MSA and community, which includes reported HFCs. Other industrial processes were not identified for the MSA or community, and therefore, these GHGs were not included in the inventory.

Total emissions were reported as metric tons of carbon dioxide equivalent (mt CO₂e).

Global Warming Potentials

All GHGs have different global warming potentials (GWP). The GWP of a GHG defines its contribution to global warming (i.e., the ability of each gas to trap heat in the atmosphere), where a GWP of one is equal to the impacts of one unit of CO₂. The effect of a non-CO₂ GHG or the combination of different GHGs is expressed as CO₂e or carbon dioxide equivalent. This allows for easy comparison of the absolute effects of different gases.

GWPs have been sourced from the Intergovernmental Panel on Climate Change (IPCC) Fifth Assessment Report.⁴ Table 1 shows the GHGs included in the inventory and their GWPs. Methane and N₂O are converted to CO₂e by multiplying their value by the 100-year GWP coefficient.

⁴ For more information see: https://www.ipcc.ch/report/ar6/wg1/downloads/report/IPCC_AR6_WGI_Chapter07_SM.pdf, page 16.

Table 1. Global Warming Potentials.

Common Name	Formula	GWP
Carbon Dioxide	CO ₂	1
Methane	CH ₄	28
Nitrous Oxide	N ₂ O	265

Lotus recommends that for each new inventory the most recent IPCC 100-year GWP values are used.

Process to Update Inventory

The total time dedicated to data collection and calculations is estimated at between 50-75 hours. Data collection typically takes up to three months.

For consistency purposes, it is good practice to request that data is provided in the same format it was provided for this inventory. In the zipped folder of data provided by Lotus, portable document format (pdf) copies of email chains with each data contact are included, where applicable. The text in the emails/pdfs can be used to save time when sending out new emails to collect data. The Inventory Data Checklist should also be referenced for a comprehensive list of all the data contacts and their contact information.

The Excel workbook is set up to ensure that it is clear which cells need to be updated on a regular basis. See Table 2 for an overview and example of how color-coding was used to help the user identify which data should be updated regularly versus less often. Some blue cells may have raw data that was manipulated to get it into the format required for the inventory. See the Assumptions section of the Data Entry tabs in the inventory and the assumption notes in the source data sections of this Inventory Management Plan for more information.

Table 2. Cell Color Guide.

Cell Color	Meaning	Notes
	Do not edit	Do not edit; this is a formula.
	May need to be updated	These cells may need to be updated depending on if a protocol releases new emission factors, large infrastructure

		changes, etc. These cells are not updated regularly.
	Needs to be updated each inventory	Blue cells require manual entry each year since data updates on an annual basis.

Cheyenne Workbook Overview

The Cheyenne MSA inventory workbook includes different color tabs, as described below:

- **Summary Tabs:** The **gray** tabs provide an overview of the spreadsheet including 1) Workbook Intro and 2) Visual Summary.
- **Community Emission Summary Tabs:** These **green** tabs include community indicators that are unique to the City of Cheyenne and the Cheyenne MSA, as well as community-specific visual summary data. The community indicators also allow for the ability to identify trends over time (e.g., population increase or decline).
- **Conversion Factors and GWPs** is a **black** tab that provides data used throughout the workbook.
- **Source Data Tabs:** The various **blue** tabs store the raw data for each source and calculate the resulting GHG emissions. All data in light blue cells within these tabs will need to be updated annually or as often as a new inventory is completed.

Summary Tabs

Workbook Introduction

The workbook introduction provides an overview of each tab within the workbook and is intended to guide the user. This does not need to be updated with each new inventory unless changes are made to the spreadsheet by adding or subtracting tabs.

Visual Summary

This tab provides visual and emission summaries from the workbook and aggregates the data in tables, charts, and graphs. No changes will need to be made annually.

Community Emission Summary Tabs

Individual data tabs for each boundary summarize all the data collected, including community indicators. The data from this spreadsheet draws from the blue source tabs which will need to be updated with each subsequent inventory. Emission Summary tabs include:

- Cheyenne Community (City of Cheyenne)
- Cheyenne MSA (Laramie County)

The purpose of community indicators is to normalize emissions and understand trends and drivers behind changes in the community's total emissions over time. This section of these tabs will need to be updated with each new inventory.

Conversion Factors and GWPs Tab

This black tab provides conversion factors that will not change over time.⁵ However, the GWPs will need to be updated when the most recent IPCC 100-year values are made available. See the **Global Warming Potentials** subsection for more information.

Source Tabs

These tabs range in color from dark blue to light blue. Within each tab, there are four sections:

- Emissions Summary: The emission summary summarizes all emissions from the tab by scope.
- Notes and Assumptions: Explains each data source and all assumptions.
- Emission Factors: Provides a list of all emission factors used in calculations. **Appendix A: Emission Factors** provides all the emissions factors.
- Data Calculations: Provides an overview of the data calculations.

Lotus and City of Cheyenne staff collected data from various sources to complete the 2023 emissions inventory. Whenever possible, actual source data on emissions-generating activity were used to calculate emissions. In some cases, it was necessary to make assumptions about the data received. The assumptions made for each sector are listed below along with a description of the data and calculations for each tab. These assumptions are also included in the notes section on each sector's data tabs. Note that some calculations listed for each sector include unit conversions. Check the units of

⁵ Conversion factors are standard conversions between two units of measure. These are collected from Google searches of necessary conversion factors.

emission factors before calculating emissions for each source to verify if unit conversions are needed.

Stationary Energy Data Tab

Each energy provider delivers data in a unique reporting format. As such, Lotus made assumptions when necessary to ensure the usability and consistency of data. The assumptions used in this section are explained for each energy source below.

Electricity

There are several utility companies that operate in the inventory boundary, so Lotus used utility maps from the Wyoming Public Service Commission to determine which utilities needed to be accounted for. Black Hills Energy electricity data were provided by Carmalee Rose. Due to the 15/15 rule, industrial electricity data were excluded.

High West Energy signed a nondisclosure agreement for their electricity usage data. Lotus maintained these files outside of Cheyenne's filing system. In order to keep these data indistinguishable, Lotus calculated emissions for each utility and combined emissions from all utilities by sector. Electricity usage was also combined by sector but emissions were not calculated from this aggregated total due to each utility's different grid emission factors.

WyRulec, an electricity provider, supplied electricity usage data for the following sectors: residential, commercial/agriculture, and irrigation. Their service territory does not reach Cheyenne, so all kWh were attributed to the MSA.

Wheatland REA provided the total number of customers they served in 2023 within Laramie County. Lotus applied the proportion of customers in each sector and the kWh/customer found in WyRulec's data

Pine Bluffs Electric Utility was not able to provide any customer or energy use information. Lotus took the 2023 residential population of Pine Bluffs from the Wyoming Administration & Information's Economic Analysis Division and applied the proportion of customers found in WyRulec to the rest of the sectors. The same kWh/customer as found in WyRulec was then applied to each sector.

Electricity consumed by electric vehicles was subtracted from residential and commercial totals for buildings. See section *Electric Vehicles* for more information. It was assumed that electric vehicle charging systems were metered with the building's consumption and were therefore removed from

the building totals so that emissions can be calculated in the appropriate sector.

A portion of electricity that is transported through the grid is lost in the transmission process—this is known as transmission and distribution (T&D) loss. A T&D loss factor was calculated based on the U.S. Energy Information Administration (EIA) State Electricity Profile for Wyoming.⁶ To calculate the T&D loss factor, go to the EIA page linked in the footnotes, scroll past the table and click the option for “Full data tables 1-17.” Download the Excel file. Once the Excel file is opened, navigate to the tab for *Table 10: Supply and Disposition of Electricity*. In Table 10, find the rows for Total Supply, Direct Use, and Estimated Losses. See the equation below for calculating T&D losses.

T&D loss percentage = Estimated losses / (Total supply – Direct use)

GHG emissions from T&D losses = MWh * T&D loss factor * emission factor * GWP

GHG emissions from electricity = MWh * emission factor * GWP

The following assumptions were made for electricity:

- Assume that industrial electricity usage in Cheyenne is accounted for in EPA FLIGHT.
- A population-based approach was used to estimate electricity usage emissions from Pine Bluffs Electricity Utility customers.

Natural Gas

Natural gas data from Black Hills Energy were provided by Carmalee Rose. Due to the 15/15 rule, Black Hills was not able to provide industrial natural gas data. Given the small number of industries that Black Hills serves throughout Colorado, South Dakota, and Wyoming (84 total in 2023 according to the Edison Electric Institute's ESG Disclosures), it is assumed that the two facilities listed in EPA Flight with Subpart C: General Stationary Fuel Combustion data comprise the majority of redacted industrial natural gas consumption. Natural gas emission factors were sourced from ICLEI's U.S. Community Protocol for Accounting and Reporting of Greenhouse Gas Emissions – Appendix C.⁷

GHG emissions from natural gas = therms * emission factor * GWP

⁶ U.S. Energy Information Administration State Electricity Profile for Wyoming.
<https://www.eia.gov/electricity/state/wyoming/>.

⁷ For more information see: <https://icleiusa.org/publications/us-community-protocol/>.

No assumptions were made for natural gas.

Stationary Diesel

Stationary diesel data was collected from the Wyoming Dept. of Environmental Quality's IMPACT and Open Air tool. Stationary diesel generators used by the City of Cheyenne were not listed in the Open Air Tool, so Shannon Keslar, Fleet/Building Manager for the police department, reported their generator fuel use in 2023.

GHG emissions from stationary diesel = annual hours of generator use * gallons/hour based on generator specs * emission factor * GWP

GHG emissions from stationary diesel = gallons of diesel * emission factor * GWP

No assumptions were made for stationary diesel.

Propane

Matt Hanson with FerrellGas supplied propane data, which was used to estimate commercial propane emissions. Only commercial data from FerrellGas were available for Cheyenne. Blakeman Propane provided additional residential and commercial propane data for Laramie County.

GHG emissions from propane = gallons * emission factor * GWP

No assumptions were made for propane.

Residential Wood

Cheyenne does not track residential wood usage, so Lotus developed a methodology to estimate usage. First, using U.S. Census House Heating Fuel data, Lotus collected the number of houses that use wood stoves for heating in each jurisdiction. Per the US Energy Information Administration, the average btu used per household in the US West for wood as a primary heat is 69.9 million btu per household. The average BTU estimate was multiplied by the number of households using wood stoves and multiplied by the emissions factors for wood (as a stationary fuel) from the EPA Emission Factors Hub. Note that CO₂ emissions from the burning of wood are classified as biogenic CO₂ emissions and are not included in the CO₂e emissions total.

GHG emissions from residential wood = # of households using wood for fuel * 69.6 million BTU per household * emissions factors (kg/MMBTU) / 1,000 kg/mt *GWP

No assumptions were made for residential wood.

Renewable Energy

Renewable energy generation data were collected for on-site hydroelectric and Black Hills Energy's Renewable Ready program. Subscribers of this program receive their energy from the 52.5 MW Corriedale Wind Energy Project. Black Hills Energy also has purchased power agreements with the Duke Energy Happy Jack Wind (30 MW) and Silver Sage Wind (30 MW) Projects, and the Platte River Power Authority Silver Sage Wind Project (12 MW).

Avoided GHG emissions from renewable energy = $-(\text{MWh} * \text{emission factor} * \text{GWP})$

No assumptions were made for renewable energy.

Fugitive Emissions Tab

Monthly oil and natural gas production data for Laramie County in 2023 were taken from the Wyoming Oil & Gas Conservation Commission. Emissions were calculated for fugitive emissions from natural gas distribution, delivery, production, gathering, processing, transmission, storage, refining, and transportation. Additional emissions from oil wells were calculated. A report on fugitive emissions by the Environmental Defense Fund⁸ was used for emission calculations.

GHG emissions from fugitive natural gas = $(\text{th}/(1 - \text{leakage rate})) * 100,000/\text{natural gas energy density}^9 * 0.02832 \text{ cubic meters/cubic foot} * \text{density of natural gas} * \text{percent of CO}_2 \text{ (or CH}_4\text{)} * \text{leakage rate}$

No assumptions were made for fugitive emissions.

On-Road Vehicles Tab

Gasoline, Diesel, and Ethanol Vehicles

A combination of data inputs is used to calculate emissions from on-road transportation including vehicle miles traveled (VMT), vehicle distribution, and fuel efficiency. Fuel emission factors were provided by the U.S. Environmental Protection Agency (EPA).¹⁰ For CH₄ and N₂O emission factors,

⁸ Utilized the default value of 0.3% (three tenths of one percent) from EDF User Guide for Natural Gas Leakage Rate Modeling Tool. <https://www.edf.org/sites/default/files/US-Natural-Gas-Leakage-Model-User-Guide.pdf>

⁹ Energy Density for Wyoming:

https://www.eia.gov/dnav/ng/ng_cons_heat_a_epg0_vgth_btucf_a.htm.

¹⁰ See EPA estimates as recommended by ICLEI: https://www.epa.gov/system/files/documents/2023-03/ghg_emission_factors_hub.pdf

Lotus weighted the emissions factors by vehicle make year from 1991-2021 for each EPA vehicle distribution type. Final emission calculations for fossil fuel-based vehicles exclude electric vehicles, see section *Electric Vehicles* for more information.

VMT data came from Google Environmental Insights Explorer (Google EIE). Using 2021 vehicle type breakdown by percentage data from the EPA State Inventory Tool (SIT) Mobile Combustion Module for Wyoming, daily VMT was broken into different vehicle classes.¹¹ The classes are as follows:

- Light Duty Gas Vehicle
- Light Duty Diesel Vehicle
- Light Duty Gas Truck
- Light Duty Diesel Truck
- Heavy Duty Gas Truck
- Heavy Duty Diesel Truck
- Ethanol Light Duty Vehicle
- Ethanol Light Duty Truck
- Ethanol Heavy Duty Truck

To understand the fuel spread by vehicle type, Lotus obtained data on the number of registered vehicles of all fuel types within the county from Kris Gisse, Motor Vehicle Supervisor and Deputy Treasurer for Laramie County. The percentage of each fuel type was used in calculations for VMT of each fuel and vehicle type combination, for example light duty gas vehicles versus light duty diesel vehicles.

Vehicle efficiencies were also taken from EPA state inventory tool. See Table 5.

Table 3. Vehicle fuel efficiency by vehicle type.

Vehicle Type	Fuel Efficiency (miles per gallon)
Gas light duty vehicles	24.1
Gas light duty trucks	18.5

¹¹ For more information visit EPA's State Greenhouse Gas Inventory and Projection Tools -- Mobile Combustion Module at: <https://www.epa.gov/statelocalenergy/download-state-inventory-and-projection-tool>. Fuel economy for buses was assumed to be the same as diesel heavy duty trucks.

Diesel light duty vehicles	32.4
Diesel light duty trucks	22.1
Gas heavy duty trucks	7.1
Diesel heavy duty trucks	6.6
Gas motorcycles	50.0

Gallons of fuel consumed by fuel type was calculated using the following example formulas for gasoline light duty vehicles:

Gasoline Light Duty Vehicle VMT = Annual VMT for light duty vehicles * % light duty that are gasoline * (1 - 10% ethanol)

Gallons of gasoline consumed in light duty vehicles = Gasoline light duty vehicle annual VMT / fuel efficiency per vehicle type

These calculations should be repeated for each vehicle type.

CO₂ emissions from gasoline, diesel, and ethanol are calculated using gallons of fuel consumed. CH₄ and N₂O emissions are calculated using annual VMT. Total emissions can be calculated with the following formulas:

CO₂ emissions from on-road transportation = (gallons consumed * emission factor)/1,000 kg/mt

CH₄ and N₂O emissions from on-road transportation = (Annual VMT * emission factor)/1,000,000 g/mt * GWP

These calculations should be repeated for each fuel type across all vehicle types.

The following assumptions were made about gasoline, diesel, and ethanol vehicles:

- Lotus assumed all gasoline used contained 10 percent ethanol. Ethanol fuel content is based on data at the fuel pump, which states that typical gasoline contains 10 percent ethanol. It is assumed that regular diesel contains no biodiesel.

- Motorcycles and heavy duty buses were not provided in the vehicle registration data. They were included with light duty vehicles and heavy duty trucks, respectively.

Electric Vehicles

Average annual VMT per vehicle for electric vehicles within Cheyenne and the MSA were not available, so a 2019 survey from the California Energy Commission for residential driving patterns was used as a proxy to estimate all battery electric and plug-in hybrid electric vehicle VMT.

It was assumed that electricity used for electric vehicle charging stations was included in existing building meters. Therefore, to avoid double counting, electricity consumed by electric vehicles was subtracted from residential and commercial building electricity consumption totals. Transmission and distribution (T&D) losses from the electricity used to charge EVs were estimated and included as scope 3 emissions. See the *Stationary Energy Data Tab* section for the emissions calculation for T&D losses.

GHG emissions from electric vehicles = number of community vehicles that are EVs * community EV VMT * vehicle efficiency * electric emission factor/1,000 kWh/MWh

Repeat the calculation for each electric vehicle type.

The following assumptions were made about electric vehicles:

- Electric vehicle fuel efficiencies are from the U.S. Department of Energy: <https://afdc.energy.gov/vehicles/electric-emissions-sources>.
- Based on a report by the Idaho National Laboratory titled *Plugged In: How Americans Charge Their Electric Vehicles*, it is assumed that 85 percent of charging occurs at charging stations in residential buildings and 15 percent of charging occurs at stations in commercial buildings.¹²
- The proportion of U.S. hybrid, plug-in hybrid, and full electric vehicle sales comes from the U.S. Bureau of Transportation Statistics.

Transit Tab

Kim Serumgard, Transit Manager for the City of Cheyenne, provided VMT and fuel totals for Cheyenne's public transit buses. All buses use gasoline. The transit system briefly goes outside of the City boundary, so QGIS was used to

¹²

For more information see: <https://avt.inl.gov/sites/default/files/pdf/arra/PluggedInSummaryReport.pdf>.

find the length of transit routes that fall within the boundary; only routes within the boundary count towards the City's emissions.

CO₂ emissions from non-electric transit = gallons of fuel * emission factor / 1,000 kg/mt

CH₄ or N₂O emissions from non-electric transit = miles traveled * emission factor / 1,000,000 g/mt * GWP

No assumptions were made for transit.

Aviation Tab

Jet fuel and aviation gasoline data for the Cheyenne Regional Airport were provided by Kyle Malacina, Director of Properties and Compliance. The number of flights for each aircraft type was taken using information from the FAA's Air Traffic Activity System.¹³ Air taxi and civil aircraft were determined to mostly use aviation gas due to the smaller aircraft in these categories. Air carrier, general aviation, and military aircraft were determined to mostly use jet fuel due to the larger size of the aircraft. These determinations were made using information from the FAA.¹⁴

GHG emissions from aviation = (Aviation gasoline usage * emission factors * GWP) + (jet fuel gasoline usage * emission factors * GWP)

No assumptions were made for the Aviation tab.

Off-Road Data Tab

Off-road data was taken from the EPA National Emissions Inventory (NEI). The NEI categorizes off-highway vehicle emissions by fuel type: compressed natural gas, diesel, gasoline, and liquified petroleum gas. Quantities of these fuels are used to calculate GHG emissions and are categorized into the following use types as determined by the EPA: agricultural equipment, commercial equipment, construction equipment, industrial equipment, lawn & garden equipment, logging equipment, and recreational equipment.

Emissions are reported at the county level, so emissions were scaled to the City level using population. The categories of agricultural, logging, and recreational equipment do not pertain to the city and therefore are omitted. Emissions from pleasure craft were determined to be non-applicable to both jurisdictions given the lack of navigable waterways in the boundary. Off-road fuel usage data from the Cheyenne Regional Airport and the Board of Public

¹³ See: <https://aspm.faa.gov/opsnet/sys/Airport.asp>.

¹⁴ See: https://aspm.faa.gov/aspmhelp/index/OPSNET_Reports_Definitions_of_Variables.html.

Utilities' vehicles were both reported as information-only in this inventory to avoid double counting.

The NEI gives CO₂, CH₄, and N₂O quantities so all that needs to be done is convert these into CO₂ equivalents.

GHG emissions from off-road vehicles = $\text{mt CO}_2 + (\text{mt CH}_4 * \text{GWP}) + (\text{mt N}_2\text{O} * \text{GWP})$

No assumptions were made for the Off-Road tab.

Railways Data Tab

Emissions data from railway activity were taken from the EPA's 2020 National Emissions Inventory (NEI). The length of railways in the City and County were calculated in GIS and were used to scale the data to the appropriate boundaries. The EPA NEI reports emissions in US tons; these values were converted to metric tons in the inventory workbook.

The NEI gives CO₂, CH₄, and N₂O quantities so all that needs to be done is convert these into CO₂ equivalents.

GHG emissions from railway activity = $\text{mt CO}_2 + (\text{mt CH}_4 * \text{GWP}) + (\text{mt N}_2\text{O} * \text{GWP})$

No assumptions were made for the Railways tab.

Waste and Recycling Data Tab

Most of the waste from the City of Cheyenne is sent to the Cheyenne Landfill. Other rural residents use private haulers who transport waste to the RAM trucking transfer station, where it is sent to Waste Management's North Weld landfill in Ault, Colorado. The Cheyenne Landfill does not have landfill gas collection whereas the North Weld landfill does. Landfill tonnage data for the Cheyenne Landfill came from Samuele Quattrini, Cheyenne Landfill Manager. A fifty-fifty split of residential to commercial waste was used in the analysis.

RAM trucking was not able to provide tonnage data for the waste they transport to the North Weld landfill. Therefore, waste landfilled outside of the county was estimated by applying the percentage of households and commercial square footage in Cheyenne versus the MSA to the tonnage of waste reported within Cheyenne.

Compost is processed at the Cheyenne Landfill and data were provided by Samuele Quattrini. Tonnage data were not available, so it was assumed that each load was three cubic yards and an EPA volume-to weight ratio was applied. Electronics recycling data were also provided by Sam in cubic yards.

EPA's volume-to-weight conversion for computer-related electronics was used to convert cubic yards into pounds. The EPA's 2018 MSW recycling characterization was used to calculate avoided emissions from recycling.

Waste characterization estimates were taken from the Wyoming Solid Waste Diversion Study commissioned by the Wyoming Dept. of Environmental Quality in 2013.

GHG emissions from landfilled waste = (sum (tons of waste * percent of waste type * emission factor)) * (1 – oxidation factor) * (1 – landfill gas collection efficiency) * GWP

GHG emissions from composted waste = tons of compost * (mt CH₄/ton * GWP) * (mt N₂O/ton * GWP)

Avoided GHG emissions from recycling = tons of recycling for each recyclable * avoided emission factor

The following assumptions were made for the Waste and Recycling tab:

- Assumed a fifty-fifty split of residential and commercial waste.
- Assumed that each compost load was three cubic yards.

Wastewater Data Tab

There are wastewater treatment facilities, septic systems, and sewage lagoons within the inventory boundary. Data on wastewater treatment facilities came from Matt Buelow, Water Reclamation Division Manager for the Cheyenne Board of Public Utilities. The population served by these facilities includes the population of Cheyenne and the South Cheyenne Sewer and Water District. Dennis Lewis, the Southeast District Engineer for Wyoming DEQ Water Quality Division, provided data on the sewage lagoons in the area. Emissions were calculated for the three public lagoons (Pine Bluffs, Burns, and Albin) and a low industrial factor of one was used due to the small population size served by the lagoons and the lack of industrial practices in the area. Tiffany Gaertner at the Cheyenne Laramie County Public Health provided data on the number of septic tank permits in Laramie County.

Emissions from wastewater treatment plants occur from processes, gas combustion and flaring, and effluent discharge. The emissions factor during the process phase depends on whether the treatment method uses nitrification or denitrification. The wastewater treatment plant in Cheyenne uses nitrification and denitrification therefore their emission factor is 7 g N₂O/person/year. Effluent discharge emissions factors are the same independent of process type.

Wastewater emissions for the following sources were calculated using GPC's wastewater calculations:

- Combustion gas emissions.
- Flared gas emissions.
- Process N₂O emissions for WWTPs with nitrification and denitrification.
- Fugitive N₂O emissions from effluent discharge.
- Emissions from septic tanks.
- Emissions from sewage lagoons.

GHG emissions from combusted gas = daily digester gas produced * % methane content in digester gas * natural gas energy density * 365 * emission factor / 1,000,000 Btu/MMBtu / 1,000 kg/mt * GWP

GHG emissions from flared gas = daily digester gas flared * % methane content in digester gas * 662 g/m³ / 1,000,000 g/mt * 0.03 m³/ft³ * 365 * (1 - methane destruction efficiency) * GWP

GHG emissions from process emissions = population served * industrial commercial discharge multiplier * emission factor / 1,000,000 g/mt * GWP

GHG emissions from effluent discharge = total nitrogen discharged * days per year * molecular weight ratio of N₂O to N₂ * emission factor / 1,000 kg/mt * GWP

GHG emissions from septic tanks = ((number of septic tanks * average household size) * CH₄ production capacity * methane correction factor * days per year * amount of BOD₅ treated per day) / 1,000 kg/mt * GWP

No assumptions were made for the Wastewater Data tab.

Industrial Processes

The EPA Facility Level Information on Greenhouse gases Tool (FLIGHT) is used to collect data on industrial processes by county. Emissions from these activities are attributed to the MSA in the inventory and are not attributed to the City. Emissions from the Cheyenne Plant (from ammonia manufacturing and nitric acid production) are accounted for in the inventory. FLIGHT also collects data for stationary combustion and electricity generation facilities; however, those emissions are already accounted for in the Building Energy tab. Thus, data from FLIGHT on the emissions from Cheyenne Prairie Generating Station are included as information-only in the Building Energy Use tab to avoid double counting. The FLIGHT tool also reports landfill data, which is accounted for in the Waste and Recycling tab.

Refrigerants

Refrigerants are not tracked by the City of Cheyenne or the MSA, however many buildings do have air conditioning. Therefore, Lotus estimated emissions from refrigerant loss based on the percent of commercial square footage that is refrigerated and standard assumptions and emission factors for commercial refrigerant leakage. Laramie County Assessor's supplied the square footage data for the City and the MSA. The constants and emission factors used in the calculations are pulled from the IPCC.¹⁵

GHG Emissions from Refrigerant Loss = % of square footage with refrigeration * square footage / 300 sf/ton of cooling * amount of refrigerant/ton of cooling * 5% refrigerant loss / 1000 kg/metric ton * GWP

Agriculture, Forestry, and Other Land Uses (AFOLU)

Agriculture

To calculate the emissions from livestock and agricultural practices within the Cheyenne MSA, Lotus used census data from the United States Department of Agriculture (USDA)¹⁶ and the EPA SIT Agriculture Module.¹⁷

Several USDA census tables were used, including this Fertilizers and Chemicals Applied table in Figure 2. Lotus highlighted any data used from the tables in the GHG. The following tables are required:

- Table 11. Cattle and Calves
- Table 12. Hogs and Pigs
- Table 13. Sheep and Lambs
- Table 14. All Goats
- Table 18. Equine
- Table 19. Poultry
- Table 40. Fertilizers and Chemicals Applied

¹⁵ Source (IPCC): https://www.ipcc.ch/publications_and_data/ar4/wg1/en/ch2s2-10-2.html.

¹⁶ United States Department of Agriculture. Census of Agriculture. 2022 Census Volume 1, Chapter 2: County Level Data. Wyoming. https://www.nass.usda.gov/Publications/AgCensus/2022/Full_Report/Volume_1_Chapter_2_County_Level/Wyoming/

¹⁷ United States Environmental Protection Agency. State Inventory Tool. Agriculture Module. Version 2024.1. <https://www.epa.gov/statelocalenergy/download-state-inventory-and-projection-tool>

Table 40. Fertilizers and Chemicals Applied: 2022 and 2017

(For meaning of abbreviations and symbols, see introductory text.)

Item	Wyoming	Albany	Big Horn	Campbell	Carter	Comstock
Commercial fertilizer, acid, and all other fertilizers						
Items, 2002	3,864	5.1	270	38	173	48
2017	2,094	85	233	37	15	64
Items, 2002	127,396	18,781	58,186	8,441	29,508	12,887
2017	782,395	38,226	71,943	8,148	29,759	18,744

Item	Cook	Fishers	Goshute	Hot Springs	Johnson	Laramie
Commercial fertilizer, acid, and all other fertilizers						
Items, 2002	48	356	267	51	80	167
2017	48	425	277	52	47	204
Items, 2002	42,305	41,684	80,821	3,355	8,091	10,611
2017	78,502	15,478	77,057	3,730	8,530	18,744
Disposal of fertilizer, except organic products	25	275	244	43	62	118

Figure 2. USDA Census data on fertilized acres in Wyoming and Laramie County.

Calculations for urea fertilization and synthetic and organic fertilizer use come from the EPA SIT Agriculture Module. A copy of this spreadsheet is saved in the Cheyenne folder; however, the latest version should be downloaded with each GHG inventory update in case of emission updates. Figure 3 is a screenshot of the opening tab within EPA SIT. Make sure that all content is enabled to be able to click within the spreadsheet. Select Wyoming as the state and click the “Check/Uncheck All” button for each section category within that tab. This is a large spreadsheet so sometimes you need to do a slow double click for the program to register the action.

State Inventory Tool - Carbon Dioxide, Methane, and Nitrous Oxide Emissions from Agriculture Module
Version 2024.1

This module provides default activity data, emission factors, and other information for use in the State Inventory Tool. It is designed to be used in conjunction with the State Inventory Tool. It is designed to be used in conjunction with the State Inventory Tool. It is designed to be used in conjunction with the State Inventory Tool.

1. Choose a State: Wyoming

Make sure Wyoming is

Steps 2 - 6. Fill in the emission factors that are used throughout the module and then proceed to the corresponding sector worksheet to input activity data. (Either type in the values or click the default box. The default GWP's used in this module are from the IPCC Fifth Assessment Report (AR5) but can be adjusted in the Controls tab to reflect other sources.)

2. Enter Emission Factors

Consult User's Guide

Reset All

Animal Group	Default Factor (kg/animal/yr)	Factor Used	Use Default? (Check for Yes)
Dairy Cattle			
Dairy Cows	varies by year	varies by year	☒
Dairy Replacement Heifers	varies by year	varies by year	☒
Replacements 0-12 mos.	varies by year	varies by year	☒
Replacements 12-24 mos.	varies by year	varies by year	☒
Beef Cattle			

Check | Uncheck All

Check all to use defaults for each category in the Control

Figure 3. Opening page of EPA SIT.

To understand the CH₄ and N₂O emissions from manure management, first go to the “CH₄ from Manure Management” tab. Scroll down to the year being considered in the GHG inventory and enter in the number of animals from the USDA Census for each animal group available. See Figure 4. Note that the number should be entered in the thousands of head so, for example, 48 chickens would be entered as 0.048. Take the values from the metric tons of CH₄ column and enter these into the blue cells in the GHG inventory under Manure Management heading. Extracting the N₂O emissions from manure management is very similar to that of CH₄. Go to the “N₂O from Manure Management” tab and scroll down to the correct inventory year. The number

of animals should transfer over from the CH₄ tab, so just take the kg of N₂O emissions and enter those into the blue cells in the GHG workbook. See Figure 5. The equation in the workbook will convert these emissions to metric tons.

CH ₄ from Manure Management											
2023											
	Number of Animals (1000 head)	Typical Animal Mass (TAR) (kg)	Excreta Solids (VS) (kg VS/head/year)	Total VS (kg/yr)	Manure CH ₄ Emissions (kg CH ₄ /kg VS)	Weighted MCP	Emissions (kg ² CH ₄)	Emissions (kg ² CH ₄)	Emissions (kg ² CH ₄)	Emissions (kg ² CH ₄)	Emissions (kg ² CH ₄)
Cows											
Dairy Cows	100	600	2,900.0	290,000	0.25	0.00	72,500	72,500	0.000	0.000	0.000
Dairy Replacement Heifers	100	400	1,200.0	120,000	0.11	0.00	13,200	13,200	0.000	0.000	0.000
Beef Cattle											
Feedlot Heifers	100	400	1,200.0	120,000	0.25	0.00	30,000	30,000	0.000	0.000	0.000
Feedlot Steers	100	400	1,200.0	120,000	0.25	0.00	30,000	30,000	0.000	0.000	0.000
Beef Cows	100	600	2,900.0	290,000	0.25	0.00	72,500	72,500	0.000	0.000	0.000
Beef Replacement Heifers	100	400	1,200.0	120,000	0.11	0.00	13,200	13,200	0.000	0.000	0.000
Other Heifers	100	400	1,200.0	120,000	0.25	0.00	30,000	30,000	0.000	0.000	0.000
Other Steers	100	400	1,200.0	120,000	0.25	0.00	30,000	30,000	0.000	0.000	0.000
Swine											
Brooding Sows	100	250	1,200.0	120,000	0.25	0.00	30,000	30,000	0.000	0.000	0.000
Market Hogs 60-90 lbs	100	80	1,200.0	120,000	0.25	0.00	30,000	30,000	0.000	0.000	0.000
Market Hogs 90-120 lbs	100	120	1,200.0	120,000	0.25	0.00	30,000	30,000	0.000	0.000	0.000
Market Hogs 120-150 lbs	100	150	1,200.0	120,000	0.25	0.00	30,000	30,000	0.000	0.000	0.000
Market Hogs 150-180 lbs	100	180	1,200.0	120,000	0.25	0.00	30,000	30,000	0.000	0.000	0.000
Poultry											
Layers	100	2	10.0	2,000	0.25	0.00	500	500	0.000	0.000	0.000
Broilers	100	2	10.0	2,000	0.25	0.00	500	500	0.000	0.000	0.000
Turkeys	100	2	10.0	2,000	0.25	0.00	500	500	0.000	0.000	0.000
Other											
Sheep on Feed	100	25	1,200.0	120,000	0.25	0.00	30,000	30,000	0.000	0.000	0.000
Sheep Not on Feed	100	25	1,200.0	120,000	0.25	0.00	30,000	30,000	0.000	0.000	0.000
Goats	100	25	1,200.0	120,000	0.25	0.00	30,000	30,000	0.000	0.000	0.000
Other	100	25	1,200.0	120,000	0.25	0.00	30,000	30,000	0.000	0.000	0.000
TOTAL							224,900	224,900	0.000	0.000	0.000

Figure 4. CH₄ Emissions from Manure Management calculations in EPA SIT.

N ₂ O from Manure Management											
2023											
	Number of Animals (1000 head)	Total Nitrogen Excreted (kg)	Unassimilated N from Manure in Excreta (kg)	Unassimilated N from Manure in Solid Storage, Slurry, & Other Systems (kg)	Emissions from Manure in Excreta (kg N ₂ O/kg N)	Emissions from Manure in Solid Storage, Slurry, & Other Systems (kg N ₂ O/kg N)	Total N ₂ O Emissions (kg N ₂ O)	Emissions (kg N ₂ O)	Emissions (kg N ₂ O)	Emissions (kg N ₂ O)	Emissions (kg N ₂ O)
Cows											
Dairy Cows	100	2,400	1,700	400	1	0	1,700	1,700	0.000	0.000	0.000
Dairy Replacement Heifers	100	1,200	800	200	1	0	800	800	0.000	0.000	0.000
Beef Cattle											
Feedlot Heifers	100	1,200	800	200	1	0	800	800	0.000	0.000	0.000
Feedlot Steers	100	1,200	800	200	1	0	800	800	0.000	0.000	0.000
Swine											
Brooding Sows	100	1,200	800	200	1	0	800	800	0.000	0.000	0.000
Market Hogs 60-90 lbs	100	1,200	800	200	1	0	800	800	0.000	0.000	0.000
Market Hogs 90-120 lbs	100	1,200	800	200	1	0	800	800	0.000	0.000	0.000
Market Hogs 120-150 lbs	100	1,200	800	200	1	0	800	800	0.000	0.000	0.000
Market Hogs 150-180 lbs	100	1,200	800	200	1	0	800	800	0.000	0.000	0.000
Poultry											
Layers	100	2	10	2	1	0	10	10	0.000	0.000	0.000
Broilers	100	2	10	2	1	0	10	10	0.000	0.000	0.000
Turkeys	100	2	10	2	1	0	10	10	0.000	0.000	0.000
Other											
Sheep on Feed	100	25	10	2	1	0	10	10	0.000	0.000	0.000
Sheep Not on Feed	100	25	10	2	1	0	10	10	0.000	0.000	0.000
Goats	100	25	10	2	1	0	10	10	0.000	0.000	0.000
Other	100	25	10	2	1	0	10	10	0.000	0.000	0.000
TOTAL		24,000	17,000	4,000	1	0	17,000	17,000	0.000	0.000	0.000

Figure 5. N₂O Emissions from Manure Management calculations in EPA SIT.

Then navigate to the Urea Fertilization tab. See Figure 6. Again, click the “Check All” button at the top to activate the default values. The orange cells should populate with a metric tons value. Use the most recent year’s value and enter into the GHG inventory. The EPA SIT value is for all of Wyoming so the equation in the GHG inventory takes the proportional amount for Laramie County using the values for fertilizer applied found in the USDA Census.

7. CO₂ from Urea Fertilization in Wyoming

Click here to find possible data sources.

The use of urea as a fertilizer results in CO₂ emissions that were previously found during the industrial production process. The amount of urea applied to soil is multiplied by the carbon emission factor, and then converted to million metric tons carbon dioxide equivalent. For more information, please refer to the Agriculture Chapter of the User's Guide.

Return to Control Sheet

Check All

Clear All Data

Year	Total Urea Applied to Soil (Metric Tons)	Emission Factor (Tee C/Tee urea)	Carbon Dioxide Emissions (MTCO ₂ e)	Carbon Dioxide Emissions (MMTCO ₂ e)	
1990	6,276	0.20	1,255	0.005	☒ Default Activity Data?
1991	7,786	0.20	1,557	0.006	☒ Default Activity Data?
1992	1,062	0.20	212	0.001	☒ Default Activity Data?
2020	147,339	0.20	29,468	0.128	☒ Default Activity Data?
2021	146,915	0.20	29,383	0.128	☒ Default Activity Data?
2022		0.20	0	0.000	☒ Default Activity Data?
2023		0.20	0	0.000	☒ Default Activity Data?
2024		0.20	0	0.000	☒ Default Activity Data?
2025		0.20	0	0.000	☒ Default Activity Data?

Urea Fertilization

Figure 6. EPA SIT calculations for urea fertilization in Wyoming.

Next navigate to the Ag Soils-Plant-Fertilizer tab. Click the “Check All Boxes” button to use the default values and find the most recent year with data. Enter the circled fields from Figure 7 into the GHG inventory. Use the metric tons of N₂O emissions instead of the MMTCO₂e so that the most up to date global warming potentials can be applied in the GHG workbook. The direct and indirect N₂O emissions proportioned of Laramie County from Wyoming using the same USDA census information as the urea fertilizer.

Lotus selected the most recent time period for the analysis. The LEARN tool analyzes land cover changes over 2-3-year time periods based on available land cover data layer availability. See screenshot below for available years for analysis.

The screenshot shows the 'Inventory Period' configuration window of the LEARN tool. On the left is a sidebar with icons for 'MAP LAYERS', 'SELECT AREA', and other functions. The main panel has a title bar with a close button (X). Below the title, instructions state: 'Move the sliders to select the start and end years that define your inventory analysis period.' There are three data layers with sliders:

- Forest and Land Cover Change:** 30m land cover change (NLCD, national coverage). The slider shows years from 2000 to 2021, with the end year set to 2021.
- Trees Outside Forests*:** 30m tree canopy (NLCD, national coverage). The slider shows years from 2001 to 2021, with the end year set to 2021.
- Mature and Old-growth (MOG) forests:** Identify and map MOG forests and protection status, and calculate carbon stocks. This layer has a toggle switch that is currently turned on.

A green 'NEXT' button is located at the bottom of the main panel.

After the time period was chosen, Lotus selected a comparable municipality to analyze the sequestration potential of Cheyenne's urban tree canopy. Casper, Wyoming was selected due to its proximity geographically and similar climate to Cheyenne. See the screenshot below.

Lotus selected the same time period for the City analysis as seen in the screenshot below.

Inventory Period

Move the sliders to select the start and end years that define your inventory analysis period.

Forest and Land Cover Change
30m land cover change (NLCD, national coverage)

2000 2004 2006 2008 2011 2013 2015 2019 2021

Trees Outside Forests*
30m tree canopy (NLCD, national coverage)

2011 2013 2015 2019 2021

Mature and Old-growth (MOG) forests

Identify and map MOG forests and protection status, and calculate carbon stocks.

NEXT

Lotus chose Casper, Wyoming as a comparable municipality again in the next step. See the screenshot below.

Current Forest Region

Rocky Mountain South

Select a City for Carbon Calculation:

Use this tool to find a city with a similar environment to yours to help estimate the carbon impact of trees outside forests. By choosing a city that has a climate and location similar to your area, you'll get more accurate results on how much carbon trees absorb or release.

The tool uses standard data for U.S. urban areas, but selecting a nearby city with a similar climate can give more precise information for your community.

Select a Similar Location

Casper, WY

NEXT

Category	Carbon Stock (tC/ha)	Carbon Stock (tC/ha)
Disturbed Forest	-37	
Fire	0	
Forest/Grassland	0	
Harvest/Clear	0	
Trees outside of forests	1,091	
Forest to Grassland	0	
Forest to Cropland	0	
Forest to Wetland	0	
Forest to Settlement	0	
Forest to Other	0	
Harvested Wood Products	0	
Trees outside of forests	1,521	
Land lost to forest	0	
TOTAL	1,584	2,791
Net C/S balance	1,217	

View Full Report **Download Data**

In the downloaded spreadsheet, Lotus copy and pasted the third section of data that contained the emissions/removals and GHG Flux into the Cheyenne GHG inventory workbook.

Process Improvement Plan

Every inventory sheds light on possible improvements for the next iteration. The Cheyenne team did an excellent job providing data, helping track down difficult datasets, and brainstorming data assumptions. Future inventories could benefit from the following process improvements:

- Check in the future to see if Pine Bluffs Electric Utility is able to provide their energy usage data.
- Consider finding a way to track residential wood usage so that estimates do not need to be used for emission calculations.
- See if electric vehicle data are able to be tracked in the future; this is important in accurately estimating emissions.
- Tracking down tonnage data for RAM trucking would improve the accuracy of waste emission estimates. Additionally, finding tonnage data for compost and recycling would improve those emission estimates.

Conclusion

The 2023 GHG emissions inventory provides the City of Cheyenne and the Cheyenne MSA with a robust and defensible update of carbon emissions with which to can be used to inform future policies and programs. Cheyenne can effectively monitor and track progress against carbon reduction goals and will be better informed when choosing specific sustainability efforts. Greenhouse gas emission inventories should be updated at regular intervals to capture changes over time, and the IMP contained herein, alongside the GHG emissions inventory workbook, will enable City staff to update GHG emissions in the future.

Appendix A: Emission Factors

See Table 4 for an overview of the emission factors that were used for calculations throughout the inventory. The Notes column provides details as to which emission factors need updated regularly.

Table 4. Summary of Emission Factors.

Stationary Energy Emission Factors—Electricity & Natural Gas					
Emission Source	GHG	Value	Unit	Source	Notes
Electricity	CO ₂	Black Hills: 0.690	mt CO ₂ /MWh	Email from Carmalee Rose from Black Hills Energy. See spreadsheet labeled "BHE Response to 2023 City of Cheyenne and Laramie County Usage Request" on file.	Verify with each inventory. Likely that the CO ₂ emission factor will change annually.
		Tri-State: 0.649	mt CO ₂ /MWh	See "Issue Brief Tri-State Carbon Rate_May 2024" on file.	
	CH ₄	0.00005	mt CH ₄ /MWh	EPA's eGrid: eGRID 2024 summary tables, table 6, sub region RMPA. https://www.epa.gov/system/files/documents/2024-02/ghg-emission-factors-hub-2024.pdf	Verify with each inventory.
	N ₂ O	0.000006	mt N ₂ O/MWh		
Natural Gas	CO ₂	0.005306	mt CO ₂ /therm	2013 ICLEI US Community Protocol, Appendix C	

	CH ₄	0.000001	mt CH ₄ /the rm	
	N ₂ O	0.00000 01	mt N ₂ O/the rm	
Stationary Diesel	CO ₂	0.01	mt CO ₂ /gall on	2024 EPA Emission Factors for Greenhouse Gas Inventories. <a href="https://www.epa.gov/system/files/documen
ts/2024-02/ghg-emission-factors-hub-
2024.pdf">https://www.epa.gov/system/files/documen ts/2024-02/ghg-emission-factors-hub- 2024.pdf . Assumes distillate fuel oil number 2 and that diesel is primarily used in generators by the industrial sector.
	CH ₄	0.00000 04	mt CH ₄ /gal lon	
	N ₂ O	0.00000 01	mt N ₂ O/gal lon	
Propane	CO ₂	0.01	mt CO ₂ /gall on	2024 EPA Emission Factors for Greenhouse Gas Inventories. <a href="https://www.epa.gov/system/files/documen
ts/2024-02/ghg-emission-factors-hub-
2024.pdf">https://www.epa.gov/system/files/documen ts/2024-02/ghg-emission-factors-hub- 2024.pdf
	CH ₄	0.00000 03	mt CH ₄ /gal lon	
	N ₂ O	0.00000 01	mt N ₂ O/gal lon	

Wood	Biogen ic CO ₂	93.8	kg CO ₂ b/M MBTU	2024 EPA Emission Factors for Greenhouse Gas Inventories. <a href="https://www.epa.gov/system/files/documen
ts/2024-02/ghg-emission-factors-hub-
2024.pdf">https://www.epa.gov/system/files/documen ts/2024-02/ghg-emission-factors-hub- 2024.pdf	Verify with each inventory
	CH ₄	7.2	g CH ₄ /M MBTU		
	N ₂ O	3.6	g N ₂ O /MMBT U		
	Averag e wood consu mptio n per house hold	69.6	mmBtu / househ old	Average btu used per household in west for wood as a primary heat found here: <a href="https://www.eia.gov/consumption/residenti
al/data/2015/c&e/pdf/ce7.2.pdf">https://www.eia.gov/consumption/residenti al/data/2015/c&e/pdf/ce7.2.pdf	
Transportation Emission Factors—Ethanol, Gasoline, and Diesel					
Emission Source	GHG	Value	Unit	Source	Notes
Gasoline	CO ₂	8.78	kg CO ₂ /gal	EPA estimates (<a href="https://www.epa.gov/system/files/docume
nts/2023-
03/ghg_emission_factors_hub.pdf">https://www.epa.gov/system/files/docume nts/2023- 03/ghg_emission_factors_hub.pdf).	Should remain constant but verify with each inventory.
	CH ₄		g/mile		

	N ₂ O	Varies by vehicle		Lotus analysis using EPA emission factors by vehicle make year and vehicle distributions.	
Diesel	CO ₂	10.21	kg CO ₂ /gal	EPA estimates (https://www.epa.gov/system/files/documents/2023-03/ghg_emission_factors_hub.pdf).	
	CH ₄	Varies by vehicle	g/mile	Lotus analysis using EPA emission factors by vehicle make year and vehicle distributions.	
	N ₂ O				
Ethanol	CO ₂	5.75	kg CO ₂ /gal	EPA estimates (https://www.epa.gov/system/files/documents/2023-03/ghg_emission_factors_hub.pdf).	
	CH ₄	Varies by vehicle	g/mile	Lotus analysis using EPA emission factors by vehicle make year and vehicle distributions.	
	N ₂ O				
Transit Emission Factors					
Emission Source	GHG	Value	Unit	Source	Notes
Gasoline	CO ₂	8.78	kg CO ₂ /gal	EPA Emission Factors for GHG Inventories: https://www.epa.gov/system/files/documents/2024-02/ghg-emission-factors-hub-2024.pdf .	Should remain constant but verify with each inventory.

	CH ₄	0.044	g CH ₄ /mile	Lotus analysis using EPA emission factors by vehicle make year and vehicle distributions.	
	N ₂ O	0.036	g N ₂ O/mile		
Aviation Emission Factors					
Emission Source	GHG	Value	Unit	Source	Notes
Jet fuel	CO ₂	9.75	kg CO2/gal	EPA Emission Factors for GHG Inventories: https://www.epa.gov/system/files/documents/2024-02/ghg-emission-factors-hub-2024.pdf .	Should remain constant but verify with each inventory.
	CH ₄	0	g CH4/gal		
	N ₂ O	0.3	g N2O/gal		
Aviation Gasoline	CO ₂	8.31	kg CO2/gal	EPA Emission Factors for GHG Inventories: https://www.epa.gov/system/files/documents/2024-02/ghg-emission-factors-hub-2024.pdf .	
	CH ₄	7.06	g CH4/gal		

	N ₂ O	0.11	g N2O/gal		
Off-Road Emission Factors					
Emission Source	GHG	Value	Unit	Source	Notes
Gasoline	CO ₂	8.78	kg CO2/gal	EPA Emission Factors for GHG Inventories: https://www.epa.gov/system/files/documents/2024-02/ghg-emission-factors-hub-2024.pdf .	Should remain constant but verify with each inventory
	CH ₄	Varies by equipment type	g/gallon	Since it is unknown whether the equipment used is 2- or 4-stroke, the average emissions factor was used for both the Industrial/Commercial Equipment category and the Recreational Equipment category.	
	N ₂ O				
Diesel	CO ₂	10.21	kg CO2/gal	EPA Emission Factors for GHG Inventories: https://www.epa.gov/system/files/documents/2024-02/ghg-emission-factors-hub-2024.pdf .	
	CH ₄	Varies by equipment type	g/gallon		
	N ₂ O				
Waste Emission Factors					
Emission Source	GHG	Value	Unit	Source	Notes

Municipal Solid Waste	CH ₄	Varies	N/A	EPA FLIGHT (subpart HH) for the Cheyenne Landfill Documentation for Greenhouse Gas Emissions and Energy Factors Used in the Waste Reduction Model (WARM)_See Exhibit 6-7: CH4 Yield for Solid Waste Components. https://www.epa.gov/system/files/documents/2024-01/warm_management_practices_v16_dec.pdf .	Verify with each inventory.
Composted Waste	CH ₄	0.0005	mt CH ₄ /ton waste	Documentation for Greenhouse Gas Emissions and Energy Factors Used in the Waste Reduction Model (WARM): https://www.epa.gov/system/files/documents/2024-01/warm_management_practices_v16_dec.pdf . Exhibit 4-3 and Exhibit 4-4. Emission factors are adjusted to from CO2e to CH4 and N2O using AR4 values that EPA uses in their reports.	
	N ₂ O	0.00022	mt N ₂ O/ton waste		
Wastewater	CH ₄	Varies by treatment	Varies	ICLEI's U.S. Community Protocol for Accounting and Reporting of Greenhouse Gas Emissions (Community Protocol) –	

Treatment Facilities	N ₂ O	Varies by treatment	Varies	Appendix F: Wastewater and Water Emission Activities and Sources, Version 1.1, July 2013: http://icleiusa.org/ghg-protocols/ .	
Septic Tanks	CH ₄	0.6	CH ₄	ICLEI's U.S. Community Protocol for Accounting and Reporting of Greenhouse Gas Emissions (Community Protocol) – Appendix F: Wastewater and Water Emission Activities and Sources, Version 1.1, July 2013: http://icleiusa.org/ghg-protocols/ .	

Appendix E: BAU Modeling Methodologies



In This Appendix

Introduction

Business-as-usual Methodology

Strategy Impact Methodology



Cheyenne Greenhouse Gas Modeling Methodology

Introduction

This document details modeling assumptions for calculating greenhouse gases (GHGs) for Cheyenne's 2026 Comprehensive Climate Action Plan (CCAP) impact model.

This methodology document will detail modeling assumptions for the following scenarios:

- **Business-as-usual (BAU):** The BAU is the modeled baseline for expected emissions out to 2050. This model incorporates existing policies, population growth, and economic growth and their estimated impact on future emissions.
- **Strategy impact:** The strategy impact is the modeled emissions reductions associated with a suite of strategies out to 2050. The suite of strategies modeled were collaboratively chosen from the CCAP mitigation strategies.

Protocols

This analysis uses the **Greenhouse Gas Protocol Policy and Action Standard**. This protocol provides guidance for:

- Determining what policies or drivers should be included in the baseline state.
- Identifying impacted activities from modeled actions.
- Determining the boundaries of modeled actions.

Business-as-usual Methodology

A BAU model was developed to forecast baseline GHG emissions in the Cheyenne MSA region from 2023 through 2050. The BAU models a forecasted baseline for GHG emissions and activity data assuming operations continue as they exist today with no additional implemented policies and no technological breakthroughs achieved. All emissions factors, outside of electric grid emissions, are assumed to remain constant over time. The results from the BAU model are used to compare GHG emission impacts from the proposed strategies in the CCAP.

For each sector, GHG activity and emissions data were collected, calculated, and forecasted based on the 2023 CCAP GHG Inventory for the Cheyenne MSA region.

Forecast parameters

Key forecast parameters used to forecast activity data included population growth and non-residential square footage. Population growth is utilized to forecast residential activity data while non-residential square footage is utilized to forecast commercial and industrial activity data, as relevant.

Key data sources

- Population growth is obtained from the [Wyoming Department of Administration and Information's Economic Analysis Division](#).
- Non-residential square footage growth from the [Cheyenne MPO Transportation Impact Fee Study Draft Report](#).

Sector-based forecast data

Energy and buildings

Residential electricity, natural gas, and propane activity data are forecasted based on population growth. Commercial and industrial electricity, natural gas, and propane activity data are forecasted based on non-residential square footage. Stationary diesel and wood consumption are assumed to remain constant over time due to uncertainty in how consumption will change in the future.

It is assumed that all emission factors remain constant over time other than the electricity emissions factor. Utility-specific emissions reduction goals are taken into consideration to adjust the emissions factor applied to activity

data. The Cheyenne MSA is serviced by Tri-State, which aims to reduce emissions by 26% in 2025, 36% in 2026, 46% in 2027, and 80% by 2030 compared to a 2005 baseline, and Black Hills Energy, which aims to reduce emissions by 40% by 2030 and 70% by 2040.

Key data sources

- [Tri-State's website](#) is utilized to obtain the utility's emissions reduction goals. Issue briefs from Tri-State are utilized to understand current and past emissions factors for carbon dioxide emissions.
- Lotus modified the Tri-State emission factor to include methane and nitrous oxide emissions using emissions factors from the [EPA's eGrid summary tables](#).
- [Black Hills Energy 2024 Sustainability Report](#) is utilized to obtain the utility's emissions reduction goals along with current and past emissions factors.

Transportation

Vehicle miles traveled (VMT) from on-road transportation is forecasted based on population growth. Activity and emissions from public transit, aviation, railways, and off-road equipment are assumed to remain constant over time due to uncertainty in how these sources will change over time.

The forecast assumes a baseline rate of EV adoption in passenger vehicles over time, displacing the use of gasoline and diesel vehicles. Fuel efficiencies (i.e., miles per gallon or kWh per mile) are assumed to improve over time with technological improvements.

Key data sources

- [RMI's Energy Policy Simulator for Wyoming](#) is utilized to estimate passenger EV uptake over time.
- Internal combustion engine fuel efficiency year-over-year improvements are estimated from the [Energy Information Administration's \(EIA\) Annual Energy Outlook](#).
- The [National Laboratory of the Rockies Electrification Futures Study](#) is utilized to project EV efficiencies over time.

Other sectors

Residential municipal solid waste, compost, and wastewater are projected based on population. Commercial municipal solid waste, and commercial

refrigerants are projected based on non-residential square footage. Emissions and activity from the Cheyenne Plant, agriculture, and forestry are estimated to remain constant due to uncertainty in how emissions will change over time.

Strategy Impact

Methodology

The strategy impact scenario models emissions reductions through 2050 from mitigation strategies in the CCAP on top of the BAU results. The model was built to be flexible and based on user inputs including a target, a year to start implementation, and a year to reach the target.

Transportation

T1: Reduce cars on the road while ensuring it is safe to bike and walk.

This strategy is modeled using a user-specified percent of vehicle trips that are two miles or less that are targeted to be shifted to walking and/or bicycle trips. It is assumed that only light-duty vehicle trips may be shifted so medium- and heavy-duty trips are excluded.

The user specifies the target share of trips under two miles that are shifted to active transportation modes. Additionally, the user enters the start year and target year for when that target is reached. The target is linearly interpolated from the start year to the target year. Avoided light-duty VMT is then calculated by multiplying each year's target share of trips to shift by the total VMT from trips two miles or less.

Avoided VMT is then distributed by vehicle and fuel type. The alternative fuel strategy (T2) is accounted for in this distribution to prevent double counting. The share of vehicle and fuel types after implementation of that strategy is calculated. Fuel and electricity use avoided due to mode shifting is then calculated from avoided VMT. Emissions reductions are then calculated from avoided VMT and fuel use.

Key data sources

- The 2022 National Household Travel Survey (NHTS) data is utilized to determine the percentage of trips that are 2 miles or less, the average length of those trips, and total vehicle trips in 2025. This data is filtered to the Mountain MSA with less than 1,000,000 people.

T2: Promote the adoption of alternative fuel vehicles.

This strategy is modeled using a user-specified target share of vehicles on the road that will be electric vehicles (EVs) or plug-in hybrid vehicles (PHEVs). The model excludes medium- and heavy-duty vehicles and only focuses on light-duty vehicle electrification.

Some vehicle electrification is included in the BAU scenario. This strategy only accounts for impacts on top of any electrification in the BAU. The model linearly interpolates the user-entered target share of vehicles to be EVs and PHEVs between the entered start and target years. BAU electrification is subtracted from each year's corresponding share to avoid double counting.

Avoided internal combustion engine vehicle VMT is calculated using the additional target electrification after removing BAU adoption. This VMT is distributed across light-duty vehicle types using the vehicle type share from the GHG inventory. Avoided fuel usage and additional electricity use are estimated by applying fuel economy forecasts to corresponding vehicle types. Emissions reductions are then calculated by comparing avoided fossil fuel vehicle emissions to additional EV and PHEV emissions.

Key data sources

- Federal Highway Administration data for average annual VMT by highway category and vehicle type is utilized to determine the average VMT traveled by passenger vehicles.
- California Air Pollution Control Officers Association (CAPCOA) assumptions are utilized to determine the average electric versus gas fuel share in a PHEV.

T3: Increase connectivity across the county and across the region into Colorado and Nebraska.

This strategy is modeled using a user-specific target shift in commuting habits from passenger vehicle to bus transit. This strategy includes commuters from Colorado and excludes commuters from Nebraska as Nebraska commuters were unknown or considered to be de minimis. A max percentage of modal shift is utilized to avoid overestimating potential impacts. This value is 23% per the CAPCOA Handbook for Western States.

Trips attributable to the Cheyenne MSA from commuting trips starting and ending in Colorado are calculated by multiplying total light-duty vehicle trips by the following values:

- The share of vehicle trips that are commuting.
- The share of Laramie county workers who live in Colorado.
- The share of Laramie county residents who work in Colorado.
- The share of commute trips over 40 miles with a personal vehicle.
- The average share of workdays where commuting occurs.

VMT from these trips are calculated by multiplying impacted trips by the target share of trips to reduce, the two-way trip distance between Cheyenne and Fort Collins, and 50% (the share of cross-boundary trips that can be attributed to Cheyenne).

Avoided VMT is distributed to light-duty vehicle and fuel types accounting for EV and PHEV adoption from strategy T2. Additional transit VMT is estimated by dividing avoided VMT by a bus load factor to estimate the difference in trips between single-occupancy vehicle trips and bus trips. Additional diesel fuel use is then estimated from this VMT. Increased emissions from additional transit trips are netted against reduced emissions from reduced passenger vehicle travel to determine emissions reduction.

Key data sources

- The 2022 NHTS data is used to determine the share of trips that are from commuting, the share of commute trips over 40 miles with a personal vehicle, the average share of workdays commuting occurs, and the number of 2025 vehicle trips. This data is filtered to the Mountain MSA with less than 1,000,000 people.
- The 2018 Wyoming Interstate and Intercounty Commuting Report is used to determine the percent of Laramie County workers live in Colorado and the percent of Laramie County residents who work in Colorado.
- The Greenhouse Gas Protocol for Community-Scale Greenhouse Gas Inventories is utilized to determine the VMT attributable to the Cheyenne MSA from commuting trips starting or ending in Colorado.
- Google maps is utilized to determine number of miles between Fort Collins, Colorado and Cheyenne, Wyoming.
- The CAPCOA handbook for Western States is utilized to set a cap for modal shift percent from transit measures to avoid overestimating possible impact.

T4: Shift commuter patterns from single-occupancy vehicles to transit.

This strategy is modeled by determining the total passenger vehicle person-miles traveled based on a vehicle load factor and shifting a user-input target share of those person-miles to transit. This strategy is focused on local commuting patterns and excludes medium- and heavy-duty vehicle travel. A max percentage of modal shift is utilized to avoid overestimating potential impacts. While this strategy is focused on commuter patterns, the model does not isolate commuter trips for this strategy as any improvements in transit are assumed to impact other trip types. The same avoided fuel use, additional transit activity, and emissions calculations approach as described for T3 is used for this strategy.

Key data sources

- ICLEI ClearPath 2.0's Transition Elements Framework is utilized to determine the average passenger vehicle and transit bus load factors.
- The CAPCOA handbook for Western States is utilized to set a cap for modal shift percent from transit measures to avoid overestimating possible impact.

Energy and buildings

EB1: Increase energy efficiency and electrification in residential buildings.

This strategy is modeled in three tiers:

- a) Building code improvements.
- b) Electrification retrofits in existing homes.
- c) Energy efficiency improvements in existing residential homes.

Building code improvements

Building code improvements are modeled with a user-specified percent energy efficiency improvement in new homes as a result of energy code updates. Code-compliant new residential square footage is isolated from existing to determine emissions savings from reduced electricity and natural gas consumption using the 2018 International Energy Conservation Code (IECC) as the baseline.

Key data sources

- Baseline energy consumption for new homes if they were built under the 2018 baseline is estimated using an energy intensity factor from the [Department of Energy's Energy Savings Analysis: 2018 IECC for Residential Buildings](#).
- The [National Laboratory of the Rockies' \(NLR\) ResStock tool](#) is used to determine:
 - The average square footage of a housing unit in Laramie County, WY.
 - The share of residential energy use that is electricity and natural gas.
- The average code compliance percentage is obtained from [a report published by the Institute for Market Transformation](#).

Electrification retrofits in existing homes

Electrification retrofits are modeled with a user-specified heat pump adoption rate in existing homes not impacted by updated code. The model uses a stock approach to isolate existing homes that are replacing equipment then applies the heat pump adoption percentage to these homes. Increased electricity emissions from heat pump adoption are netted against decreased natural gas emissions to determine emissions savings from this strategy.

Key data sources

- Equipment turnover rate is estimated from [RMI's Energy Policy Simulator](#).
- The [Residential Energy Consumption Survey \(RECS\)](#) from the EIA is used to:
 - Estimate average annual fuel consumption for space and water heating by fuel type.
 - Average fuel splits by end use in residential homes.
- Equipment efficiencies to determine additional electricity consumption from heat pump adoption are obtained from:
 - [NLR's Building Energy Optimization \(BEopt\) tool](#).
 - [EIA minimum efficiency requirements](#).
 - [ENERGY STAR criteria](#).

Energy efficiency improvements in existing homes

Energy efficiency in existing homes is modeled with a user-specified increase in efficiency. This increase in energy efficiency is applied to baseline electricity

and natural gas consumption to determine decreased emissions from efficiency improvements.

Key data sources

No new data sources are used in this strategy. See the BAU section to see how baseline energy use for existing homes is calculated.

EB2: Increase energy efficiency and electrification in commercial and industrial buildings.

This strategy is modeled in three tiers:

- a) Building code improvements.
- b) Electrification retrofits in existing commercial buildings.
- c) Energy efficiency improvements in existing commercial buildings.

Building code improvements

Building code improvements are modeled with a user-specified percent energy efficiency improvement in new commercial buildings as a result of energy code updates. Code-compliant new non-residential square footage is isolated from existing to determine emissions savings from reduced electricity and natural gas consumption using the 2018 IECC as the baseline.

Key data sources

- Baseline energy consumption for new non-residential square footage if they were built under the 2018 baseline is estimated using an energy intensity factor from the [Pacific Northwest National Laboratory's Energy and Energy Cost Savings Analysis of the 2018 IECC for Commercial Buildings](#).
- Share of commercial energy use that occurs from electricity and natural gas is estimated from the [NLR ComStock](#) dataset for Wyoming, filtered to climate zone 6B and buildings built from 2012-2018.
- The average code compliance percentage is obtained from [a report published by the Institute for Market Transformation](#).

Electrification retrofits in existing commercial buildings

Electrification retrofits are modeled using a user-input percent of heating equipment replaced with electric equipment. The model uses a stock approach to isolate existing non-residential square footage that is replacing equipment then applies the heat pump adoption percentage to that square footage. Increased electricity emissions from heat pump adoption are netted

against decreased natural gas emissions to determine emissions savings from this strategy.

Key data sources

- Equipment turnover rate is estimated from [RMI's Energy Policy Simulator](#).
- Data from the [EIA Commercial Buildings Energy Consumption Survey \(CBECS\)](#) is used to determine fuel usage by fuel type in commercial heating and cooling equipment.
- Inventory data are used to derive energy use intensity rates to apply to replaced equipment to calculate consumption changes.
- Equipment efficiencies to determine additional electricity consumption from heat pump adoption are obtained from:
 - o [NLR's Building Energy Optimization \(BEopt\) tool](#).
 - o [EIA minimum efficiency requirements](#).
 - o [ENERGY STAR criteria](#).

Energy efficiency improvements in existing commercial buildings.

Energy efficiency in existing commercial buildings is modeled with a user-specified increase in efficiency. This increase in energy efficiency is applied to baseline electricity and natural gas consumption to determine decreased emissions from efficiency improvements.

Key data sources

No new data sources are used in this strategy. See the BAU section to see how baseline energy use for existing homes is calculated.

EB3: Encourage adoption of clean energy

This strategy is modeled as a user-defined increase in installed solar capacity (MW-DC) with user-defined start and target years. The strategy offsets electricity emissions from the grid with solar generated energy.

Key data sources

- Expected solar energy from installed solar capacity is obtained from the [NLR PVWATTS tool](#).
- Annual degradation rate of solar generation is applied from [NLR's Annual Technology Baseline](#).

Waste and Water

WW1: Increase waste diversion.

This strategy is modeled as an increase in the proportion of municipal solid waste that is recycled and composted. The user defines the target landfill diversion rate, start year, and year to reach the target diversion rate. Diversion rates are linearly interpolated between the start and year to reach the target.

The model determines the baseline diversion rate using data from the inventory, then the strategy linearly increases diversion from the baseline to the target. This strategy focused on municipal solid waste and does not consider diversion of construction and demolition waste. It is also assumed that hard to divert items such as non-recyclable paper, used oil, anti-freeze, and other miscellaneous waste will continue to be landfilled in the future.

Emissions reductions from decreased tonnage sent to the landfill and increased emissions from increased compost tonnage are considered to calculate the net emissions reduction achieved from diversion. The inventory reports emissions using the waste in place methodology while the model utilizes methane commitment to estimate reduced emissions. The percent decrease in emissions from the methane commitment method is applied to baseline emissions to determine waste in place reductions.

Key data sources

- Wyoming Solid Waste Diversion Study is used to categorize waste and determine what tonnage is eligible for recycling or compost.
- Material emissions factors for landfilling and composting are obtained from the EPA's Waste Reduction Model (WARM).
- Oxidation and collection efficiency rates are obtained from landfill-specific EPA FLIGHT reports. Where a factor was not available, defaults from ICLEI's U.S. Community Protocol are used.

Natural and working lands

NWL3: Improve the management of natural spaces

This strategy is modeled as an increase in preserved natural lands in Cheyenne's MSA region. The user can define the target share of natural lands preserved, the year to begin preserving land, and the year to reach the preservation target.

The share of natural lands preserved is set to at least 100%, meaning that natural lands are maintained and no additional land is preserved. An increase in land preservation, anything over 100%, results in a decrease in emissions

due to an increase in carbon sequestration. Sequestration emissions are estimated using a weighted average of natural land use types in Cheyenne from the inventory. Land types considered include forest, grasslands, and wetlands.

Key data sources

- ICLEI's Land Emissions and Removals Navigator (LEARN) tool is used to:
 - o Determine baseline natural land area.
 - o Calculate a sequestration factor weighted by land use type.